



US012399889B1

(12) **United States Patent**
Hensley et al.

(10) **Patent No.: US 12,399,889 B1**

(45) **Date of Patent: Aug. 26, 2025**

(54) **MODIFYING A DATABASE USING A LARGE LANGUAGE MODEL**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(21) Appl. No.: **18/893,710**

Primary Examiner — Marcin R Filipczyk

(22) Filed: **Sep. 23, 2024**

(74) Attorney, Agent, or Firm — Baker & McKenzie LLP

(51) **Int. Cl.**

G06F 16/23 (2019.01)

G06F 16/2457 (2019.01)

G06F 16/28 (2019.01)

G06F 18/2111 (2023.01)

G06F 40/30 (2020.01)

(57)

ABSTRACT

Methods and systems for producing and transmitting recommendations associated with a database using a large language model (LLM). An exemplary method includes receiving a set of computing prompts from a source. Receiving vector embeddings for the set of computing prompts from an LLM and determining using the vector embeddings, files from a database that correspond with a first computing prompt from the set of computing prompts or a second computing prompt from the set of computing prompts. The system receives first data from the LLM based on the first and second computing prompts, and the files from the database. The method generates second data comprising the first data and computing indicators. Recommendations associated with the database are generated based on the second data and transmitted to a system maintaining the database.

(52) **U.S. Cl.**

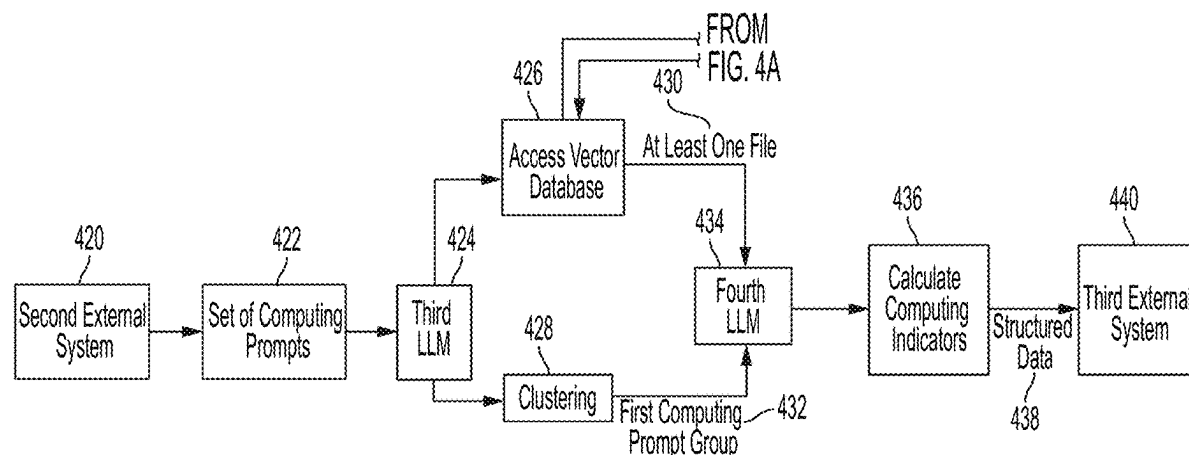
CPC **G06F 16/2379** (2019.01); **G06F 16/24578** (2019.01); **G06F 16/285** (2019.01); **G06F 18/2111** (2023.01); **G06F 40/30** (2020.01)

(58) **Field of Classification Search**

CPC G06F 16/2379; G06F 16/24578; G06F 18/2111; G06F 18/2115; G06F 18/214; G06F 40/30; G06N 20/00; G06V 10/82; G06T 2207/20081

See application file for complete search history.

22 Claims, 20 Drawing Sheets



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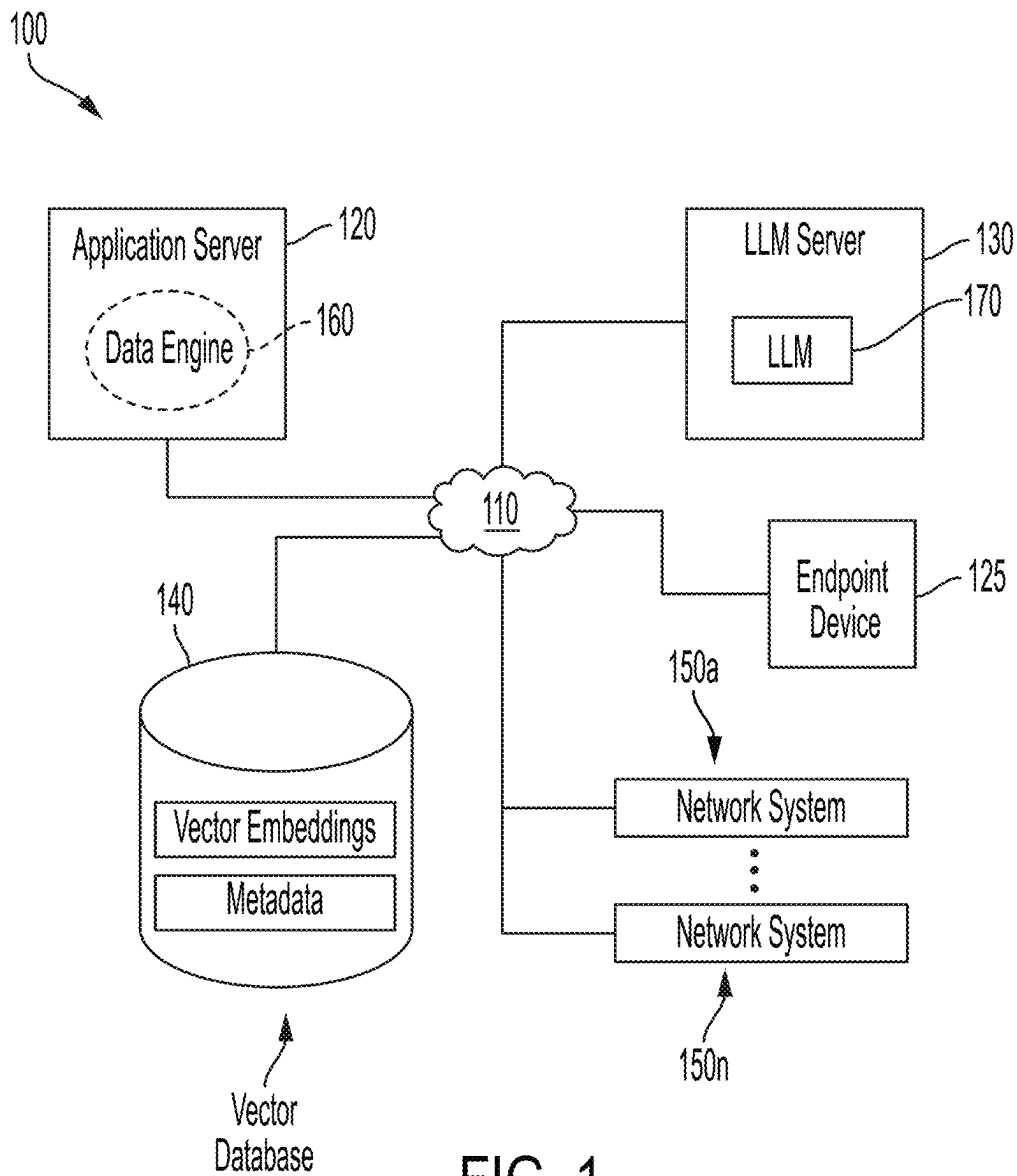


FIG. 1

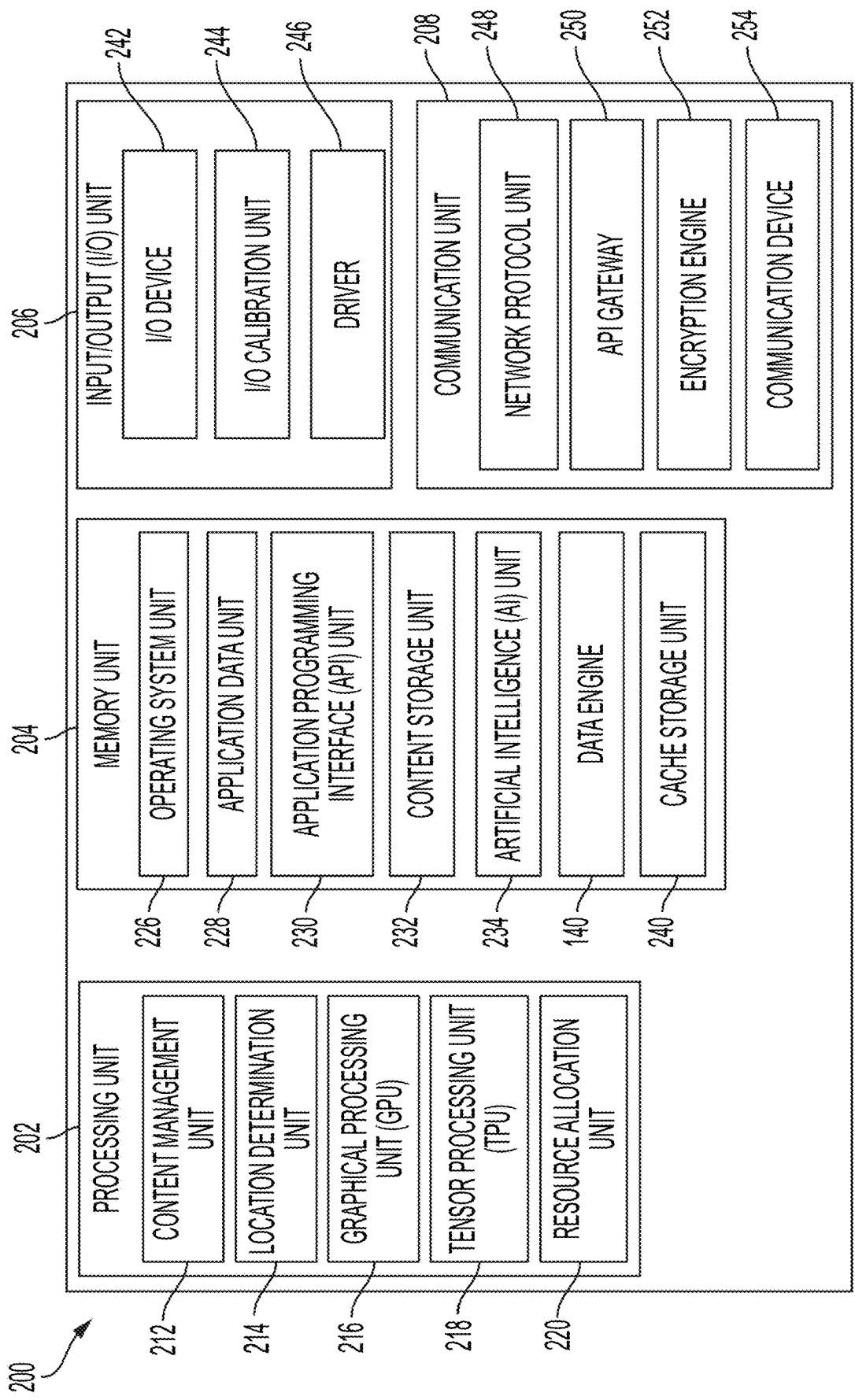


FIG. 2

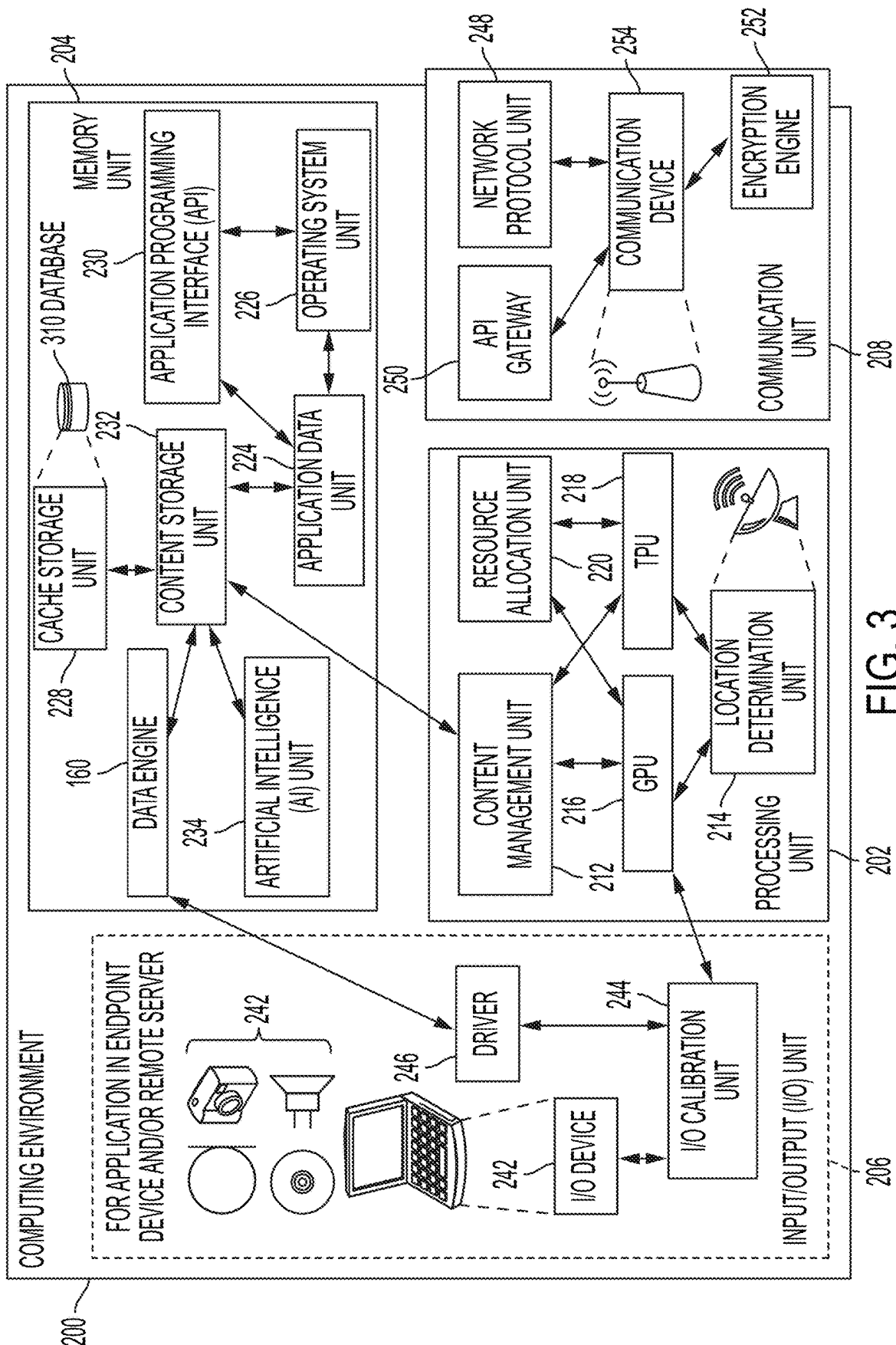


FIG. 3

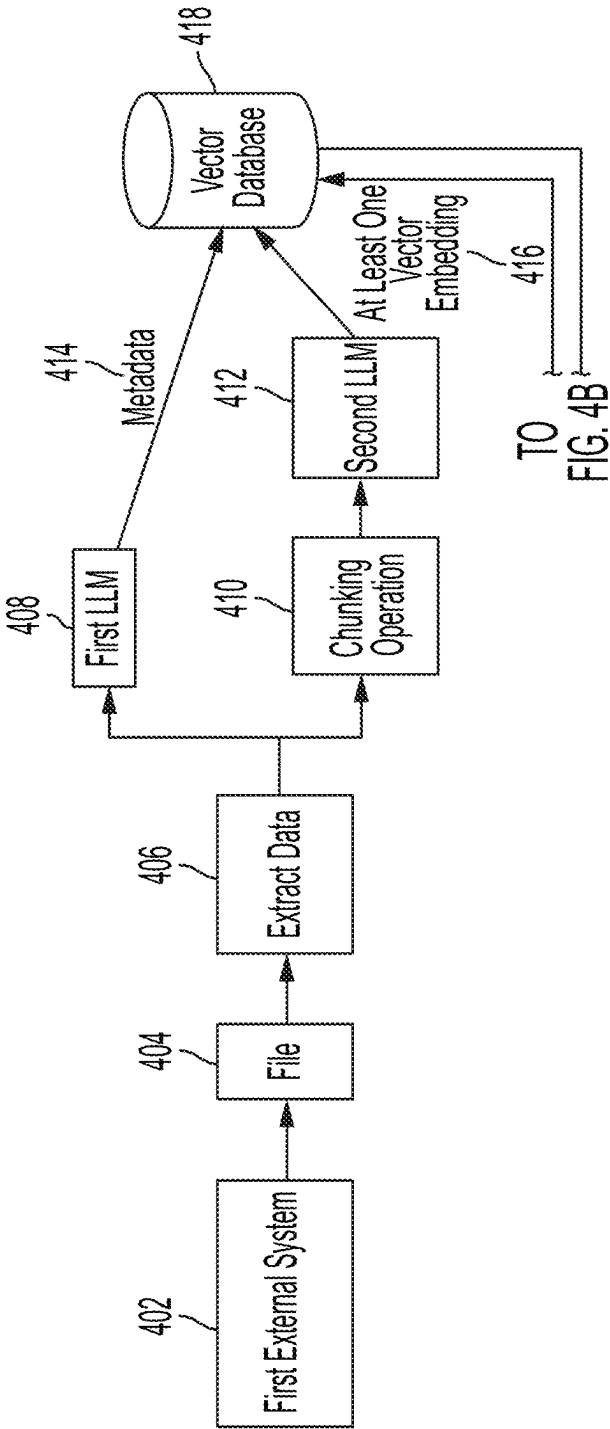


FIG. 4A

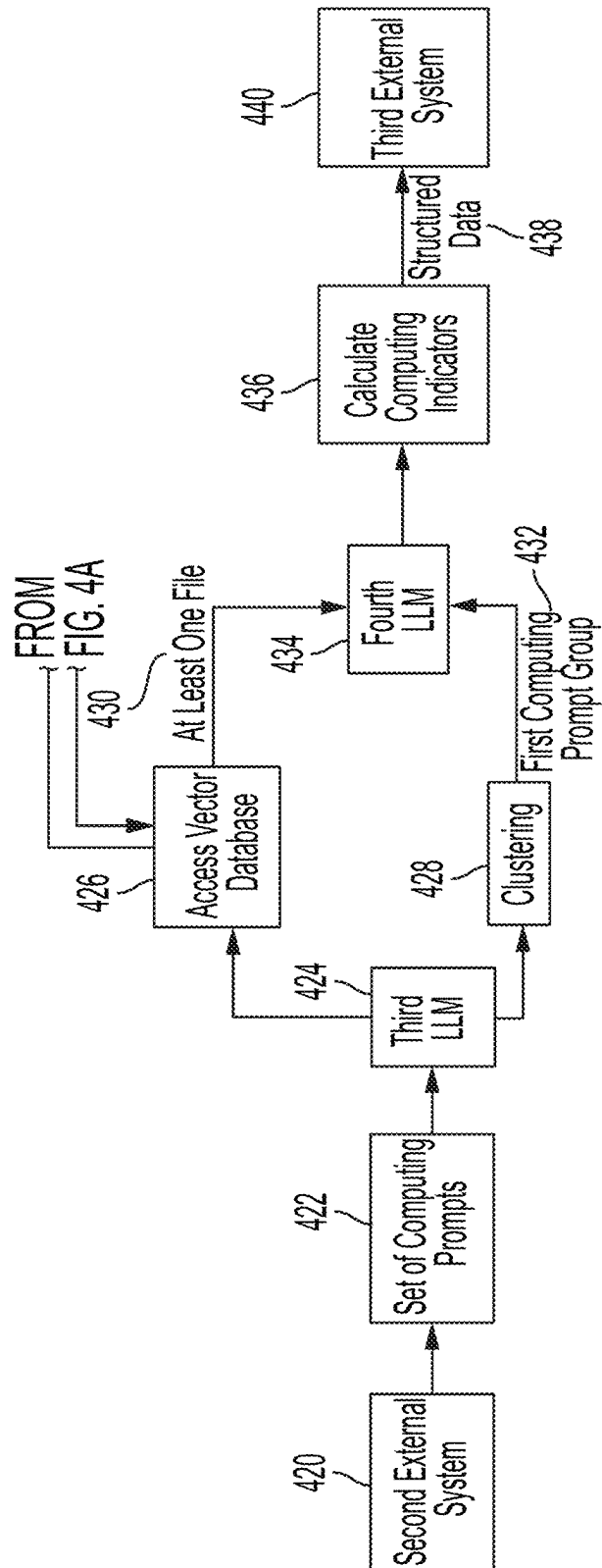


FIG. 4B

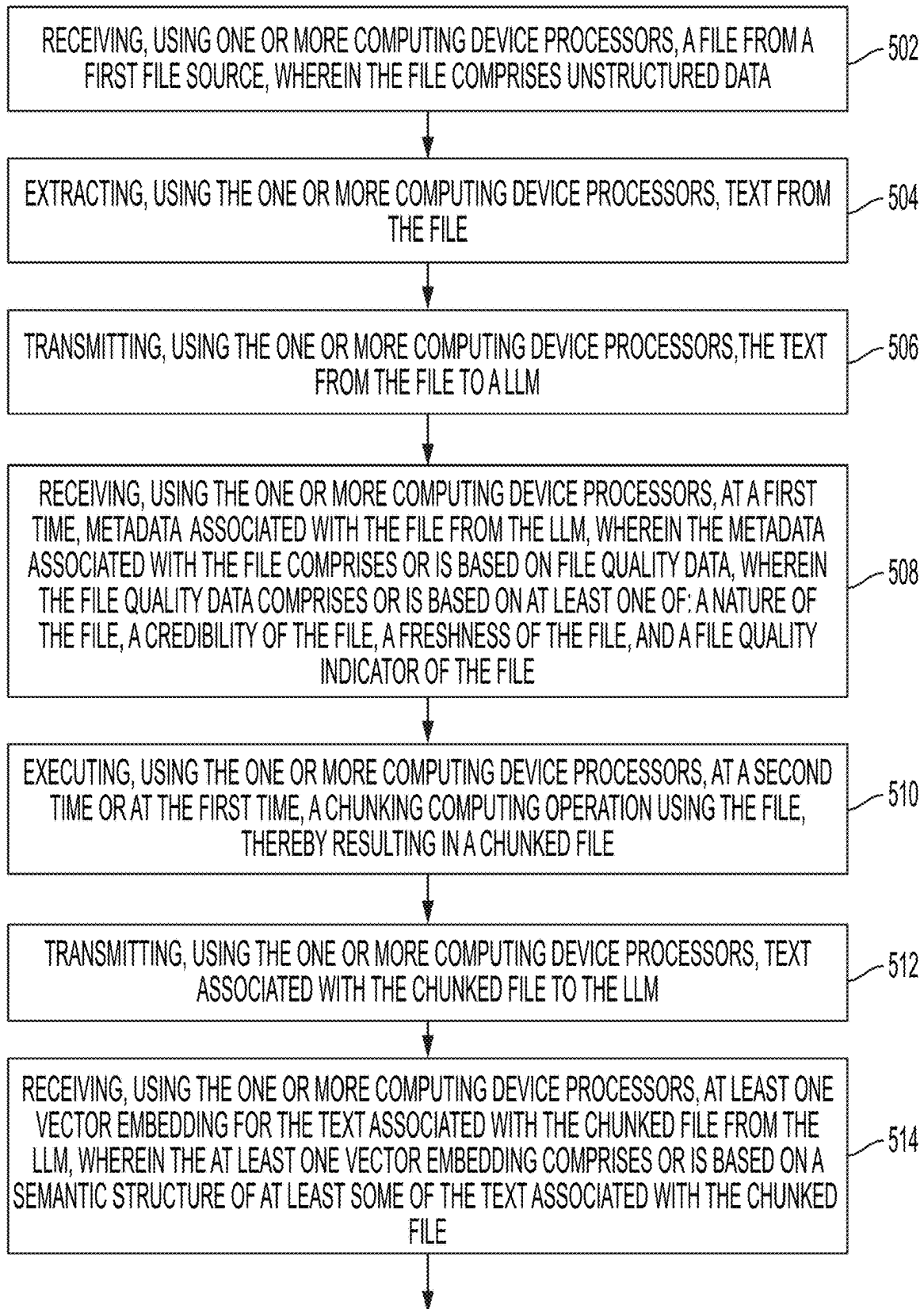


FIG. 5A

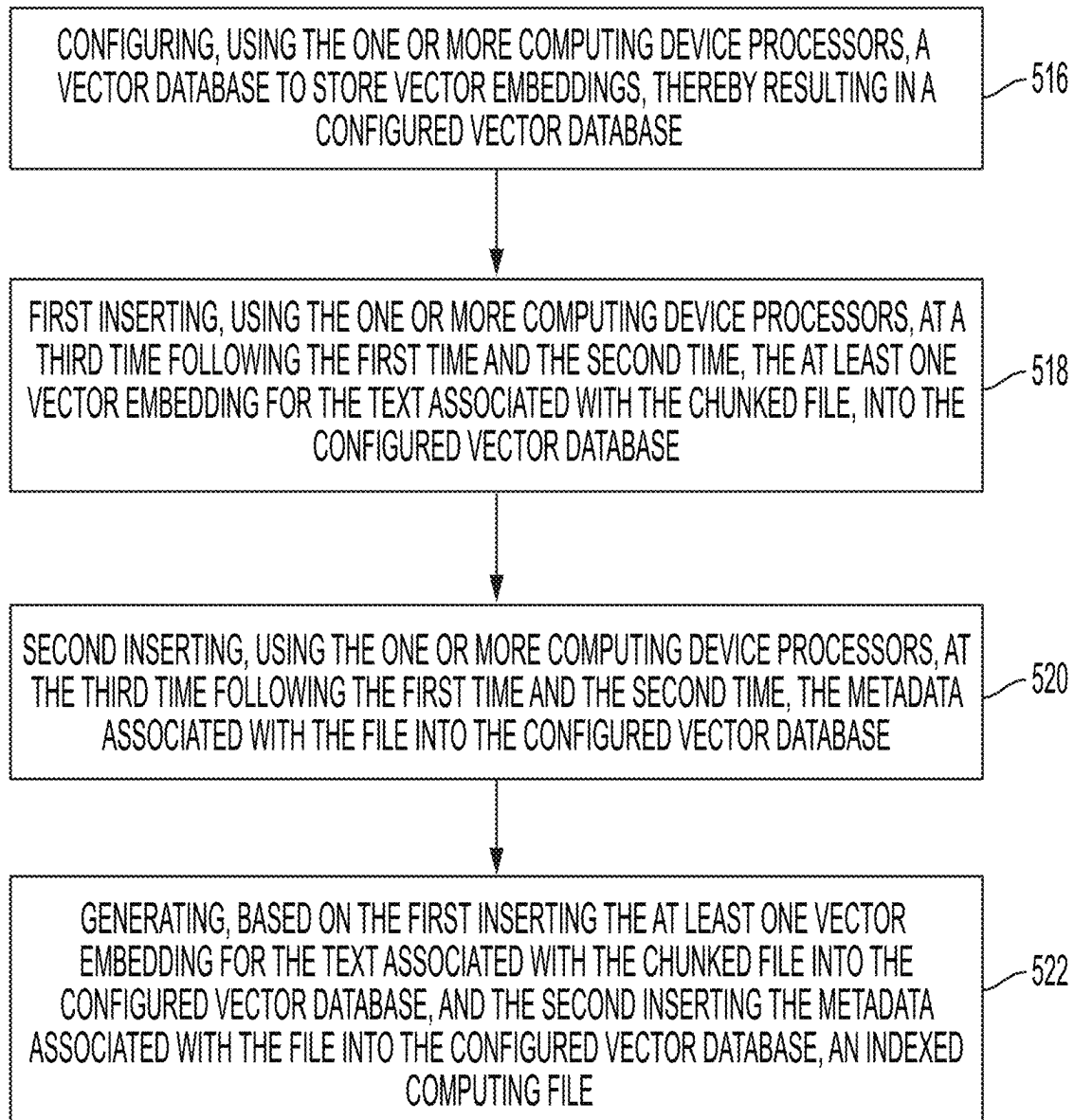


FIG. 5B

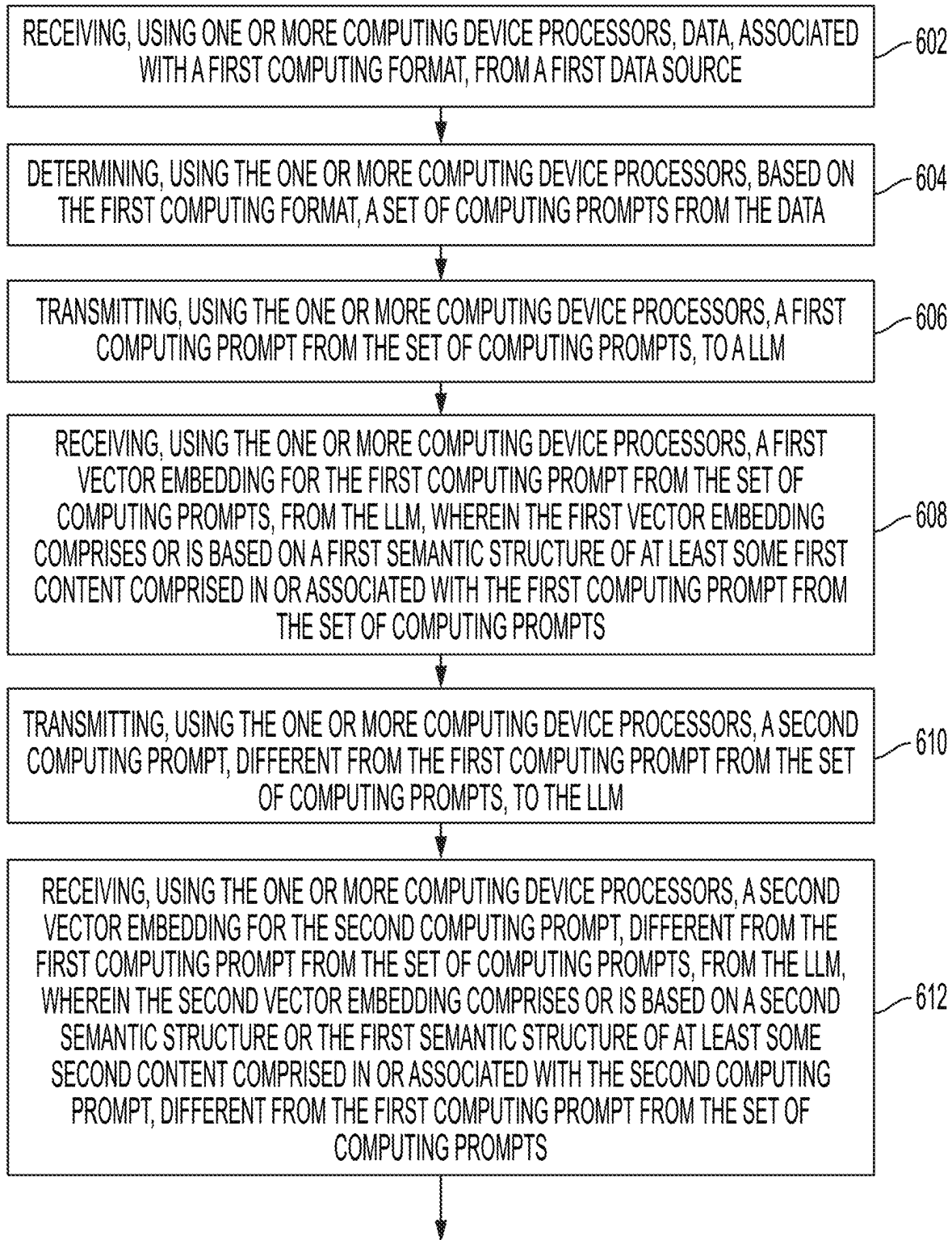


FIG. 6A

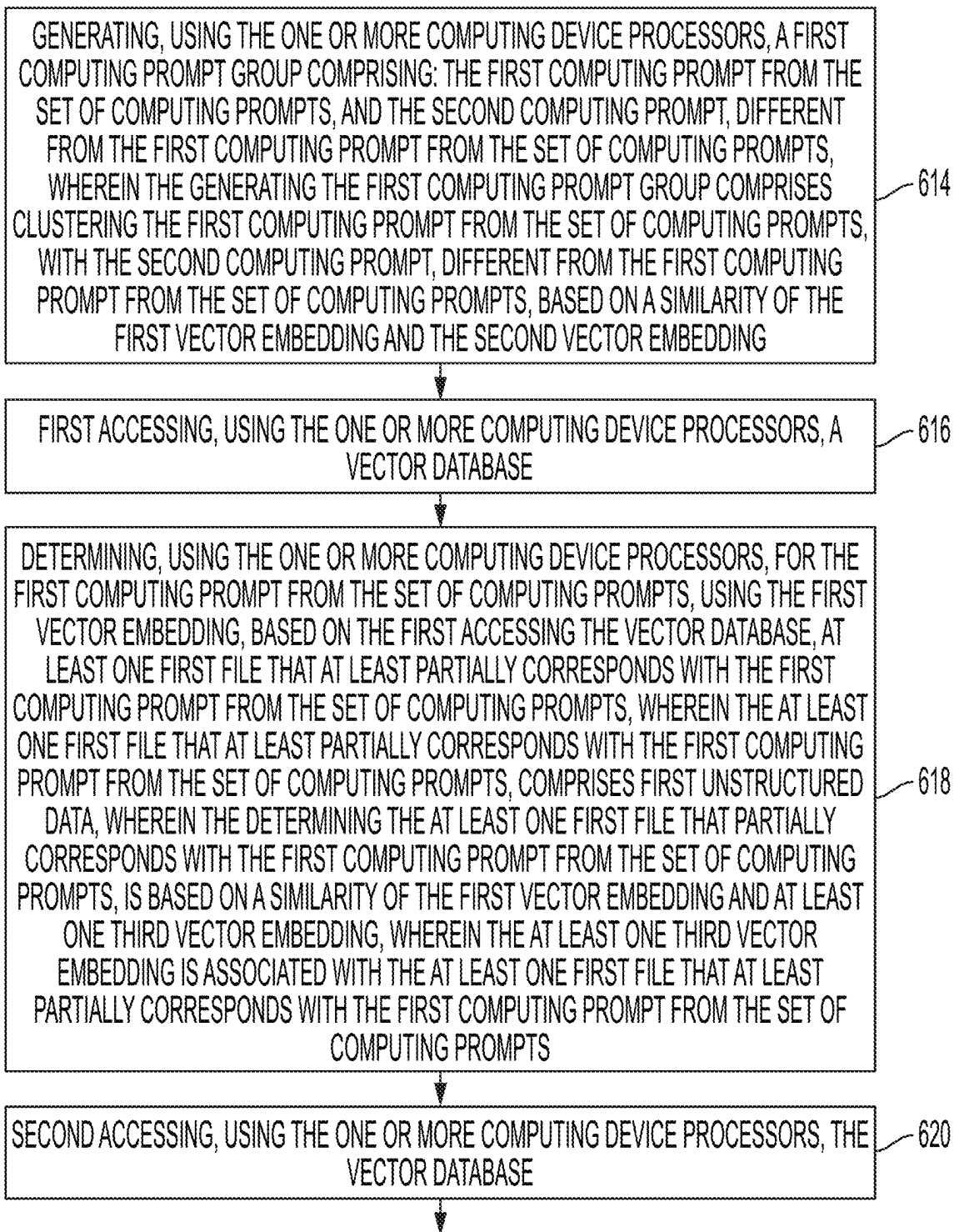


FIG. 6B

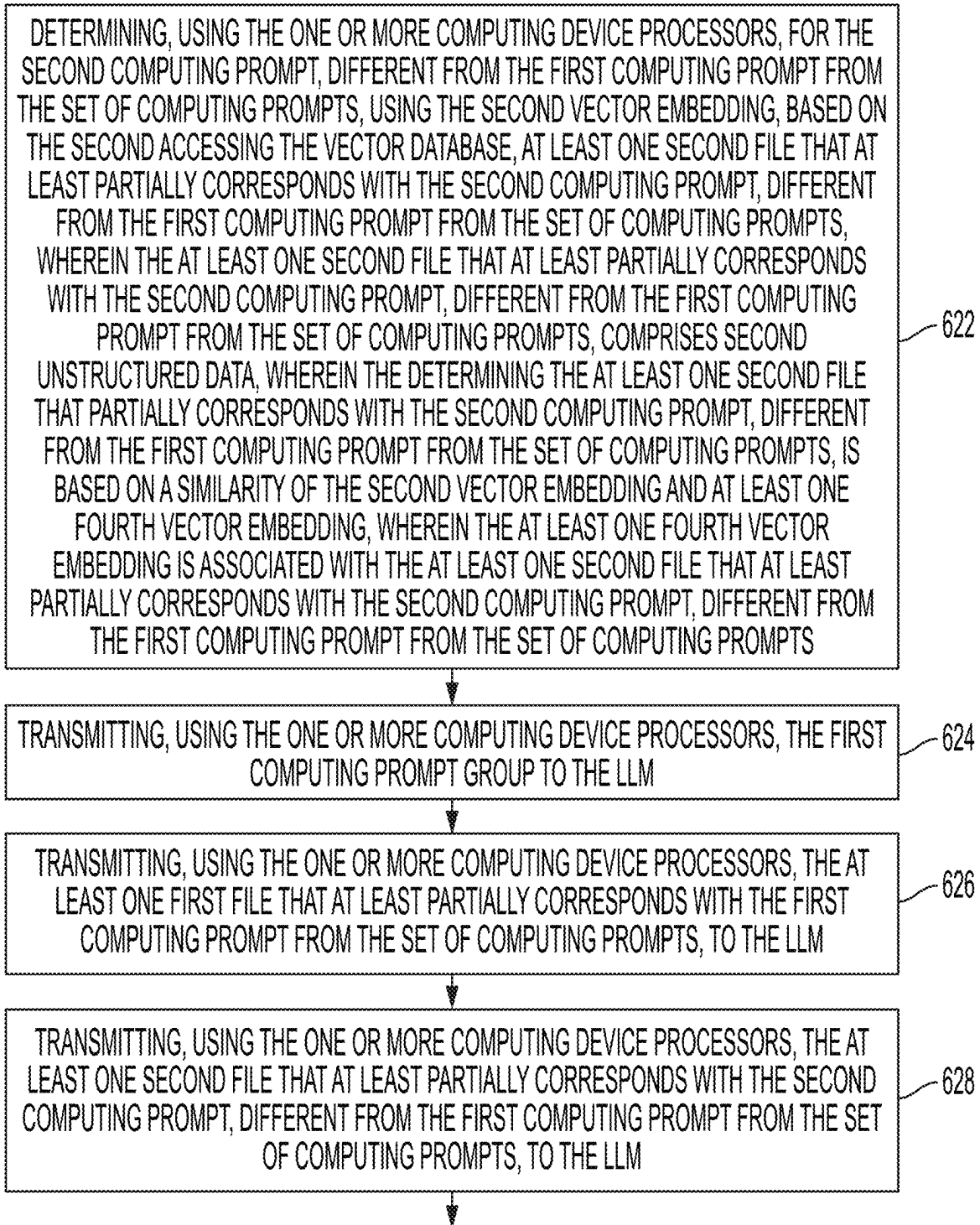


FIG. 6C

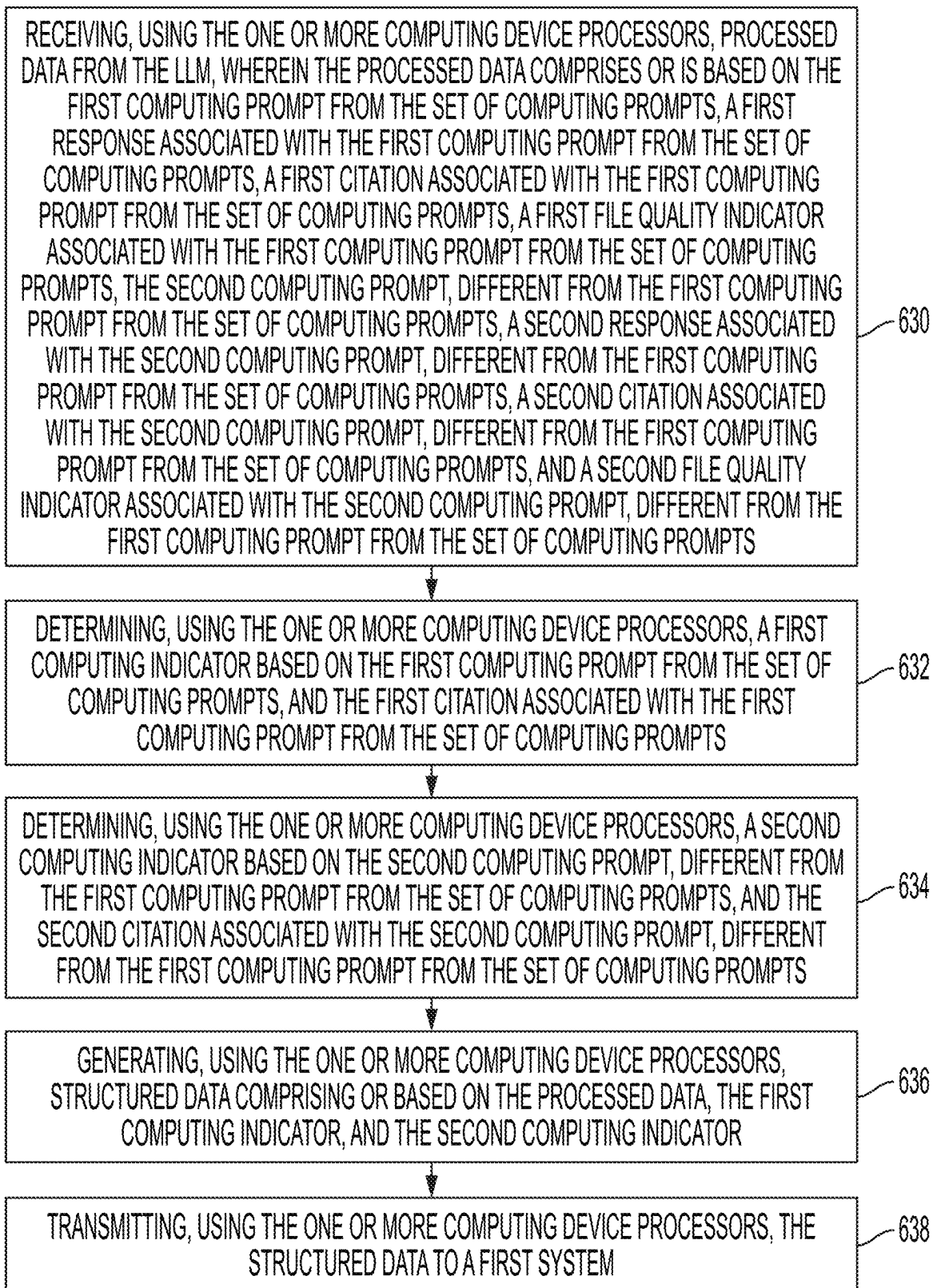


FIG. 6D

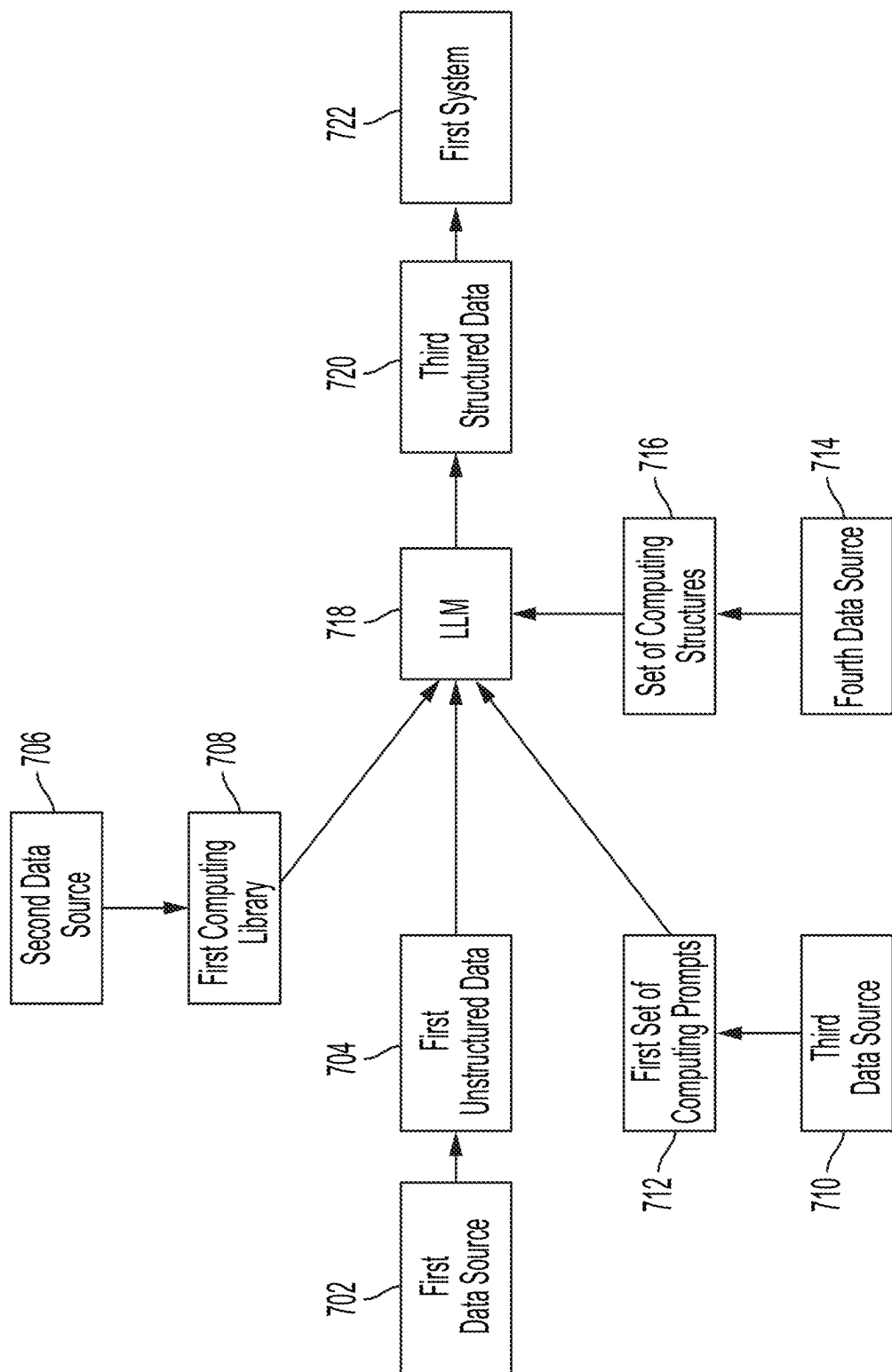


FIG. 7

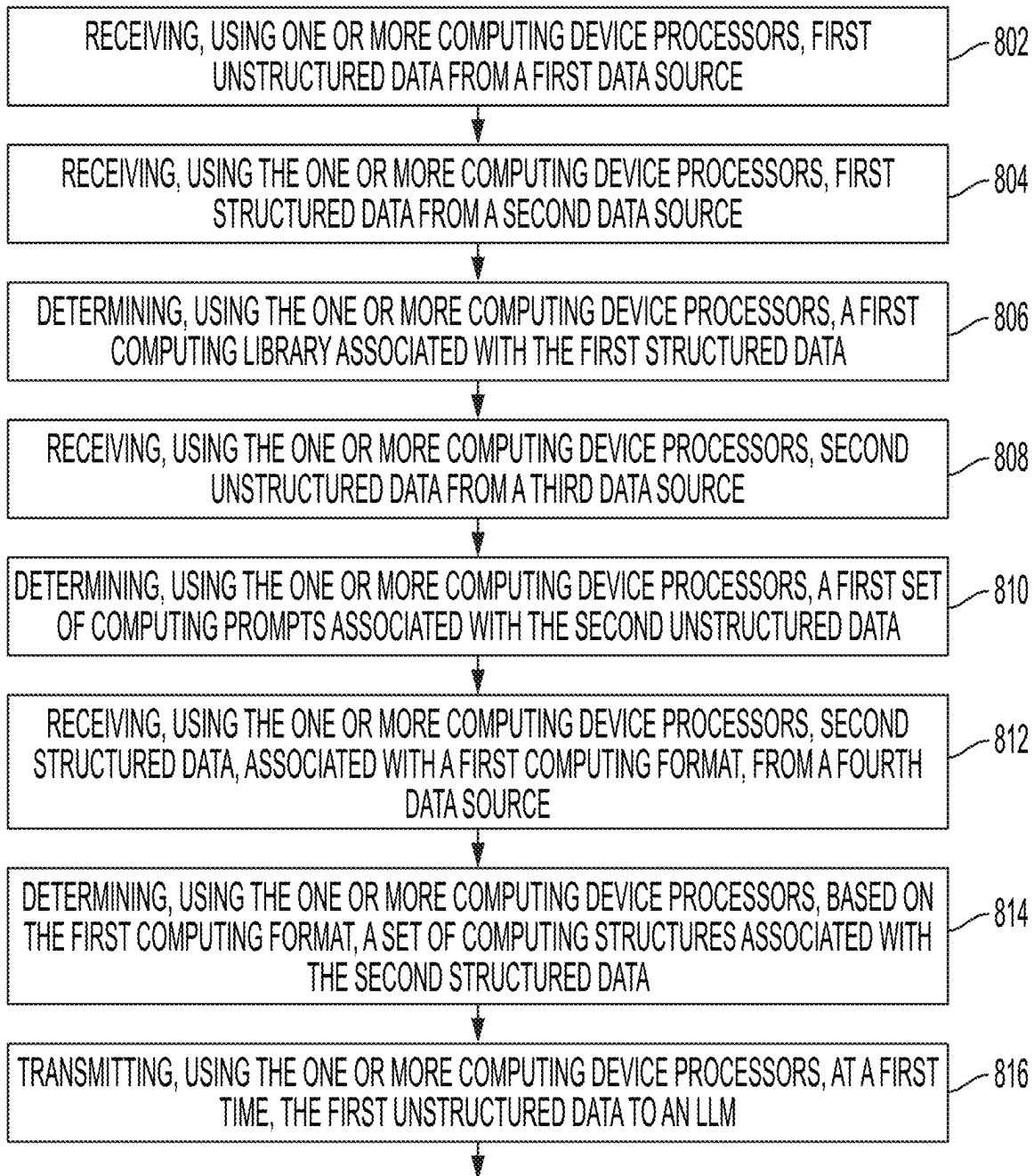


FIG. 8A

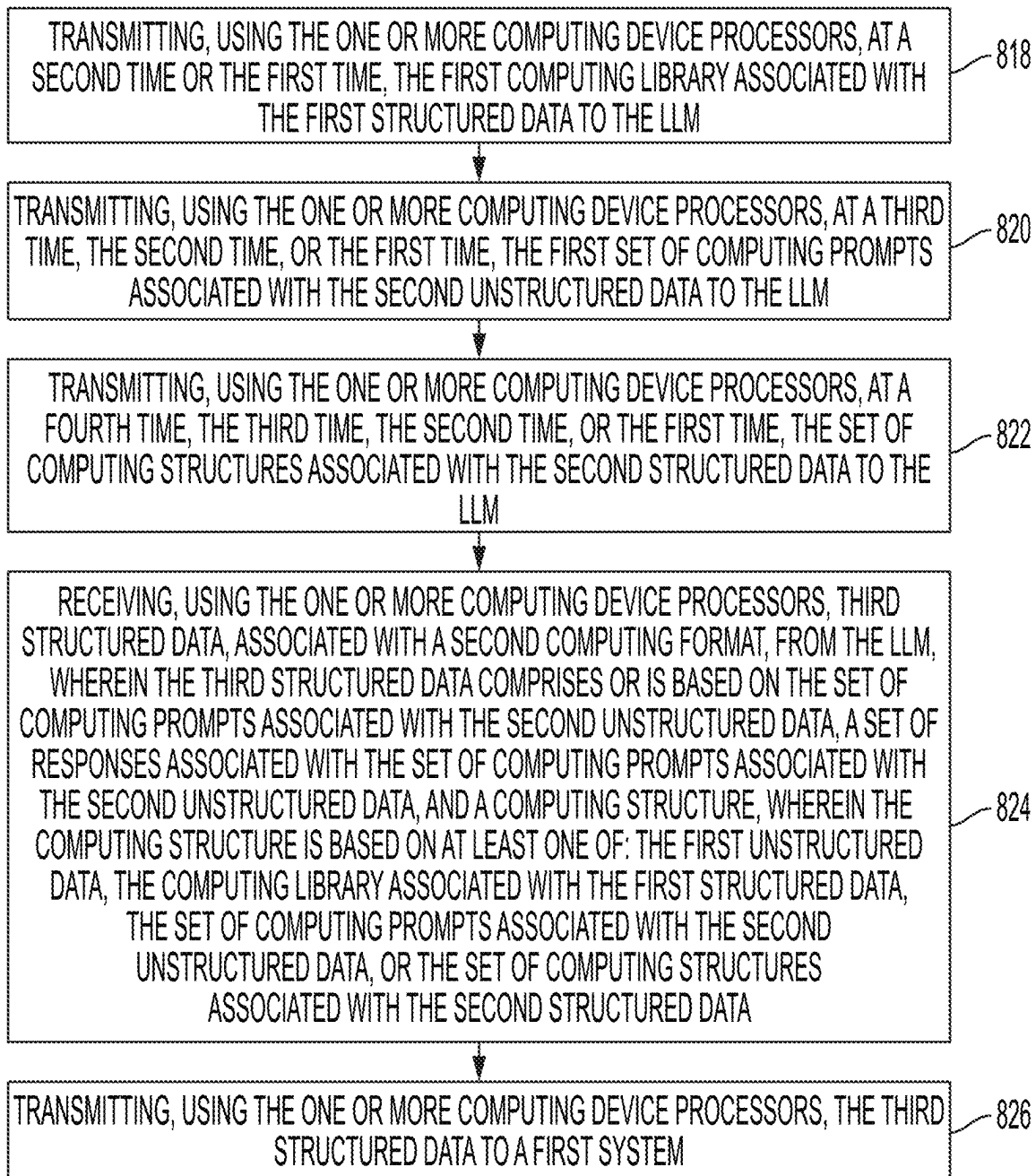
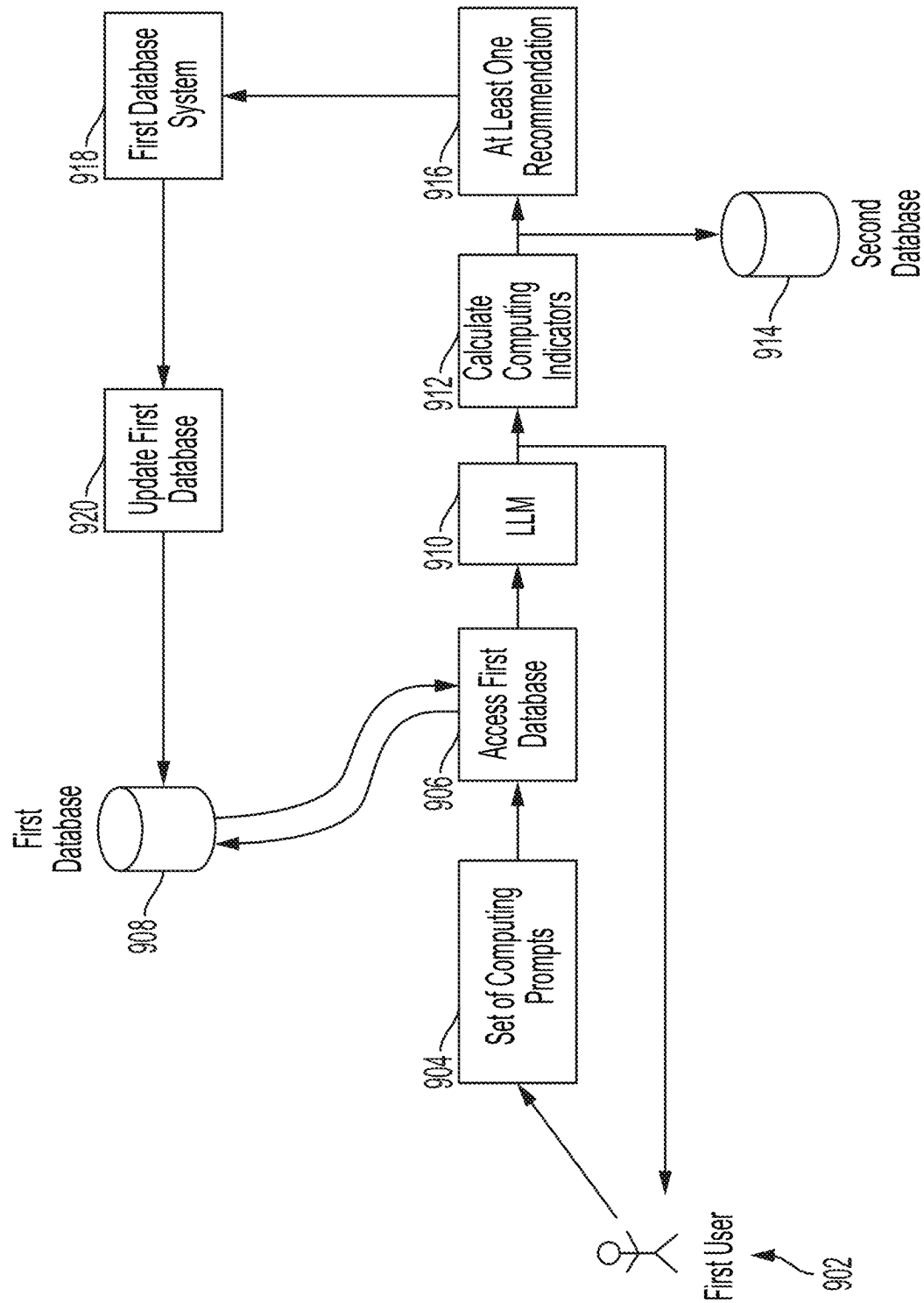


FIG. 8B



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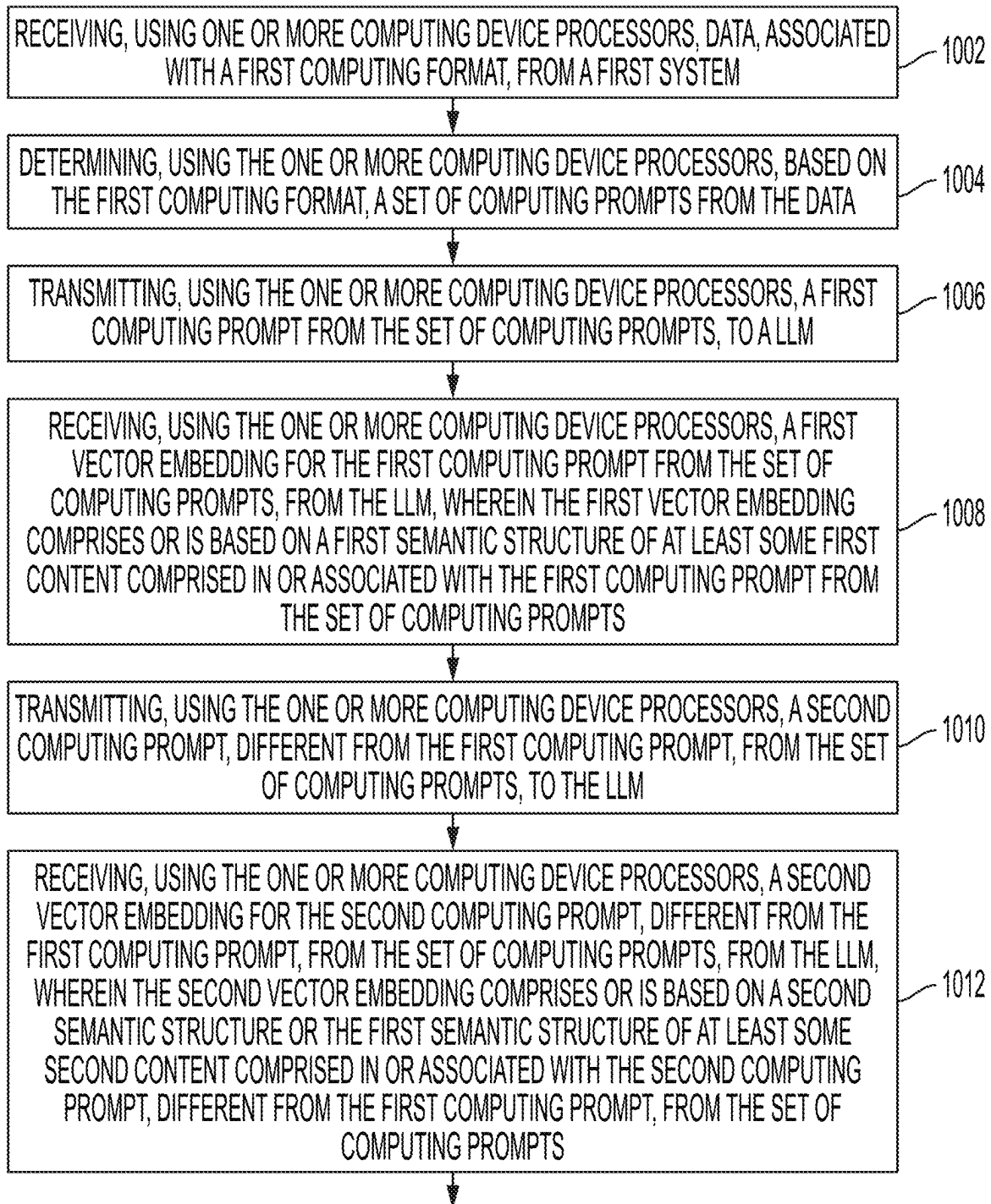


FIG. 10A

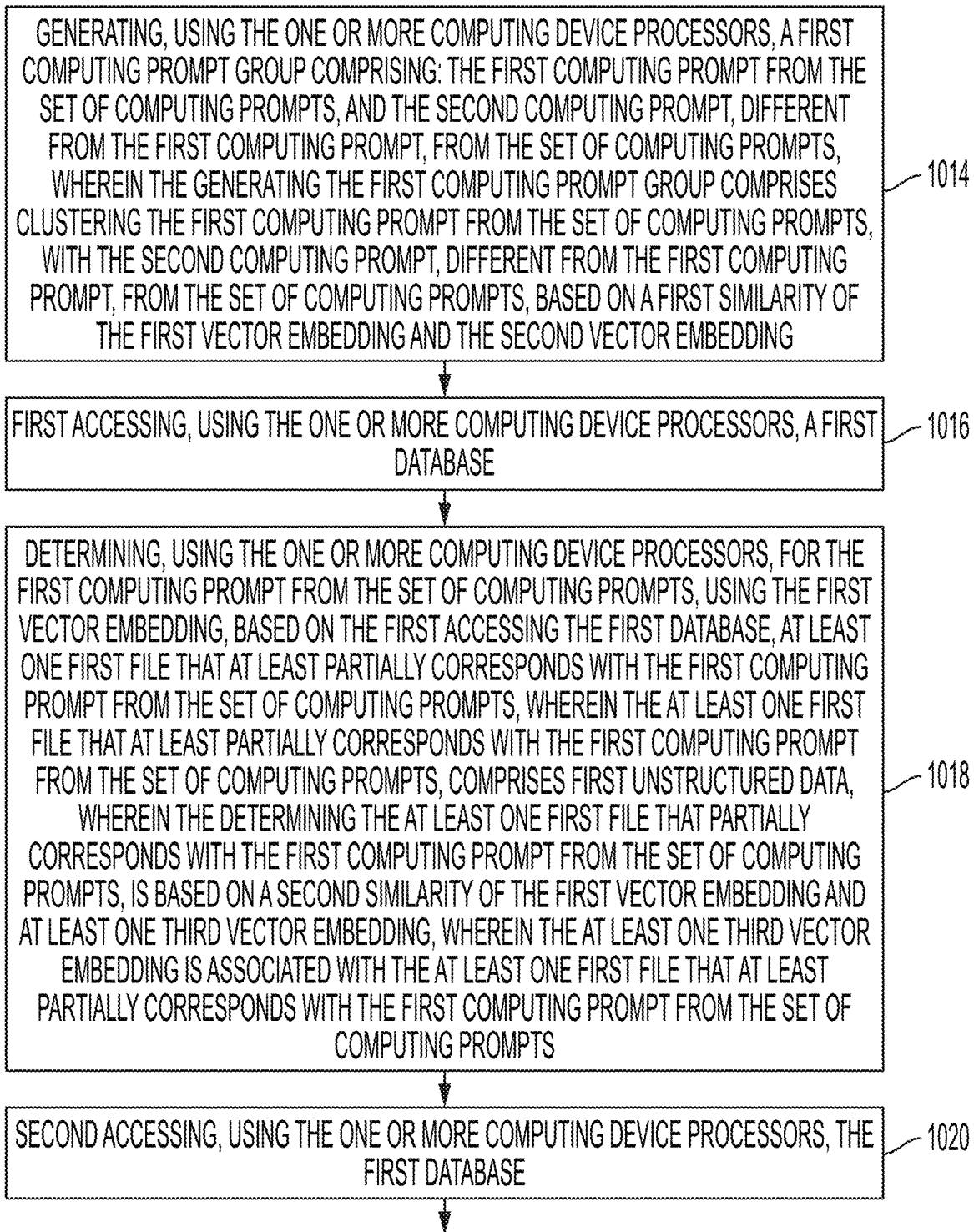


FIG. 10B

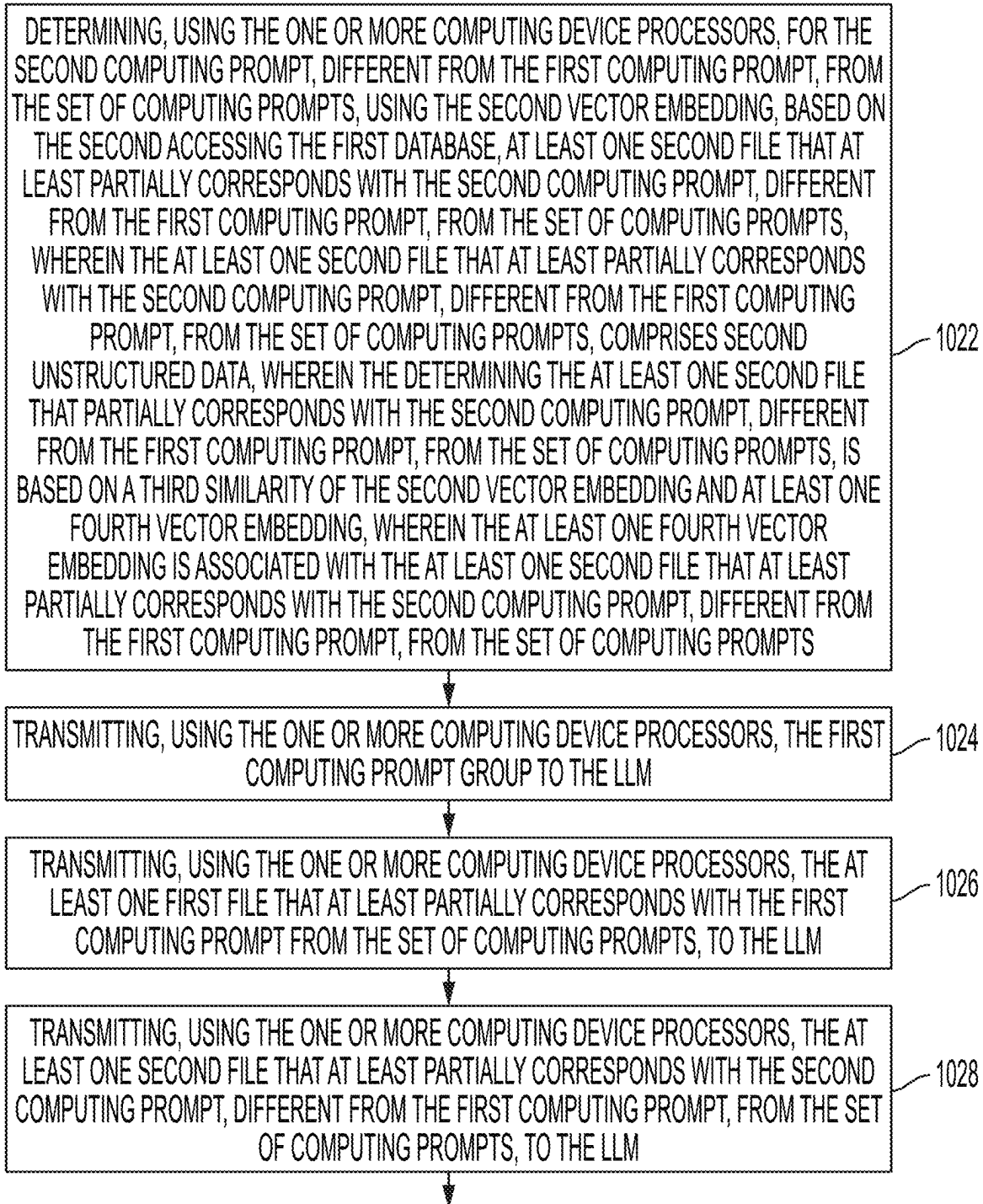


FIG. 10C

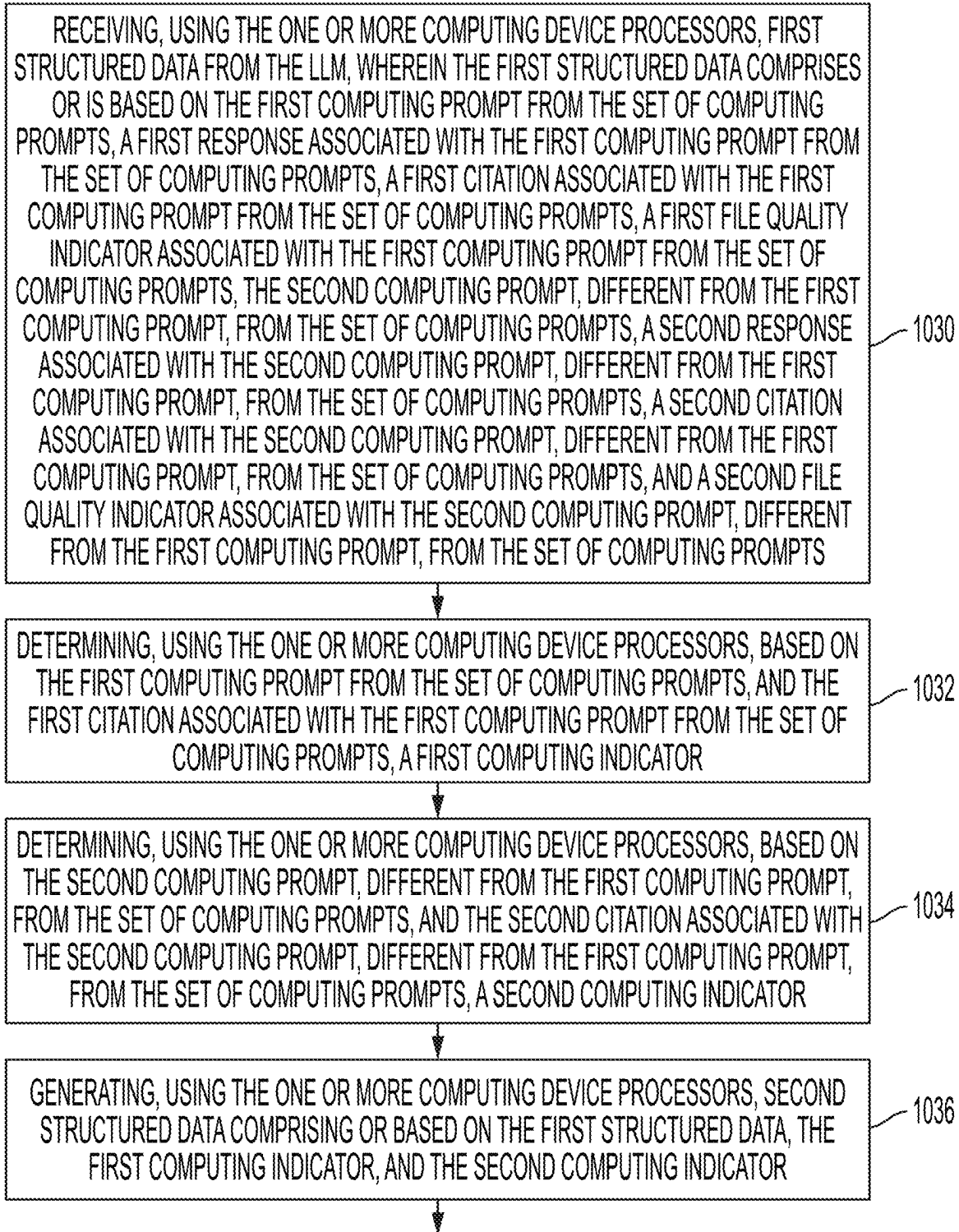


FIG. 10D

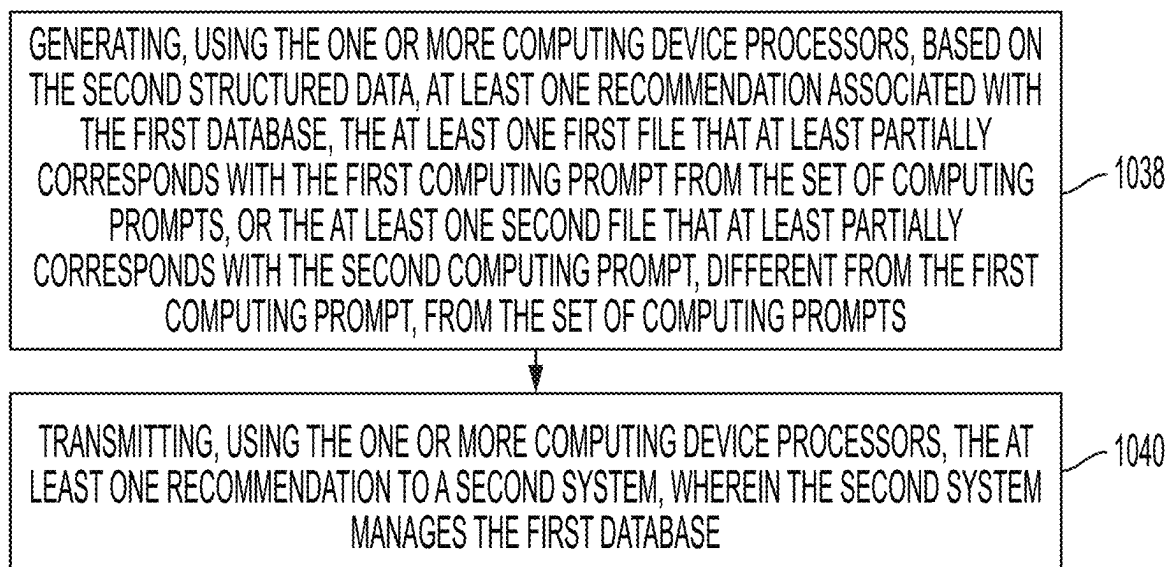


FIG. 10E

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MODIFYING A DATABASE USING A LARGE LANGUAGE MODEL**TECHNICAL FIELD**

The present disclosure relates to databases.

BACKGROUND

There is a need for a method for updating and improving data in a database for various software applications.

SUMMARY

Disclosed are methods and systems for modifying a database using a large language model (LLM). An exemplary method includes: receiving a set of computing prompts from a source; receiving vector embeddings for the set of computing prompts from an LLM; determining, using the vector embeddings, files from a database that correspond with at least one of: a first computing prompt from the set of computing prompts, or a second computing prompt from the set of computing prompts; receiving first data from the LLM based on the first computing prompt, the second computing prompt, and the files from the database; generating second data comprising the first data and computing indicators; generating, based on the second data, recommendations associated with the database; and transmitting the recommendations to a system maintaining the database or a user of the system maintaining the database.

The present disclosure is directed to methods, systems, and computer program products that generate structured data from files comprising unstructured data, and transmit recommendations, based on the structured data, to a system storing the files, using a large language model (LLM), generate structured data from a file comprising unstructured data based on accessing an LLM, generate an indexed computing file based on inserting vector embeddings and metadata, received from an LLM, into a configured vector database, and generate a computing structure based on both unstructured data and structured data using an LLM. According to one embodiment, a method for generating structured data from files comprising unstructured data, and transmitting recommendations, based on the structured data, to a system storing the files, using a large language model (LLM) comprises: receiving, using one or more computing device processors, data, associated with a first computing format, from a first system; determining, using the one or more computing device processors, based on the first computing format, a set of computing prompts from the data; transmitting, using the one or more computing device processors, a first computing prompt from the set of computing prompts, to an LLM; receiving, using the one or more computing device processors, a first vector embedding for the first computing prompt from the set of computing prompts, from the LLM, wherein the first vector embedding comprises or is based on a first semantic structure of at least some first content comprised in or associated with the first computing prompt from the set of computing prompts; transmitting, using the one or more computing device processors, a second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM; receiving, using the one or more computing device processors, a second vector embedding for the second computing prompt, different from the first computing prompt, from the set of computing prompts, from the LLM, wherein the second vector embedding comprises or is based on a

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second semantic structure or the first semantic structure of at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts; generating, using the one or more computing device processors, a first computing prompt group comprising: the first computing prompt from the set of computing prompts, and the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the generating the first computing prompt group comprises clustering the first computing prompt from the set of computing prompts, with the second computing prompt, different from the first computing prompt, from the set of computing prompts, based on a first similarity of the first vector embedding and the second vector embedding; first accessing, using the one or more computing device processors, a first database; determining, using the one or more computing device processors, for the first computing prompt from the set of computing prompts, using the first vector embedding, based on the first accessing the first database, at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, wherein the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, comprises first unstructured data, wherein the determining the at least one first file that partially corresponds with the first computing prompt from the set of computing prompts, is based on a second similarity of the first vector embedding and at least one third vector embedding, wherein the at least one third vector embedding is associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts; second accessing, using the one or more computing device processors, the first database; determining, using the one or more computing device processors, for the second computing prompt, different from the first computing prompt, from the set of computing prompts, using the second vector embedding, based on the second accessing the first database, at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, comprises second unstructured data, wherein the determining the at least one second file that partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, is based on a third similarity of the second vector embedding and at least one fourth vector embedding, wherein the at least one fourth vector embedding is associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts; transmitting, using the one or more computing device processors, the first computing prompt group to the LLM; transmitting, using the one or more computing device processors, the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, to the LLM; transmitting, using the one or more computing device processors, the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM; receiving, using the one or more computing device processors, first structured data from the LLM, wherein the first structured data comprises or is based

on the first computing prompt from the set of computing prompts, a first response associated with the first computing prompt from the set of computing prompts, a first citation associated with the first computing prompt from the set of computing prompts, a first file quality indicator associated with the first computing prompt from the set of computing prompts, the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second response associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, and a second file quality indicator associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts; determining, using the one or more computing device processors, based on the first computing prompt from the set of computing prompts, and the first citation associated with the first computing prompt from the set of computing prompts, a first computing indicator; determining, using the one or more computing device processors, based on the second computing prompt, different from the first computing prompt, from the set of computing prompts, and the second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second computing indicator; generating, using the one or more computing device processors, second structured data comprising or based on the first structured data, the first computing indicator, and the second computing indicator; generating, using the one or more computing device processors, based on the second structured data, at least one recommendation associated with at least one of: the first database, the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, or the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts; and transmitting, using the one or more computing device processors, the at least one recommendation to a second system or a user associated with the second system, wherein the second system manages the first database.

In some embodiments, the method further comprises transmitting, using the one or more computing device processors, the set of computing prompts and a set of responses associated with the set of computing prompts, to the first system.

In other embodiments, the method further comprises: accessing, using the one or more computing device processors, a second database; and transmitting, using the one or more computing device processors, the second structured data to the second database.

In one embodiment, the first similarity of the first vector embedding and the second embedding, is based on a semantic similarity of the first vector embedding and the second vector embedding.

Furthermore, in some embodiments, the semantic similarity of the first vector embedding and the second vector embedding is calculated based on one or more of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity.

In another embodiment, the second similarity of the first vector embedding and the at least one third vector embed-

ding further comprises a semantic similarity of the first vector embedding and the at least one third vector embedding.

According to some embodiments, an entry in the first database comprises a vector embedding associated with a file and metadata associated with the file.

In other embodiments, a system and a computer program can include or execute the method described above. These and other implementations may each optionally include one or more of the following features.

In some embodiments, the system comprises or is comprised in one or more computing systems associated with one or more locations

According to one embodiment, the first computing prompt group comprises at least three computing prompts from the set of computing prompts.

In some cases, the first structured data comprises the set of computing prompts, a set of responses associated with the set of computing prompts, a set of citations associated with the set of computing prompts, and a set of file quality indicators associated with the set of computing prompts.

In one embodiment, the LLM is hosted on a third-party server.

In another embodiment, the LLM is hosted on a local server.

In some embodiments, the method comprises: receiving, using one or more computing device processors, data, associated with a first computing format, from a first system; determining, using the one or more computing device processors, based on the first computing format, a set of computing prompts from the data; transmitting, using the one or more computing device processors, a first computing prompt from the set of computing prompts, to an LLM; receiving, using the one or more computing device processors, a first vector embedding for the first computing prompt from the set of computing prompts, from the LLM, wherein the first vector embedding comprises or is based on a first semantic structure of at least some first content comprised in or associated with the first computing prompt from the set of computing prompts; transmitting, using the one or more computing device processors, a second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM; receiving, using the one or more computing device processors, a second vector embedding for the second computing prompt, different from the first computing prompt, from the set of computing prompts, from the LLM, wherein the second vector embedding comprises or is based on a second semantic structure or the first semantic structure of at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts; generating, using the one or more computing device processors, a first computing prompt group comprising: the first computing prompt from the set of computing prompts, and the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the generating the first computing prompt group comprises clustering the first computing prompt from the set of computing prompts, with the second computing prompt, different from the first computing prompt, from the set of computing prompts, based on a first similarity of the first vector embedding and the second vector embedding; first accessing, using the one or more computing device processors, a first database; determining, using the one or more computing device processors, for the first computing prompt from the set of computing prompts, using the first vector embedding, based on the first accessing

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the first database, first file data, associated with at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, wherein the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, comprises first unstructured data, wherein the determining the first file data, associated with the at least one first file that partially corresponds with the first computing prompt from the set of computing prompts, is based on a second similarity of the first vector embedding and at least one third vector embedding, wherein the at least one third vector embedding is associated with or comprised in the first file data; second accessing, using the one or more computing device processors, the first database; determining, using the one or more computing device processors, the first computing prompt, different from the first computing prompt, from the set of computing prompts, using the second vector embedding, based on the second accessing the first database, second file data, associated with at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, is based on a third similarity of the second vector embedding and at least one fourth vector embedding, wherein the at least one fourth vector embedding is associated with or comprised in the second file data; transmitting, using the one or more computing device processors, the first computing prompt group to the LLM; transmitting, using the one or more computing device processors, the first file data, associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, to the LLM; transmitting, using the one or more computing device processors, the second file data, associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM; receiving, using the one or more computing device processors, first structured data from the LLM, wherein the first structured data comprises the first computing prompt from the set of computing prompts, a first response associated with the first computing prompt from the set of computing prompts, a first citation associated with the first computing prompt from the set of computing prompts, a first file quality indicator associated with the first computing prompt from the set of computing prompts, the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second response associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, and a second file quality indicator associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts; determining, using the one or more computing device processors, based on the first computing prompt from the set of computing prompts, and the first citation associated with the first computing prompt from the set of computing prompts, a first computing indicator; determining,

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using the one or more computing device processors, based on the second computing prompt, different from the first computing prompt, from the set of computing prompts, and the second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second computing indicator; generating, using the one or more computing device processors, second structured data comprising or based on the first structured data, the first computing indicator and the second computing indicator; accessing, using the one or more computing device processors, a second database; transmitting, using the one or more computing device processors, the second structured data to the second database; generating, using the one or more computing device processors, based on the second structured data, at least one recommendation associated with the first database; and transmitting, using the one or more computing device processors, the at least one recommendation to a second system, wherein the second system manages the first database.

According to one embodiment, the unstructured data comprises raw information or information without a predetermined structure or format.

In some cases, the first citation, associated with the first computing prompt from the set of computing prompts, comprises at least one of: at least some first text from the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, at least some second text from the second file data associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, a file name corresponding with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts or the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, and a page number corresponding with the at least some first text from the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, or the at least some second text from the second file data associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts.

The first file quality indicator, in some embodiments, is based on metadata associated with the first citation.

In one embodiment, the determining, using the one or more computing device processors, based on the first computing prompt from the set of computing prompts, and the first citation, associated with the first computing prompt from the set of computing prompts, the first computing indicator, is based on a semantic similarity of the first vector embedding and a fifth vector embedding, wherein the fifth vector embedding is associated with the first citation.

According to some embodiments, the fifth vector embedding comprises or is comprised in the at least one third vector embedding or the at least one fourth vector embedding.

In one embodiment, the one or more computing device processors are comprised in one or more computing systems, wherein the one or more computing systems are located in one or more locations.

According to some embodiments, the at least one recommendation associated with the first database comprises at least one of: at least one recommendation associated with

the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, at least one recommendation associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, or at least one recommendation associated with at least one third file that at least partially corresponds with a third computing prompt, different from the first computing prompt and the second computing prompt, from the set of computing prompts.

In other embodiments, the method further comprises updating or modifying, based on the at least one recommendation, the first database, wherein the updating or modifying the first database comprises or is based on at least one of: at least one request for an additional file to be inserted into the first database, wherein the additional file comprises or is associated with an improved file quality indicator or the additional file comprises or is associated with second data not comprised in the first database, or at least one request for an existing file in the first database to be updated.

In another embodiment, a method for generating structured data from a file comprising unstructured data, based on accessing an LLM comprises: receiving, using one or more computing device processors, data, associated with a first computing format, from a first data source; determining, using the one or more computing device processors, based on the first computing format, a set of computing prompts from the data; transmitting, using the one or more computing device processors, a first computing prompt from the set of computing prompts, to an LLM; receiving, using the one or more computing device processors, a first vector embedding for the first computing prompt from the set of computing prompts, from the LLM, wherein the first vector embedding comprises or is based on a first semantic structure of at least some first content comprised in or associated with the first computing prompt from the set of computing prompts; transmitting, using the one or more computing device processors, a second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM; receiving, using the one or more computing device processors, a second vector embedding for the second computing prompt, different from the first computing prompt, from the set of computing prompts, from the LLM, wherein the second vector embedding comprises or is based on a second semantic structure or the first semantic structure of at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts; generating, using the one or more computing device processors, a first computing prompt group comprising: the first computing prompt from the set of computing prompts, and the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the generating the first computing prompt group comprises clustering the first computing prompt from the set of computing prompts, with the second computing prompt, different from the first computing prompt, from the set of computing prompts, based on a similarity of the first vector embedding and the second vector embedding; first accessing, using the one or more computing device processors, a vector database; determining, using the one or more computing device processors, for the first computing prompt from the set of computing prompts, using the first vector embedding, based on the first accessing the vector database, at least one first file that at least partially corresponds with the first computing prompt from the set of computing

prompts, wherein the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, comprises first unstructured data, wherein the determining the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, is based on a similarity of the first vector embedding and at least one third vector embedding, wherein the at least one third vector embedding is associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts; second accessing, using the one or more computing device processors, the vector database; determining, using the one or more computing device processors, for the second computing prompt, different from the first computing prompt, from the set of computing prompts, using the second vector embedding, based on the second accessing the vector database, at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, comprises second unstructured data, wherein the determining the at least one second file that partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, is based on a similarity of the second vector embedding and at least one fourth vector embedding, wherein the at least one fourth vector embedding is associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts; transmitting, using the one or more computing device processors, the first computing prompt group to the LLM; transmitting, using the one or more computing device processors, the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, to the LLM; transmitting, using the one or more computing device processors, the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM; receiving, using the one or more computing device processors, processed data from the LLM, wherein the processed data comprises or is based on the first computing prompt from the set of computing prompts, a first response associated with the first computing prompt from the set of computing prompts, a first citation associated with the first computing prompt from the set of computing prompts, a first file quality indicator associated with the first computing prompt from the set of computing prompts, the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second response associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, and a second file quality indicator associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts; determining, using the one or more computing device processors, a first computing indicator based on the first computing prompt from the set of computing prompts, and the first citation associated with the first computing prompt from the set of computing prompts; determining, using the one or more computing device processors, a second computing indicator based on the second

computing prompt, different from the first computing prompt, from the set of computing prompts, and the second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts; generating, using the one or more computing device processors, structured data comprising or based on the processed data, the first computing indicator, and the second computing indicator; and transmitting, using the one or more computing device processors, the structured data to a first system.

In some embodiments, the method further comprises: receiving, using the one or more computing device processors, filter data from the first data source or a second data source, and executing, using the one or more computing device processors, based on the filter data, a filtering operation on the vector database, thereby limiting entries associated with the vector database.

Furthermore, the filter data can comprise one or more of: a nature of the file, a credibility of the file, a freshness of the file, a file quality indicator of the file, a name of the file, and a source of the file.

In some embodiments, the first computing format comprises JavaScript Object Notation (JSON) format.

According to some embodiments, the similarity of the first vector embedding and the second vector embedding, is based on a semantic similarity of the first vector embedding and the second vector embedding.

Furthermore, the semantic similarity is calculated based on one or more of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity.

In some embodiments, the similarity of the first vector embedding and the at least one third vector embedding further comprises a semantic similarity of the first vector embedding and the at least one third vector embedding.

In other embodiments, a system and a computer program can include or execute the method described above. These and other implementations may each optionally include one or more of the following features.

According to some embodiments, the system comprises or is comprised in one or more computing systems associated with one or more locations.

The first computing prompt group, in some embodiments, further comprises at least three computing prompts.

In some cases, the processed data comprises the set of computing prompts, a set of responses associated with the set of computing prompts, a set of citations associated with the set of computing prompts, and a set of file quality indicators associated with the set of computing prompts.

In one embodiment, an entry in the vector database comprises at least one of: a vector embedding and metadata associated with an indexed file, or the vector embedding and metadata associated with a file comprising unstructured data.

In some embodiments, the unstructured data comprises raw information or information without a predetermined structure or format.

In other embodiments, the method comprises: receiving, using one or more computing device processors, data, associated with a first computing format, from a first data source; determining, using the one or more computing device processors, based on the first computing format, a set of computing prompts from the data; transmitting, using the one or more computing device processors, a first computing prompt from the set of computing prompts, to an LLM; receiving, using the one or more computing device processors, a first vector embedding for the first computing prompt

from the set of computing prompts, from the LLM, wherein the first vector embedding comprises or is based on a first semantic structure of at least some first content comprised in or associated with the first computing prompt from the set of computing prompts; transmitting, using the one or more computing device processors, a second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM; receiving, using the one or more computing device processors, a second vector embedding for the second computing prompt, different from the first computing prompt, from the set of computing prompts, from the LLM, wherein the second vector embedding comprises or is based on a second semantic structure or the first semantic structure of at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts; generating, using the one or more computing device processors, a first computing prompt group comprising: the first computing prompt from the set of computing prompts, and the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the generating the first computing prompt group comprises clustering the first computing prompt from the set of computing prompts, with the second computing prompt, different from the first computing prompt, from the set of computing prompts, based on a similarity of the first vector embedding and the second vector embedding; first accessing, using the one or more computing device processors, a vector database; determining, using the one or more computing device processors, for the first computing prompt from the set of computing prompts, using the first vector embedding, based on the first accessing the vector database, first file data, associated with at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, wherein the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, comprises first unstructured data, wherein the determining the first file data, associated with the at least one first file that partially corresponds with the first computing prompt from the set of computing prompts, is based on a similarity of the first vector embedding and at least one third vector embedding, wherein the at least one third vector embedding is associated with or comprised in the first file data; second accessing, using the one or more computing device processors, the vector database; determining, using the one or more computing device processors, for the second computing prompt, different from the first computing prompt, from the set of computing prompts, using the second vector embedding, based on the second accessing the vector database, second file data, associated with at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, comprises second unstructured data, wherein the determining the second file data, associated with the at least one second file that partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, is based on a similarity of the second vector embedding and at least one fourth vector embedding, wherein the at least one fourth vector embedding is associated with or comprised in the second file data; transmitting, using the one or more computing device processors, the first computing prompt group

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to the LLM; transmitting, using the one or more computing device processors, the first file data, associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, to the LLM; transmitting, using the one or more computing device processors, the second file data, associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM; receiving, using the one or more computing device processors, processed data from the LLM, wherein the processed data comprises the first computing prompt from the set of computing prompts, a first response associated with the first computing prompt from the set of computing prompts, a first citation associated with the first computing prompt from the set of computing prompts, a first file quality indicator associated with the first computing prompt from the set of computing prompts, the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second response associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, and a second file quality indicator associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts; determining, using the one or more computing device processors, a first computing indicator based on the first computing prompt from the set of computing prompts, and the first citation associated with the first computing prompt from the set of computing prompts; determining, using the one or more computing device processors, a second computing indicator based on the second computing prompt, different from the first computing prompt, from the set of computing prompts, and the second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts; generating, using the one or more computing device processors, structured data comprising or based on the processed data, the first computing indicator and the second computing indicator; and transmitting, using the one or more computing device processors, the structured data to a first system.

In one embodiment, the structured data is associated with a second computing format, wherein the second computing format comprises JSON format.

According to some embodiments, the first citation, associated with the first computing prompt from the set of computing prompts, comprises at least one of: at least some first text from the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, at least some second text from the second file data associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, a file name corresponding with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts or the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, and a page number corresponding with the at least some first text from the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, or the at least

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some second text from the second file data associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts.

In other embodiments, the first file quality indicator is based on metadata associated with the first citation.

In some cases, the determining, using the one or more computing device processors, the first computing indicator based on the first computing prompt from the set of computing prompts, and the first citation, associated with the first computing prompt from the set of computing prompts, is based on a semantic similarity of the first vector embedding and a fifth vector embedding, wherein the fifth vector embedding is associated with the first citation.

In one embodiment, the fifth vector embedding comprises or is comprised in the at least one third vector embedding or the at least one fourth vector embedding.

In some embodiments, the one or more computing device processors are comprised in one or more computing systems, wherein the one or more computing systems are located in one or more locations.

According to yet another embodiment, a method for generating an indexed computing file based on inserting vector embeddings and metadata, received from a large language model (LLM), into a configured vector database, comprises: receiving, using one or more computing device processors, a file from a first file source, wherein the file comprises unstructured data; extracting, using the one or more computing device processors, text from the file; transmitting, using the one or more computing device processors, the text from the file to an LLM; receiving, using the one or more computing device processors, at a first time, metadata associated with the file from the LLM, wherein the metadata associated with the file comprises or is based on file quality data, wherein the file quality data comprises or is based on at least one of: a nature of the file, a credibility of the file, a freshness of the file, and a file quality indicator of the file; executing, using the one or more computing device processors, at a second time or at the first time, a chunking computing operation using the file, thereby resulting in a chunked file; transmitting, using the one or more computing device processors, text associated with the chunked file to the LLM; receiving, using the one or more computing device processors, at least one vector embedding for the text associated with the chunked file from the LLM, wherein the at least one vector embedding comprises or is based on a semantic structure of at least some of the text associated with the chunked file; configuring, using the one or more computing device processors, a vector database to store vector embeddings and metadata, thereby resulting in a configured vector database; first inserting, using the one or more computing device processors, at a third time following the first time and the second time, the at least one vector embedding for the text associated with the chunked file, into the configured vector database; second inserting, using the one or more computing device processors, at the third time following the first time and the second time, the metadata associated with the file into the configured vector database; and generating, based on the first inserting the at least one vector embedding for the text associated with the chunked file into the configured vector database, and the second inserting the metadata associated with the file into the configured vector database, an indexed computing file.

In some embodiments, the file quality data comprising or being based on the at least one of: the nature of the file, the credibility of the file, the freshness of the file, and the file quality indicator of the file, further comprises or is based on

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at least two of: the nature of the file, the credibility of the file, the freshness of the file, and the file quality indicator of the file.

In other embodiments, the file quality data comprising or being based on the at least one of: the nature of the file, the credibility of the file, the freshness of the file, and the file quality indicator of the file, further comprises or is based on at least three of: the nature of the file, the credibility of the file, the freshness of the file, and the file quality indicator of the file.

In yet other embodiments, the file quality data comprising or being based on the at least one of: the nature of the file, the credibility of the file, the freshness of the file, and the file quality indicator of the file, further comprises or is based on: the nature of the file, the credibility of the file, the freshness of the file, and the file quality indicator of the file.

In one embodiment, the nature of the file comprises an indicator associated with a classification of the file.

According to some embodiments, the credibility of the file comprises an indicator associated with a source of the file.

In some cases, the freshness of the file comprises an indicator associated with a creation time of the file.

In other embodiments, a system and a computer program can include or execute the method described above. These and other implementations may each optionally include one or more of the following features.

In some embodiments, the system comprises or is comprised in one or more computing systems associated with one or more locations.

The metadata associated with the file, according to one embodiment, further comprises a citation, wherein the citation comprises: at least some text from the file, a file name corresponding with the file, and a page number associated with the at least some text from the file.

In one embodiment, the metadata associated with the file further comprises third-party source data.

In some embodiments, the LLM is hosted on a third-party server.

In other embodiments, the LLM is hosted on a local server.

According to some embodiments, the semantic structure of the at least some of the text associated with the chunked file comprises a conceptual meaning of the at least some of the text associated with the chunked file.

In some cases, the method comprises: receiving, using one or more computing device processors, a file from a first file source, wherein the file comprises unstructured data; extracting, using the one or more computing device processors, data from the file; transmitting, using the one or more computing device processors, the data from the file to an LLM; receiving, using the one or more computing device processors, at a first time, metadata associated with the file from the LLM, wherein the metadata associated with the file comprises or is based on file quality data, wherein the file quality data comprises or is based on at least one of: a nature of the file, a credibility of the file, a freshness of the file, and a file quality indicator of the file; executing, using the one or more computing device processors, at a second time or at the first time, a chunking computing operation using the file, thereby resulting in a chunked file; transmitting, using the one or more computing device processors, data associated with the chunked file to the LLM; receiving, using the one or more computing device processors, at least one vector embedding for the data associated with the chunked file from the LLM, wherein the at least one vector embedding comprises or is based on a semantic structure of at least

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some of the data associated with the chunked file; configuring, using the one or more computing device processors, a vector database to store vector embeddings and metadata, thereby resulting in a configured vector database; first inserting, using the one or more computing device processors, at a third time following the first time and the second time, the at least one vector embedding for the data associated with the chunked file, into the configured vector database; second inserting, using the one or more computing device processors, at the third time following the first time and the second time, the metadata associated with the file into the configured vector database, wherein the first inserting the at least one vector embedding for the data associated with the chunked file into the configured vector database, and the second inserting the metadata associated with the file into the configured vector database, result in an indexed computing file; and storing, using the one or more computing device processors, the indexed computing file in a file repository.

In some embodiments, the at least some of the data associated with the chunked file comprises or is based on at least one of: a word from the chunked file, a phrase from the chunked file, a sentence from the chunked file, a paragraph from the chunked file, or the chunked file.

According to one embodiment, the file quality indicator of the file is based on the nature of the file, the credibility of the file, and the freshness of the file.

In other embodiments, the one or more computing device processors are comprised in one or more computing systems, wherein the one or more computing systems are located in one or more locations.

In some cases, the data from the file comprises at least one of: text, an image, a figure, a table, and a diagram.

The file from the first file source, according to some embodiments, comprises at least one of: an audit document, a Service Organization Control (SOC) 2 report, a policy document, a 10K financial report, a technical description document, a SOC 1 report, a data security document, a corporate charter, an information technology procedure document, a financial report, a questionnaire, a 10Q report, a human resources document, a compliance report, or a screenshot of an internal system.

In still another embodiment, a method for generating a computing structure based on both unstructured data and structured data using a large language model (LLM) comprises: receiving, using one or more computing device processors, first unstructured data from a first data source; receiving, using the one or more computing device processors, first structured data from a second data source; determining, using the one or more computing device processors, a first computing library associated with the first structured data; receiving, using the one or more computing device processors, second unstructured data from a third data source; determining, using the one or more computing device processors, a first set of computing prompts associated with the second unstructured data; receiving, using the one or more computing device processors, second structured data, associated with a first computing format, from a fourth data source; determining, using the one or more computing device processors, based on the first computing format, a set of computing structures associated with the second structured data; transmitting, using the one or more computing device processors, at a first time, the first unstructured data to an LLM; transmitting, using the one or more computing device processors, at a second time or the first time, the first computing library associated with the first structured data to the LLM; transmitting, using the one or more computing

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device processors, at a third time, the second time, or the first time, the first set of computing prompts associated with the second unstructured data to the LLM; transmitting, using the one or more computing device processors, at a fourth time, the third time, the second time, or the first time, the set of computing structures associated with the second structured data to the LLM; receiving, using the one or more computing device processors, third structured data, associated with a second computing format, from the LLM, wherein the third structured data comprises or is based on the first set of computing prompts associated with the second unstructured data, a set of responses associated with the first set of computing prompts associated with the second unstructured data, and a computing structure, wherein the computing structure is not comprised in the set of computing structures associated with the second structured data, and wherein the computing structure comprises or is based on the first unstructured data, the first computing library associated with the first structured data, the first set of computing prompts associated with the second unstructured data, and the set of computing structures associated with the second structured data; and transmitting, using the one or more computing device processors, the third structured data to a first system.

In some embodiments, the first computing format comprises JavaScript Object Notation (JSON) format.

In other embodiments, the second computing format comprises JSON format.

According to one embodiment, the first computing library associated with the first structured data comprises at least one of: a second set of computing prompts, a set of attributes, a set of entities, a set of workflow task types, or a set of configured objects.

In another embodiment, the method further comprises generating, using the one or more computing device processors, a second computing library using the LLM.

According to some embodiments, the computing structure further comprises or is based on the second computing library.

In some embodiments, the one or more computing device processors are comprised in one or more computing systems, wherein the one or more computing systems are located in one or more locations.

In other embodiments, a system and a computer program can include or execute the method described above. These and other implementations may each optionally include one or more of the following features.

According to some embodiments, the system comprises or is comprised in one or more computing systems associated with one or more locations.

In some embodiments, at least one of the first unstructured data or the second unstructured data comprises raw information or information without a predetermined structure or format.

In one embodiment, the LLM is hosted on a third-party server.

In another embodiment, the LLM is hosted on a local server.

According to one embodiment, one or more of the instructions execute in a first stage and a second stage, such that fifth structured data associated with the second stage comprises or is based on fourth structured data associated with the first stage.

In some cases, the method for generating a computing structure based on both unstructured data and structured data using a large language model (LLM) comprises: receiving, using one or more computing device processors, first unstructured data from a first data source; receiving, using the one or more computing device processors, first structured data from a second data source; determining, using the

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one or more computing device processors, a computing library associated with the first structured data; receiving, using the one or more computing device processors, second unstructured data from a third data source;

determining, using the one or more computing device processors, a set of computing prompts associated with the second unstructured data; receiving, using the one or more computing device processors, second structured data, associated with a first computing format, from a fourth data source; determining, using the one or more computing device processors, based on the first computing format, a set of computing structures associated with the second structured data; transmitting, using the one or more computing device processors, at a first time, the first unstructured data to an LLM; transmitting, using the one or more computing device processors, at a second time or the first time, the computing library associated with the first structured data to the LLM; transmitting, using the one or more computing device processors, at a third time, the second time, or the first time, the set of computing prompts associated with the second unstructured data to the LLM; transmitting, using the one or more computing device processors, at a fourth time, the third time, the second time, or the first time, the set of computing structures associated with the second structured data to the LLM; receiving, using the one or more computing device processors, third structured data, associated with a second computing format, from the LLM, wherein the third structured data comprises or is based on the set of computing prompts associated with the second unstructured data, a set of responses associated with the set of computing prompts associated with the second unstructured data, and a computing structure, wherein the computing structure is based on at least one of: the first unstructured data, the computing library associated with the first structured data, the set of computing prompts associated with the second unstructured data, or the set of computing structures associated with the second structured data; and transmitting, using the one or more computing device processors, the third structured data to a first system.

In one embodiment, at least one of the first unstructured data or the second unstructured data comprises at least one of: text, an image, a figure, a table, audio, a video, a graph, or a diagram.

According to some embodiments, the first unstructured data comprises at least one of: documentation of at least one system, or documentation of at least one process.

According to one embodiment, the computing structure comprises a system configuration.

In some embodiments, the set of computing structures associated with the second structured data comprises at least one example system configuration.

In other embodiments, the set of computing prompts associated with the second unstructured data comprises at least one of: at least one requirement associated with a system configuration, or at least one capability associated with the system configuration.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure is illustrated by way of example, and not by way of limitation in the figures of the accompanying drawings in which like reference numerals are used to refer to similar elements. It is emphasized that various features may not be drawn to scale and the dimensions of various features may be arbitrarily increased or reduced for clarity of discussion. Further, some components may be omitted in certain figures for clarity of discussion.

FIG. 1 is a high-level block diagram of a complex computing network for executing the principles disclosed.

FIG. 2 is a functional block diagram of a complex computing environment associated with the disclosed techniques.

FIG. 3 is a detailed system diagram of the complex computing environment of FIG. 2.

FIGS. 4A-4B show exemplary workflows for generating an indexed computing file based on inserting vector embeddings and metadata, received from a large language model (LLM), into a configured vector database and extracting structured data from a source comprising unstructured data, respectively, in a complex computing network according to some embodiments of this disclosure.

FIGS. 5A-5B show exemplary flowcharts for methods, systems/apparatuses, and computer program products that generate an indexed computing file based on inserting vector embeddings and metadata, received from an LLM, into a configured vector database in a complex computing network such as the network of FIG. 1, in accordance with some embodiments of this disclosure.

FIGS. 6A-6D show exemplary flowcharts for methods, systems/apparatuses, and computer program products that extract structured data from a source comprising unstructured data in a complex computing network such as the network of FIG. 1, in accordance with some embodiments of this disclosure.

FIG. 7 shows an exemplary workflow for generating a computing structure based on both unstructured data and structured data using a large language model (LLM), in a complex computing network according to some embodiments of this disclosure.

FIGS. 8A-8B show exemplary flowcharts for methods, systems/apparatuses, and computer program products that generate a computing structure based on both unstructured data and structured data using a large language model (LLM), in a complex computing network such as the network of FIG. 1, according to some embodiments of this disclosure.

FIG. 9 shows an exemplary workflow for generating structured data from files comprising unstructured data, and transmitting recommendations, based on the structured data, to a system storing the files, using a large language model (LLM), in a complex computing network according to some embodiments of this disclosure.

FIGS. 10A-10E show exemplary flowcharts for methods, systems/apparatuses, and computer program products that generate structured data from files comprising unstructured data, and transmit recommendations, based on the structured data, to a system storing the files, using a large language model (LLM), in a complex computing network such as the network of FIG. 1, according to some embodiments of this disclosure.

All of these drawings are illustrations of certain embodiments. The scope of the claims is not limited to the specific embodiments illustrated in the drawings and described below.

DETAILED DESCRIPTION

System Environment

Illustrated in FIG. 1 is a high-level diagram of an exemplary system 100 for executing the principles disclosed. In the illustrated implementation, the system 100 may include an application server 120 communicatively coupled to a plurality of network systems 150a . . . 150n via a network 110. The system 100 may also include a large language

model (LLM) server 130, a vector database 140, and an endpoint device 125 communicatively coupled via the network 110. While a single LLM server 130, a single vector database 140, and a single endpoint device 125 are illustrated, the disclosed principles and techniques could be expanded to include multiple LLM servers, multiple vector databases, and multiple endpoints.

In some embodiments, the application server 120 may include a computing device such as a mainframe server, a content server, a communication server, a laptop computer, a desktop computer, a handheld computing device, a smart phone, a wearable computing device, a tablet computing device, a virtual machine, a mobile computing device, a cloud-based computing solution and/or a cloud-based service, and/or the like. The application server 120 may include a plurality of computing devices configured to communicate with one another and/or implement the techniques described herein.

The application server 120 may include various elements of a computing environment as described in association with the computing environment 200 of FIGS. 2 and 3. For example, the application server 120 may include processing unit 202, a memory unit 204, an input/output (I/O) unit 206, and/or a communication unit 208 which are discussed in association with FIGS. 2 and 3. The application server 120 may further include subunits and/or other modules for performing operations associated with generating structured data from a file comprising unstructured data based on accessing a large language model (LLM), and generating an indexed computing file based on inserting vector embeddings and metadata, received from an LLM, into a configured vector database. The application server 120 may be locally or remotely operated as the case may require.

Turning back to FIG. 1, the application server 120 may include a data engine 160. The data engine 160 may either be implemented on the application server 120 and/or on the endpoint device 125. The data engine 160 may include one or more instructions or computer logic that are executed by the one or more processors such as the processors discussed in association with FIGS. 2 and 3. In particular, the data engine 160 facilitates executing the processing procedures, methods, techniques, and workflows provided in this disclosure. Some embodiments include an iterative refinement of one or more data models (e.g., learning model, large language model) associated with the network environment 100 disclosed via feedback loops executed by one or more computing device processors and/or through other control devices or mechanisms that make determinations regarding optimization of a given action, template, or model.

In some embodiments, the one or more data engines may access an operating system of a computing device comprised in the network environment 100 in order to execute the disclosed techniques. For instance, the one or more data engines may gain access into an operating system associated with the network environment 100 to initiate the various processes disclosed.

Turning back to FIG. 1, the endpoint device 125 may be a handheld computing device, a smart phone, a tablet, a laptop computer, a desktop computer, a personal digital assistant (PDA), a smart device, a wearable device, a biometric device, a computer server, a virtual server, a virtual machine, a mobile device, and/or a communication server. In some embodiments, the endpoint device 125 may include a plurality of computing devices configured to communicate with one another and/or implement the techniques described in this disclosure.

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The other elements of the endpoint device **125** are discussed in association with the computing environment **200** of FIGS. **2** and **3**. For example, elements such as a processing unit **202**, a memory unit **204**, an input/output (I/O) unit **206**, and/or a communication unit **208** may execute one or more of the modules of endpoint device **125** shown in FIG. **1**. The endpoint device **125** may also include subunits and/or other computing instances as provided in this disclosure for performing operations associated with generating structured data from a file comprising unstructured data based on accessing an LLM, generating an indexed computing file based on inserting vector embeddings and metadata, received from an LLM, into a configured vector database, etc.

The network **110** may include a plurality of networks. For instance, the network **110** may include any wired and/or wireless communication network that facilitates communication between the application server **120**, the endpoint device **125**, the LLM server **130**, the vector database **140**, and the network systems **150a** . . . **150n**. The network **110**, in some instances, may include an Ethernet network, a cellular network, a computer network, the Internet, a wireless fidelity (Wi-Fi) network, a light fidelity (Li-Fi) network, a Bluetooth network, a radio frequency identification (RFID) network, a near-field communication (NFC) network, a laser-based network, a 5G network, and/or the like.

The network systems **150a** . . . **150n** may include one or more computing devices or servers, services, or applications that can be accessed by application server **120** and/or the vector database **140** and/or the LLM server **130** and/or the endpoint device **125** via the network **110**. In one embodiment, the network systems **150a** . . . **150n** comprises third-party applications or services that are native or non-native to either the application server **120** and/or the LLM server **130** and/or the endpoint device **125**. The third-party applications or services, for example, may facilitate receiving one or more files comprising unstructured data. According to some implementations, the applications or services associated with the network systems **150a** . . . **150n** and/or associated with the application server **120**, and/or the LLM server **130**, and/or the vector database **140** and/or the endpoint device **125** must be registered to activate or otherwise enable their usage in the network environment **100**.

Returning to FIG. **1**, the vector database **140** may comprise one or more storage devices that store data, information and instructions used by the application server **120** and/or the LLM server **130** and/or the endpoint device **125**. The stored information may include information about metadata, information about vector embeddings, information associated with a file comprising unstructured data, etc. In one embodiment, the one or more storage devices mentioned above in association with the vector database **140** can be non-volatile memory or similar permanent storage device and media. For example, the one or more storage devices may include a hard disk drive, a CD-ROM device, a DVD-ROM device, a DVD-RAM device, a DVD-RW device, a flash memory device, solid state media, or some other mass storage device known in the art for storing information on a more permanent basis.

While the vector database **140** is shown as being coupled to the application server **120**, the endpoint device **125**, and the LLM server **130** via the network **110**, the data in the vector database **140** may be replicated, in some embodiments, on the application server **120** and/or the LLM server **130** and/or the endpoint device **125**. That is to say that a local copy of the data in the vector database **140** may be stored on the application server **120** and/or the LLM server

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130 and/or the endpoint device **125**. This local copy may be synced with the vector database **140** so that when there are any changes to the information in the vector database **140**, the local copy on either the application server **120** and/or the LLM server **130** and/or the endpoint device **125** is also similarly updated or synced in real-time or in near-real-time to be consistent with the information in the vector database **140** and vice versa.

Turning back to FIG. **1**, in some embodiments, the LLM server **130** may include a computing device such as a mainframe server, a content server, a communication server, a laptop computer, a desktop computer, a handheld computing device, a smart phone, a wearable computing device, a tablet computing device, a virtual machine, a mobile computing device, a cloud-based computing solution and/or a cloud-based service, and/or the like. The LLM server **130** may include a plurality of computing devices configured to communicate with one another and/or implement the techniques described herein.

The LLM server **130** may include various elements of a computing environment as described in association with the computing environment **200** of FIGS. **2** and **3**. For example, the LLM server **130** may include processing unit **202**, a memory unit **204**, an input/output (I/O) unit **206**, and/or a communication unit **208** which are discussed in association with FIGS. **2** and **3**. The LLM server **130** may further include subunits and/or other modules for performing operations associated with generating structured data from a file comprising unstructured data based on accessing an LLM, generating an indexed computing file based on inserting vector embeddings and metadata, received from an LLM, into a configured vector database, etc. The LLM server **130** may be locally hosted. Additionally or alternatively, the LLM server may be hosted by a third-party.

In some embodiments, the LLM server **130** may include an LLM **170** for comprehending and generating text. The LLM **170** may be trained with at least one of: zero-shot learning, few-shot learning, and fine-tuning. The LLM **170** may comprise at least one of: GPT-4, LLAMA-3, BLOOM, PaLM, GPT-3.5, BERT, Gemini, LaMDA, and Falcon. The LLM **170** may also include multiple LLMs and therefore may be configured to perform and/or execute multiple processes in parallel. In some embodiments, the LLM server **130** may include a special chipset for processing large numbers of complex operations in a reduced amount of time.

FIGS. **2** and **3** illustrate exemplary functional and system diagrams of a computing environment **200**, according to some embodiments of this disclosure. Specifically, FIG. **2** provides a functional block diagram of the computing environment **200**, whereas FIG. **3** provides a detailed system diagram of the computing environment **200**.

As seen in FIGS. **2** and **3**, the computing environment **200** may include a processing unit **202**, a memory unit **204**, an I/O unit **206**, and a communication unit **208**. The processing unit **202**, the memory unit **204**, the I/O unit **206**, and the communication unit **208** may include one or more subunits for performing operations described in this disclosure. Additionally, each unit and/or subunit may be operatively and/or otherwise communicatively coupled with each other and to the network **110**. The computing environment **200** may be implemented on general-purpose hardware and/or specifically-purposed hardware as the case may be. In particular, the computing environment **200** and any units and/or subunits of FIGS. **2** and **3** may be included in one or more elements of system **100** as described above.

The processing unit **202** may control one or more of the memory unit **204**, the I/O unit **206**, and the communication

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unit **208** of the computing environment **200**, as well as any included subunits, elements, components, devices, and/or functions performed by the memory unit **204**, I/O unit **206**, and the communication unit **208**. The described sub-elements of the computing environment **200** may also be included in similar fashion in any of the other units and/or devices included in the system **100** of FIG. 1. Additionally, any actions described herein as being performed by a processor may be taken by the processing unit **202** of FIGS. 2 and 3 alone and/or by the processing unit **202** in conjunction with one or more additional processors, units, subunits, elements, components, devices, and/or the like. Further, while one processing unit **202** may be shown in FIGS. 2 and 3, multiple processing units may be present and/or otherwise included in the computing environment **200** or elsewhere in the overall system (e.g., system **100** of FIG. 1). Thus, while instructions may be described as being executed by the processing unit **202** (and/or various subunits of the processing unit **202**), the instructions may be executed simultaneously, serially, and/or otherwise by one or multiple processing units **202** on one or more devices.

In some embodiments, the processing unit **202** may be implemented as one or more computer processing unit (CPU) chips and/or graphical processing unit (GPU) chips and may include a hardware device capable of executing computer instructions. The processing unit **202** may execute instructions, codes, computer programs, and/or scripts. The instructions, codes, computer programs, and/or scripts may be received from and/or stored in the memory unit **204**, the I/O unit **206**, the communication unit **208**, subunits, and/or elements of the aforementioned units, other devices, and/or computing environments, and/or the like.

In some embodiments, the processing unit **202** may include, among other elements, subunits such as a content management unit **212**, a location determination unit **214**, a graphical processing unit (GPU) **216**, a tensor processing unit (TPU) **218**, and a resource allocation unit **220**. Each of the aforementioned subunits of the processing unit **202** may be communicatively and/or otherwise operably coupled with each other.

The content management unit **212** may facilitate generation, modification, analysis, transmission, and/or presentation of content. Content may be file content, exception event content, media content, security event content, tracking content, or any combination thereof. In some instances, content on which the content management unit **212** may operate includes device information, user interface data, image data, text data, themes, audio data or audio files, video data or video files, documents, and/or the like. Additionally, the content management unit **212** may control the audiovisual environment and/or appearance of application data during execution of various processes. In some embodiments, the content management unit **212** may interface with a third-party content server (e.g., third-party content server associated with the LLM server **130**), and/or specific memory locations for execution of its operations.

The location determination unit **214** may facilitate detection, generation, modification, analysis, transmission, and/or presentation of location information. Location information may include global positioning system (GPS) coordinates, an internet protocol (IP) address, a media access control (MAC) address, geolocation information, a port number, a server number, a proxy name and/or number, device information (e.g., a serial number), an address, a zip code, and/or the like. In some embodiments, the location determination unit **214** may include various sensors, radar, and/or other specifically-purposed hardware elements for the location

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determination unit **214** to acquire, measure, and/or otherwise transform location information.

The GPU **216** may facilitate generation, modification, analysis, processing, transmission, and/or presentation of content described above, as well as any data (e.g., file quality data, metadata, structured data, unstructured data, filter data, etc.) described herein. In some embodiments, the GPU **216** may be utilized to render content for presentation on a computing device. In some embodiments, the GPU **216** may be utilized to perform computations on vector embeddings. The GPU **216** may also include multiple GPUs and therefore may be configured to perform and/or execute multiple processes in parallel. In some implementations, the GPU **216** may be used in conjunction with the data engine **160**, and/or the TPU **218** and/or other subunits associated with the memory unit **204**, the I/O unit **206**, the communication unit **208**, and/or a combination thereof.

The TPU **218** may facilitate generation, modification, analysis, processing, transmission, and/or presentation of any vector embeddings described herein. In some embodiments, the TPU **218** may be utilized to perform computations comprising or based on vector embeddings. For example, the TPU may execute similarity operations (e.g., semantic similarity operations) on vector embeddings. In some instances, the similarity operations are calculated based on at least one of cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity. In some embodiments, the similarity operations comprise or are based on two vector embeddings. In other embodiments, the similarity operations comprise or are based on at least three vector embeddings.

The TPU **218** may also include multiple TPUs and therefore may be configured to perform and/or execute multiple processes in parallel. In some implementations, the TPU **218** may be used in conjunction with the data engine **160**, and/or the GPU **216**, and/or other subunits associated with the memory unit **204**, the I/O unit **206**, the communication unit **208**, and/or a combination thereof. In some embodiments, the TPU **218** may interface with a third-party content server (e.g., third-party content server associated with the LLM server **130**) for execution of its operations.

The resource allocation unit **220** may facilitate the determination, monitoring, analysis, and/or allocation of computing resources throughout the computing environment **200** and/or other computing environments. For example, the computing environment may facilitate a high volume of data (e.g., file quality data, metadata, structured data, unstructured data, filter data, etc.), to be processed and analyzed. As such, computing resources of the computing environment **200** used by the processing unit **202**, the memory unit **204**, the I/O unit **206**, and/or the communication unit **208** (and/or any subunit of the aforementioned units) such as processing power, data storage space, network bandwidth, and/or the like may be in high demand at various times during operation of the computing environment **200**. Accordingly, the resource allocation unit **220** may include sensors and/or other specially-purposed hardware for monitoring performance of each unit and/or subunit of the computing environment **200**, as well as hardware for responding to the computing resource needs of each unit and/or subunit. In some embodiments, the resource allocation unit **220** may use computing resources of a second computing environment separate and distinct from the computing environment **200** to facilitate a desired operation. For example, the resource allocation unit **220** may determine a number of simultaneous computing processes and/or requests. The

resource allocation unit **220** may also determine that the number of simultaneous computing processes and/or requests meet and/or exceed a predetermined threshold value. Based on this determination, the resource allocation unit **220** may determine an amount of additional computing resources (e.g., processing power, storage space of a particular non-transitory computer-readable memory medium, network bandwidth, and/or the like) required by the processing unit **202**, the memory unit **204**, the I/O unit **206**, the communication unit **208**, and/or any subunit of the aforementioned units for safe and efficient operation of the computing environment while supporting the number of simultaneous computing processes and/or requests. The resource allocation unit **220** may then retrieve, transmit, control, allocate, and/or otherwise distribute determined amount(s) of computing resources to each element (e.g., unit and/or subunit) of the computing environment **200** and/or another computing environment.

The memory unit **204** may be used for storing, recalling, receiving, transmitting, and/or accessing various files and/or data (e.g., file quality data, metadata, structured data, unstructured data, filter data, etc.) during operation of computing environment **200**. For example, memory unit **204** may be used for storing, recalling, and/or updating file quality data, metadata, structured data, unstructured data, and/or filter data as well as other data associated with, resulting from, and/or generated by any unit, or combination of units and/or subunits of the computing environment **200**. In some embodiments, the memory unit **204** may store instructions, code, and/or data that may be executed by the processing unit **202**. For instance, the memory unit **204** may store code that execute operations associated with one or more units and/or one or more subunits of the computing environment **200**. For example, the memory unit may store code for the processing unit **202**, the I/O unit **206**, the communication unit **208**, and for itself.

Memory unit **204** may include various types of data storage media such as solid state storage media, hard disk storage media, virtual storage media, and/or the like. Memory unit **204** may include dedicated hardware elements such as hard drives and/or servers, as well as software elements such as cloud-based storage drives. In some implementations, memory unit **204** may be a random access memory (RAM) device, a dynamic random access memory (DRAM) device, a static random access memory (SRAM) device, flash memory, read only memory (ROM) device, and/or various forms of secondary storage. The RAM device may be used to store volatile data and/or to store instructions that may be executed by the processing unit **202**. For example, the instructions stored by the RAM device may be a command, a current operating state of computing environment **200**, an intended operating state of computing environment **200**, and/or the like. As a further example, data stored in the RAM device of memory unit **204** may include instructions related to various methods and/or functionalities described herein. The ROM device may be a non-volatile memory device that may have a smaller memory capacity than the memory capacity of a secondary storage. The ROM device may be used to store instructions and/or data that may be read during execution of computer instructions. In some embodiments, access to both the RAM device and ROM device may be faster to access than the secondary storage.

Secondary storage may comprise one or more disk drives and/or tape drives and may be used for non-volatile storage of data or as an over-flow data storage device if the RAM device is not large enough to hold all working data. Secondary storage may be used to store programs that may be

loaded into the RAM device when such programs are selected for execution. In some embodiments, the memory unit **204** may include one or more databases **310** (shown in FIG. **3**) for storing any data described herein. For example, depending on the implementation, the one or more databases may be used as a local database of the network systems **150a** . . . **150n** discussed with reference to FIG. **1**. Additionally or alternatively, one or more secondary databases (e.g., the vector database **140** discussed with reference to FIG. **1**) located remotely from computing environment **200** may be used and/or accessed by memory unit **204**. In some embodiments, memory unit **204** and/or its subunits may be local to the application server **120** and/or the endpoint device **125** and/or the network systems **150a** . . . **150n** and/or remotely located in relation to the application server **120** and/or the endpoint device **125** and/or the network systems **150a** . . . **150n**.

Turning back to FIG. **2**, the memory unit **204** may include subunits such as an operating system unit **226**, an application data unit **228**, an application programming interface (API) unit **230**, a content storage unit **232**, an artificial intelligence (AI) unit **234**, a data engine **160**, and a cache storage unit **240**. Each of the aforementioned subunits of the memory unit **204** may be communicatively and/or otherwise operably coupled with each other and other units and/or subunits of the computing environment **200**. It is also noted that the memory unit **204** may include other modules, instructions, or code that facilitate the execution of the techniques described.

The operating system unit **226** may facilitate deployment, storage, access, execution, and/or utilization of an operating system utilized by computing environment **200** and/or any other computing environment described herein. In some embodiments, operating system unit **226** may include various hardware and/or software elements that serve as a structural framework for processing unit **202** to execute various operations described herein. Operating system unit **226** may further store various pieces of information and/or data associated with the operation of the operating system and/or computing environment **200** as a whole, such as a status of computing resources (e.g., processing power, memory availability, resource utilization, and/or the like), runtime information, modules to direct execution of operations described herein, user permissions, security credentials, and/or the like.

The application data unit **228** may facilitate deployment, storage, access, execution, and/or utilization of an application used by computing environment **200** and/or any other computing environment described herein. For example, the endpoint device **125** may be required to download, install, access, and/or otherwise use a software application (e.g., web application) to facilitate performance of the disclosed techniques. As such, the application data unit **228** may store any information and/or data associated with an application. The application data unit **228** may further store various pieces of information and/or data associated with the operation of an application and/or computing environment **200** as a whole, such as status of computing resources (e.g., processing power, memory availability, resource utilization, and/or the like), runtime information, user interfaces, modules to direct execution of operations described herein, user permissions, security credentials, and/or the like.

The API unit **230** may facilitate deployment, storage, access, execution, and/or utilization of information associated with APIs of computing environment **200** and/or any other computing environment described herein. For example, computing environment **200** may include one or

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more APIs for various devices, applications, units, subunits, elements, and/or other computing environments to communicate with each other and/or utilize the same data. Accordingly, API unit **230** may include API databases containing information that may be accessed and/or utilized by applications, units, subunits, elements, and/or operating systems of other devices and/or computing environments. In some embodiments, each API database may be associated with a customized physical circuit included in memory unit **204** and/or API unit **230**. Additionally, each API database may be public and/or private, and so authentication credentials may be required to access information in an API database. In some embodiments, the API unit **230** may enable the application server **120**, the endpoint device **125**, and the network systems **150a . . . 150n** to communicate with each other. It is appreciated that the API unit **230** may facilitate accessing, using the data engine **160**, one or more applications or services on the application server **120** and/or the network systems **150a . . . 150n**.

The content storage unit **232** may facilitate deployment, storage, access, and/or utilization of information associated with performance of implementing operations associated with the network environment **100** by computing environment **200** and/or any other computing environment described herein. In some embodiments, content storage unit **232** may communicate with content management unit **212** to receive and/or transmit content files (e.g., media content, file quality data content, metadata content, structured data content, unstructured data content, filter data content, etc.).

The AI unit **234** may facilitate deployment, storage, access, execution, and/or utilization of information associated with the use of AI within the computing environment **200** and/or any other computing environment described herein. For example, the network environment **100** may utilize the AI unit **234** for configuration management, and/or troubleshooting, and/or network performance. In some embodiments, the LLM server **130** may utilize the AI unit **234** for comprehending and/or generating text with the LLM **170**.

As previously discussed, the data engine **160** facilitates executing the processing procedures, methods, techniques, and workflows provided in this disclosure. In particular, the data engine **160** may be configured to execute computing operations associated with the disclosed methods, systems/apparatuses, and computer program products.

The cache storage unit **240** may facilitate short-term deployment, storage, access, analysis, and/or utilization of data. In some embodiments, cache storage unit **240** may serve as a short-term storage location for data so that the data stored in cache storage unit **240** may be accessed quickly. In some instances, cache storage unit **240** may include RAM devices and/or other storage media types for quick recall of stored data. Cache storage unit **240** may include a partitioned portion of storage media included in memory unit **204**.

The I/O unit **206** may include hardware and/or software elements for the computing environment **200** to receive, transmit, and/or present information useful for performing the disclosed processes. For example, elements of the I/O unit **206** may be used to receive input from a user of the endpoint device **125**. As described herein, I/O unit **206** may include subunits such as an I/O device **242**, an I/O calibration unit **244**, and/or driver **246**.

The I/O device **242** may facilitate the receipt, transmission, processing, presentation, display, input, and/or output of information as a result of executed processes described herein. In some embodiments, the I/O device **242** may

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include a plurality of I/O devices. In some embodiments, I/O device **242** may include a variety of elements that enable a user to interface with computing environment **200**. For example, I/O device **242** may include a keyboard, a touchscreen, a button, a sensor, a biometric scanner, a laser, a microphone, a camera, and/or another element for receiving and/or collecting input from a user. Additionally and/or alternatively, I/O device **242** may include a display, a screen, a sensor, a vibration mechanism, a light emitting diode (LED), a speaker, a radio frequency identification (RFID) scanner, and/or another element for presenting and/or otherwise outputting data to a user. In some embodiments, the I/O device **242** may communicate with one or more elements of processing unit **202** and/or memory unit **204** to execute operations associated with the disclosed techniques and systems.

The I/O calibration unit **244** may facilitate the calibration of the I/O device **242**. For example, I/O calibration unit **244** may detect and/or determine one or more settings of I/O device **242**, and then adjust and/or modify settings so that the I/O device **242** may operate more efficiently. In some embodiments, I/O calibration unit **244** may use a driver **246** (or multiple drivers) to calibrate I/O device **242**. For example, the driver **246** may include software that is to be installed by I/O calibration unit **244** so that an element of computing environment **200** (or an element of another computing environment) may recognize and/or integrate with I/O device **242** for the processes described herein.

The communication unit **208** may facilitate establishment, maintenance, monitoring, and/or termination of communications between computing environment **200** and other computing environments, third-party server systems, and/or the like (e.g., between the application server **120** and the LLM server **130** and/or the endpoint device **125** and/or the network systems **150a . . . 150n**). Communication unit **208** may also facilitate internal communications between various elements (e.g., units and/or subunits) of computing environment **200**. In some embodiments, communication unit **208** may include a network protocol unit **248**, an API gateway **250**, an encryption engine **252**, and/or a communication device **254**. Communication unit **208** may include hardware and/or other software elements.

The network protocol unit **248** may facilitate establishment, maintenance, and/or termination of a communication connection for computing environment **200** by way of a network. For example, the network protocol unit **248** may detect and/or define a communication protocol required by a particular network and/or network type. Communication protocols used by the network protocol unit **248** may include Wi-Fi protocols, Li-Fi protocols, cellular data network protocols, Bluetooth® protocols, WiMAX protocols, Ethernet protocols, powerline communication (PLC) protocols, mesh network protocols, 5G network protocols, and/or the like. In some embodiments, facilitation of communication for computing environment **200** may include transforming and/or translating data from being compatible with a first communication protocol to being compatible with a second communication protocol. In some embodiments, the network protocol unit **248** may determine and/or monitor an amount of data traffic to consequently determine which particular network protocol is to be used for establishing a secure communication connection, transmitting data, and/or performing malware scanning operations and/or other processes described herein.

The API gateway **250** may allow other devices and/or computing environments to access the API unit **230** of the memory unit **204** associated with the computing environ-

ment **200**. For example, an endpoint device **125** may access the API unit **230** of the computing environment **200** via the API gateway **250**. In some embodiments, the API gateway **250** may be required to validate user credentials associated with a user of the endpoint device **125** prior to providing access to the API unit **230** to a user. The API gateway **250** may include instructions for the computing environment **200** to communicate with another computing device and/or between elements of the computing environment **200**.

The encryption engine **252** may facilitate translation, encryption, encoding, decryption, and/or decoding of information received, transmitted, and/or stored by the computing environment **200**. Using encryption engine **252**, each transmission of data may be encrypted, encoded, and/or translated for security reasons, and any received data may be encrypted, encoded, and/or translated prior to its processing and/or storage. In some embodiments, encryption engine **252** may generate an encryption key, an encoding key, a translation key, and/or the like, which may be transmitted along with any data content.

The communication device **254** may include a variety of hardware and/or software specifically purposed to facilitate communication for computing environment **200** and or between two or more computing environments **200**. In one embodiment, communication device **254** may include one or more radio transceivers, chips, analog front end (AFE) units, antennas, processing units, memory, other logic, and/or other components to implement communication protocols (wired or wireless) and related functionality for facilitating communication for computing environment **200**. Additionally and/or alternatively, communication device **254** may include a modem, a modem bank, an Ethernet device such as a router or switch, a universal serial bus (USB) interface device, a serial interface, a token ring device, a fiber distributed data interface (FDDI) device, a wireless local area network (WLAN) device and/or device component, a radio transceiver device such as code division multiple access (CDMA) device, a global system for mobile communications (GSM) radio transceiver device, a universal mobile telecommunications system (UMTS) radio transceiver device, a long term evolution (LTE) radio transceiver device, a worldwide interoperability for microwave access (WiMAX) device, and/or other transceiver devices used for communication purposes.

EXEMPLARY EMBODIMENTS

FIG. 4A shows an exemplary workflow for generating an indexed computing file based on inserting vector embeddings and metadata, received from a large language model (LLM), into a configured vector database in a system (e.g., application server **120** in FIG. 1) within a complex computing network such as the network of FIG. 1 according to some embodiments. The various blocks of FIG. 4A may be executed in a different order from that shown in FIG. 4A. Some blocks may be optional. As shown in the figure, the system receives a file **404** from a first external system (e.g., at least one of a computing system, a database, etc.). In one embodiment, the first external system **402** is referred to as a first file source. In another embodiment, the first external system **402** may be one of the network systems **150-150n**. In some embodiments, the file **404** may comprise: an audit document, a Service Organization Control (SOC) 2 report, a policy document, a 10K financial report, a technical description document, a SOC 1 report, a data security document, a corporate charter, an information technology procedure document, a financial report, a questionnaire, a 10Q report,

a human resources document, or a screenshot of an internal system. Following receipt of the file **404**, the system extracts data **406** from the file **404**. In some embodiments, the data from the file **404** may comprise at least one of text, an image, a figure, a diagram, a graph, and/or a table.

The system transmits data from the file **404** to a first LLM **408** at a first time. In some embodiments, the first LLM **408** comprises at least one of: GPT-4, LLAMA-3, BLOOM, PaLM, GPT-3.5, BERT, Gemini, LaMDA, and Falcon. Any list of LLMs, vector databases, similarity operations, and file types in this disclosure is provided for exemplary purposes only. Other LLMs, vector databases, similarity operations, and file types may additionally or alternatively be used. In one embodiment, the first LLM **408** is hosted on a first third-party server. According to another embodiment, the first LLM **408** is hosted on a local server. The system receives metadata **414** associated with the file **404** from the first LLM **408**. In some embodiments, the metadata **414** associated with the file **404** comprises file quality data. Furthermore, the file quality data comprises or is based on at least one of: a nature of the file **404**, a credibility of the file **404**, a freshness of the file **404**, and a file quality indicator of the file **404**. In some embodiments, the system (e.g., the application server **120**, an apparatus, etc.) may comprise one or more computing systems.

In some embodiments, the file quality data comprising or being based on the at least one of: the nature of the file **404**, the credibility of the file **404**, the freshness of the file **404**, and the file quality indicator of the file **404**, further comprises or is based on at least two of: the nature of the file **404**, the credibility of the file **404**, the freshness of the file **404**, and the file quality indicator of the file **404**. In other embodiments, the file quality data comprising or being based on the at least one of: the nature of the file **404**, the credibility of the file **404**, the freshness of the file **404**, and the file quality indicator of the file **404**, further comprises or is based on at least three of: the nature of the file **404**, the credibility of the file **404**, the freshness of the file **404**, and the file quality indicator of the file **404**. In yet other embodiments, the file quality data comprising or being based on the at least one of: the nature of the file **404**, the credibility of the file **404**, the freshness of the file **404**, and the file quality indicator of the file **404**, further comprises or is based on: the nature of the file **404**, the credibility of the file **404**, the freshness of the file **404**, and the file quality indicator of the file **404**.

In one embodiment, the nature of the file **404** further comprises an indicator associated with a classification of the file **404**. Furthermore, the classification of the file **404** may be at least one of: audit from a reliable source, audit from an unreliable source, policy or procedure document, and unofficial document. In some embodiments, the indicator associated with the classification of the file **404** is numerical. In another embodiment, the credibility of the file **404** comprises an indicator associated with a source of the file **404**. In some embodiments, the indicator associated with the source of the file **404** is numerical. In yet other embodiments, the freshness of the file **404** comprises an indicator associated with a creation time of the file **404**. In some embodiments, the file quality indicator of the file **404** comprises or is based on the nature of the file **404**, the credibility of the file **404**, and the freshness of the file **404** (e.g. an average of the nature of the file **404**, the credibility of the file **404**, and the freshness of the file **404**, a median of the nature of the file **404**, the credibility of the file **404**, and the freshness of the file **404**, etc.). In one embodiment, the file quality indicator comprises an overall quality score.

In some embodiments, the metadata **414** associated with the file **404** further comprises a citation. Furthermore, the citation may comprise at least some text from the file **404**, a file name corresponding with the file **404**, and a page number associated with the at least some text from the file **404**. In some embodiments, the at least some text from the file **404** is a quote from the file **404**. The quote may comprise one of: a main idea of the file **404**, a brief summary of the file **404**, or a direct quote from the file **404**. In another embodiment, the metadata **414** associated with the file **404** further comprises third-party source data. Furthermore, the third-party source data may comprise at least one of: a source of the file **404**, a client associated with the file **404**, a tenant associated with the file **404**.

At the first time or a second time, the system executes a chunking operation **410** using the file **404**, thereby resulting in a chunked file. In one embodiment, the second time is prior to the first time. In another embodiment, the second time is after the first time. The system transmits the data associated with the chunked file to a second LLM **412**. In some embodiments, the second LLM **412** comprises at least one of: GPT-4, LLAMA-3, BLOOM, PaLM, GPT-3.5, BERT, Gemini, LaMDA, and Falcon. According to one embodiment, the second LLM **412** is hosted on a second third-party server. In some embodiments, the second third-party server comprises or is comprised in the first third-party server. In another embodiment, the second LLM **412** is hosted on the local server. In one embodiment, the second LLM **412** comprises or is comprised of the first LLM **408**. The system receives at least one vector embedding **416** for the data associated with the chunked file from the second LLM **412**.

In some embodiments, the at least one vector embedding **416** comprises or is based on a semantic structure of at least some of the data associated with the chunked file. Furthermore, the semantic structure of the at least some of the data associated with the chunked file comprises a conceptual meaning of the at least some of the data associated with the chunked file. In some embodiments, the data from the chunked file may comprise text, an image, a figure, a diagram, a graph, and/or a table. In other embodiments, the at least some of the data associated with the chunked file comprises or is based on: a word from the chunked file, a phrase from the chunked file, a sentence from the chunked file, a paragraph from the chunked file, an image from the chunked file, a figure from the chunked file, a diagram from the chunked file, a graph from the chunked file, a table from the chunked file, or the chunked file.

The system configures the vector database **418** to store vector embeddings and metadata, thereby resulting in a configured vector database. In some embodiments, the vector database **418** comprises at least one of: Chroma, Vespa, Marqo, Drant, LanceDB, Milvus, Pinecone, Weaviate, and PostgreSQL. At a third time following the first time and the second time, the system inserts the metadata **414** associated with the file **404** into the configured database. At a fourth time or the third time, the system inserts the at least one vector embedding **416** for the data associated with the chunked file into the configured vector database. In one embodiment, the fourth time is before the third time. In another embodiment, the fourth time is after the third time.

Upon insertion of the metadata **414** associated with the file **404** and the at least one vector embedding **416** for the data associated with the chunked file into the configured vector database, the system generates an indexed computing file. In some embodiments, the indexed computing file may be referred to as an entry in the vector database, and/or an

entry in the configured vector database, and/or a record in the vector database, and/or a record in the configured vector database. In some embodiments, the system may store the indexed computing file in a fourth external system. In one embodiment, the fourth external system comprises a file repository. In another embodiment, the fourth external system may be one of the network systems **150-150n**.

FIG. **4B** shows an exemplary workflow for extracting structured data from a source comprising unstructured data using an LLM in a system (e.g., application server **120** in FIG. **1**) within a complex computing network such as the network of FIG. **1** according to some embodiments. The various blocks of FIG. **4B** may be executed in a different order from that shown in FIG. **4B**. Some blocks may be optional. As shown in the figure, the system receives data associated with a first computing format from a second external system **420** (e.g., a computing system, a database, etc.). In some embodiments, the second external system **420** is referred to as a first data source. In other embodiments, the second external system **420** may be one of the network systems **150-150n**. In one embodiment, the first computing format is JavaScript Object Notation (JSON) format. Based on the first computing format, the system determines a set of computing prompts **422** from the data. In some embodiments, the system (e.g., the application server **120**, an apparatus, etc.) may be located in one or more locations. In other embodiments, the set of computing prompts comprises a set of questions.

The system transmits a first computing prompt from the set of computing prompts **422** to a third LLM **424**. In one embodiment, the first computing prompt from the set of computing prompts is a question (e.g., a questionnaire question, a survey question, etc.). In some embodiments, the third LLM **424** comprises at least one of: GPT-4, LLAMA-3, BLOOM, PaLM, GPT-3.5, BERT, Gemini, LaMDA, and Falcon. According to one embodiment, the third LLM **424** is hosted on a third third-party server. In some embodiments, the third third-party server comprises or is comprised in the first third-party server. In other embodiments, the third third-party server comprises or is comprised in the second third-party server. In another embodiment, the third LLM **424** is hosted on the local server. In one embodiment, the third LLM **424** comprises or is comprised of the first LLM **408**. In another embodiment, the third LLM **424** comprises or is comprised of the second LLM **412**.

The system receives a first vector embedding for the first computing prompt from the set of computing prompts **422** from the third LLM **424**. In some embodiments, the first vector embedding comprises or is based on a first semantic structure of at least some first content comprised in or associated with the first computing prompt from the set of computing prompts **422**. Furthermore, the first semantic structure of the at least some first content comprised in or associated with the first computing prompt from the set of computing prompts comprises a conceptual meaning of the at least some first content comprised in or associated with the first computing prompt from the set of computing prompts **422**. In some embodiments, the at least some first content comprised in or associated with the first computing prompt from the set of computing prompts **422** may comprise text, an image, a figure, a diagram, a graph, and/or a table.

The system transmits a second computing prompt, different from the first computing prompt, from the set of computing prompts **422** to the third LLM **424**. In one embodiment, the second computing prompt, different from the first computing prompt, from the set of computing prompts is a

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question (e.g., a questionnaire question, a survey question, etc.). In some embodiments, the second computing prompt, different from the first computing prompt, from the set of computing prompts 422 is transmitted to the third LLM 424 at the same time as the first computing prompt from the set of computing prompts is transmitted to the third LLM 424. In other embodiments, the second computing prompt, different from the first computing prompt, from the set of computing prompts 422 is transmitted to the third LLM 424 before the first computing prompt from the set of computing prompts is transmitted to the third LLM 424. In yet other embodiments, the second computing prompt, different from the first computing prompt, from the set of computing prompts 422 is transmitted to the third LLM 424 after the first computing prompt from the set of computing prompts is transmitted to the third LLM 424.

The system receives a second vector embedding for the second computing prompt, different from the first computing prompt, from the set of computing prompts 422 from the third LLM 424. In some embodiments, the second vector embedding comprises or is based on a second semantic structure of at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422. In one embodiment, the second semantic structure comprises or is comprised in the first semantic structure. Furthermore, the second semantic structure of the at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422 comprises a conceptual meaning of the at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422. In some embodiments, the at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422 may comprise text, an image, a figure, a diagram, a graph, and/or a table.

The system generates a first computing prompt group 432 comprising the first computing prompt from the set of computing prompts 422 and the second computing prompt, different from the first computing prompt, from the set of computing prompts 422. In some embodiments, generating the first computing prompt group comprises clustering 428 the first computing prompt from the set of computing prompts 422, with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422, based on a similarity of the first vector embedding and the second vector embedding. According to some embodiments, the similarity of the first vector embedding and the second vector embedding comprises or is based on a semantic similarity of the first vector embedding and the second vector embedding. Furthermore, the semantic similarity of the first vector embedding and the second vector embedding may be calculated based on one or more of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity. In one embodiment, the first computing prompt group 432 comprises at least three computing prompts from the set of computing prompts 422.

The system first accesses the vector database 426 from FIG. 4A at a first time. Based on first accessing the vector database 426, the system determines first file data associated with at least one first file 430 that at least partially corresponds with the first computing prompt from the set of computing prompts 422. In one embodiment, the at least one

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first file 430 that at least partially corresponds with the first computing prompt from the set of computing prompts 422, comprises first unstructured data. In some embodiments, the first unstructured data may comprise raw information or information without a predetermined structure or format.

In one embodiment, the at least one first file 430 that at least partially corresponds with the first computing prompt from the set of computing prompts 422 may at least partially match the first computing prompt from the set of computing prompts 422. In another embodiment, the at least one first file 430 that at least partially corresponds with the first computing prompt from the set of computing prompts 422 may at least partially associate with the first computing prompt from the set of computing prompts 422. According to one embodiment, determining the first file data associated with the at least one first file 430 that at least partially corresponds with the first computing prompt from the set of computing prompts may comprise or be based on a comparison between a first structure of language in the first computing prompt from the set of computing prompts and a second structure of language in the at least one first file 430 that at least partially corresponds with the first computing prompt from the set of computing prompts.

In some embodiments, the system receives filter data from a second data source or the first data source. Furthermore, the second data source may be comprised in a fifth external system. The fifth external system may be one of the network systems 150-150n. In one embodiment, the filter data comprises document source specification data. According to some embodiments, the system may execute, based on the filter data, a filtering operation on the vector database 418, thereby limiting entries in the vector database 418. In some embodiments, the filter data may comprise at least one of: a nature of the file, a credibility of the file, a freshness of the file, a file quality indicator of the file, a name of the file, a third-party associated with the file, and a source of the file. In one embodiment, the filtering operation on the vector database 418 may occur at the fifth time. In another embodiment, the filtering operation on the vector database 418 may occur after the fifth time. In yet another embodiment, the filtering operation on the vector database 418 may occur prior to the fifth time.

Furthermore, in one embodiment, the nature of the file comprises an indicator associated with a classification of the file. The classification of the file may be one of: audit from a reliable source, audit from an unreliable source, a policy or procedure document, and an unofficial document. In some embodiments, the indicator associated with the classification of the file is numerical. In another embodiment, the credibility of the file comprises an indicator associated with a source of the file. In some embodiments, the indicator associated with the source of the file is numerical. In yet other embodiments, the freshness of the file comprises an indicator associated with a creation time of the file. In some embodiments, the file quality indicator of the file comprises or is based on the nature of the file, the credibility of the file, and the freshness of the file.

In some embodiments, the determining the first file data associated with the at least one first file that partially corresponds with the first computing prompt from the set of computing prompts 422, is based on a similarity (e.g., semantic similarity) of the first vector embedding and at least one third vector embedding, wherein the at least third vector embedding is associated with or comprised in the first file data. Furthermore, the similarity of the first vector embedding and the at least one third vector embedding may be calculated based on one or more of: cosine similarity,

Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity. In some embodiments, the at least one first file may comprise: an audit document, a SOC 2 report, a policy document, a 10K financial report, a technical description document, a SOC 1 report, a data security document, a corporate charter, an information technology procedure document, a financial report, a questionnaire, a 10Q report, a human resources document, or a screenshot of an internal system.

The system second accesses the vector database 426 from FIG. 4A at a sixth time or the fifth time. In one embodiment, the sixth time is prior to the fifth time. According to another embodiment, the sixth time is after the fifth time. According to some embodiments, an entry in the vector database 418 comprises a vector embedding and metadata associated with an indexed file. In other embodiments, an entry in the vector database 418 comprises a vector embedding and metadata associated with a file comprising unstructured data. In yet other embodiments, an entry in the vector database 418 comprises at least two vector embeddings and metadata associated with an indexed file. In still other embodiments, an entry in the vector database 418 comprises at least two vector embeddings and metadata associated with a file comprising unstructured data. In one embodiment, the filtering operation on the vector database 418 may occur at the sixth time. In another embodiment, the filtering operation on the vector database may occur after the sixth time. In yet another embodiment, the filtering operation on the vector database 418 may occur prior to the sixth time.

Based on second accessing the vector database 426, the system determines second file data, associated with at least one second file 430 that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422. In one embodiment, the at least one second file 430 that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422, comprises second unstructured data. In some embodiments, the second unstructured data may comprise raw information or information without a predetermined structure or format.

In one embodiment, the at least one second file 430 that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422 may at least partially match the second computing prompt, different from the first computing prompt, from the set of computing prompts 422. In another embodiment, the at least one second file 430 that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422 may at least partially associate with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422.

In some embodiments, determining the second file data, associated with the at least one second file 430 that partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422, is based on a similarity of the second vector embedding and at least one fourth vector embedding, wherein the at least one fourth vector embedding is associated with or comprised in the second file data. According to one embodiment, the similarity of the second vector embedding and the at least one fourth vector embedding comprises or is based on a semantic similarity of the second vector embedding and the at least one fourth vector embedding. Furthermore, the semantic similarity of the first vector

embedding and the at least one third vector embedding may be calculated based on one or more of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity. In some embodiments, the at least one second file 430 that partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422 may comprise: an audit document, a SOC 2 report, a policy document, a 10K financial report, a technical description document, a SOC 1 report, a data security document, a corporate charter, an information technology procedure document, a financial report, a questionnaire, a 10Q report, a human resources document, a screenshot of an internal system, etc.

At a seventh time following the fifth time and the sixth time, the system transmits the first computing prompt group 432 to a fourth LLM 434. In some embodiments, the fourth LLM 434 comprises at least one of: GPT-4, LLAMA-3, BLOOM, PaLM, GPT-3.5, BERT, Gemini, LaMDA, and Falcon. According to one embodiment, the fourth LLM 434 is hosted on a fourth third-party server. In some embodiments, the fourth third-party server comprises or is comprised in the first third-party server. In other embodiments, the fourth third-party server comprises or is comprised in the second third-party server. In yet other embodiments, the fourth third-party server comprises or is comprised in the third third-party server. In another embodiment, the fourth LLM 434 is hosted on the local server. In one embodiment, the fourth LLM 434 comprises or is comprised of the first LLM 408. In another embodiment, the fourth LLM 434 comprises or is comprised of the second LLM 412. In yet another embodiment, the fourth LLM 434 comprises or is comprised of the third LLM 424.

The system transmits, at the seventh time or an eighth time, the first file data associated with the at least one first file 430 that at least partially corresponds with the first computing prompt from the set of computing prompts 422 to the fourth LLM 434. In one embodiment, the eighth time is before the seventh time, but still after the fifth time and the sixth time. In another embodiment, the eighth time is after the seventh time. At the seventh time, the eighth time, or a ninth time, the system transmits the second file data associated with the at least one second file 430 that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422 to the fourth LLM 434. In one embodiment, the ninth time is before the seventh time and the eighth time, but still after the fifth time and the sixth time. In another embodiment, the ninth time is after the seventh time, but before the eighth time. In yet another embodiment, the ninth time is after the seventh time and the eighth time. In still another embodiment, the ninth time is after the eighth time, but before the seventh time.

The system receives processed data from the fourth LLM 434. In some embodiments, the processed data comprises: the first computing prompt from the set of computing prompts 422, a first response associated with the first computing prompt from the set of computing prompts 422, a first citation associated with the first computing prompt from the set of computing prompts 422, a first file quality indicator associated with the first computing prompt from the set of computing prompts 422, the second computing prompt, different from the first computing prompt, from the set of computing prompts 422, a second response associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422, a second citation associated with the second computing

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prompt, different from the first computing prompt, from the set of computing prompts 422, and a second file quality indicator associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422.

In some embodiments, the first file quality indicator comprises or is based on metadata associated with the first citation. In one embodiment, the metadata associated with the first citation comprises a first file quality score. In other embodiments, the second file quality indicator comprises or is based on metadata associated with the second citation. In another embodiment, the metadata associated with the second citation comprises a second file quality score. According to one embodiment, the first response associated with the first computing prompt from the set of computing prompts may comprise an indication that the LLM could not determine a response (e.g. a response of “I don’t know.”, a response of “Insufficient information to respond.”, a response in which no file from the vector database is used in the response, etc.).

In other embodiments the processed data may comprise: the set of computing prompts 422, a set of responses associated with the set of computing prompts 422, a set of citations associated with the set of computing prompts 422, and a set of file quality indicators associated with the set of computing prompts 422. In one embodiment, the processed data is associated with a second computing format. Furthermore, the second computing format, in some embodiments, comprises JSON format.

According to some embodiments, the first citation may comprise: at least some first text from the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 422, a file name corresponding with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 422, and a page number corresponding with the at least some first text from the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 422. In one embodiment, the first citation may further comprise metadata associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 422.

In other embodiments, the first citation may comprise at least one of: at least some second text from the second file data associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422, a file name corresponding with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422, and a page number corresponding with the at least some second text from the second file data associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422. In one embodiment, the first citation may further comprise metadata associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422.

In some embodiments, the second citation may comprise: the at least some first text from the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of

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computing prompts 422, the file name corresponding with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 422, and the page number corresponding with the at least some first text from the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 422. In one embodiment, the second citation may further comprise the metadata associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 422.

According to other embodiments, the second citation may comprise at least one of: the at least some second text from the second file data associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422, the file name corresponding with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422, and the page number corresponding with the at least some second text from the second file data associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422. In one embodiment, the second citation may further comprise the metadata associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422.

Furthermore, in one embodiment, the at least some first text from the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 422 may comprise a first quote (e.g., indirect quote, direct quote, etc.) from the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts. The first quote may comprise one of: a first main idea of the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 422, a first brief summary of the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 422, or a first direct quote from the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 422. In some embodiments, the term “text” anywhere in this disclosure may additionally or alternatively include other types of data, including any data described or not described in this disclosure. In some embodiments, the term “file” anywhere in this disclosure may additionally or alternatively be referred to as or include other types of documents, including any documents described or not described in this disclosure.

According to another embodiment, the at least some second text from the second file data associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422 may comprise a second quote (e.g., an indirect quote (e.g., a non-verbatim summary), a direct quote (e.g., a verbatim quote), etc.) from the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 422. The second quote may comprise one of: a second main idea of the at least one second file that at least partially corresponds with the second computing

prompt, different from the first computing prompt, from the set of computing prompts **422**, a second brief summary of the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts **422**, or a second direct quote from the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts **422**.

The system determines a first computing indicator **436** associated with the first computing prompt from the set of computing prompts **422**. In some embodiments, determining the first computing indicator **436** is based on the first computing prompt from the set of computing prompts **422**, and the first citation associated with the first computing prompt from the set of computing prompts **422**. In one embodiment, the first computing indicator comprises a first confidence score.

In some embodiments, determining the first computing indicator **436** may be based on a similarity (e.g., semantic similarity) of the first vector embedding and a fifth vector embedding, wherein the fifth vector embedding is associated with the first citation. In one embodiment, the fifth vector embedding comprises or is comprised in the at least one third vector embedding. In another embodiment, the fifth vector embedding comprises or is comprised in the at least one fourth vector embedding. Furthermore, the similarity of the first vector embedding and the fifth vector embedding may be calculated based on one or more of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity.

The system determines a second computing indicator **436** associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts **422**. In some embodiments, determining the second computing indicator **436** is based on the second computing prompt, different from first computing prompt, from the set of computing prompts **422**, and the second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts **422**. In one embodiment, the second computing indicator comprises a second confidence score.

In other embodiments, determining the second computing indicator **436** may be based on a similarity (e.g., semantic similarity) of the second vector embedding and a sixth vector embedding, wherein the sixth vector embedding is associated with the second citation. In one embodiment, the sixth vector embedding comprises or is comprised in the at least one third vector embedding. In another embodiment, the sixth vector embedding comprises or is comprised in the at least one fourth vector embedding. Furthermore, the similarity of the second vector embedding and the sixth vector embedding may be calculated based on one or more of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity.

The system generates structured data **438**. In some embodiments, the structured data **438** comprises or is based on the processed data, the first computing indicator and the second computing indicator. In other embodiments, the structured data **438** may comprise or be based on the processed data and a set of computing indicators associated with the set of computing prompts **422**. In one embodiment, the structured data **438** is associated with a third computing format. Furthermore, the third computing format, in some embodiments, comprises JSON format. The system transmits the structured data **438** to a third external system (e.g.,

a computing system, a database, etc.) **440**. In another embodiment, the third external system **440** may be one of the network systems **150-150n**.

For any of the embodiments described herein, the large language model may refer to any language or learning computational model (e.g., an artificial neural network) of any size and is not limited to any minimum size or any minimum number of nodes.

FIG. 7 shows an exemplary workflow for generating a computing structure based on both unstructured data and structured data using a large language model (LLM) in a system (e.g., application server **120** in FIG. 1) within a complex computing network such as the network of FIG. 1 according to some embodiments. The various blocks of FIG. 7 may be executed in a different order from that shown in FIG. 7. Some blocks may be optional. As shown in the figure, the system receives first unstructured data **704** from a first data source **702** (e.g., a computing system, a database, etc.). In some embodiments, the first unstructured data **704** comprises raw information or information without a predetermined structure or format. In other embodiments, the first unstructured data **704** comprises at least one of: text, images, figures, tables, audio, videos, graphs, diagrams, etc. In yet other embodiments, the first unstructured data **704** comprises at least one of: documentation of at least one system, documentation of at least one process, documentation of at least one application, documentation of at least one apparatus, documentation of at least one procedure, etc. The system receives first structured data from a second data source **706**. In some embodiments, the second data source **706** comprises or is comprised in the first data source **702**.

The system determines a first computing library **708** associated with the first structured data. According to some embodiments, the first computing library **708** associated with the first structured data may comprise at least one of: a second set of computing prompts, a set of attributes, a set of entities, a set of workflow task types, a set of configured objects, a set of resources, a set of functions, a set of scripts (e.g., code), etc. The system receives second unstructured data from a third data source **710**. In some embodiments, the second unstructured data comprises raw information or information without a predetermined structure or format. In other embodiments, the second unstructured data **704** comprises at least one of: text, images, figures, tables, audio, videos, graphs, diagrams, etc. In one embodiment, the third data source **710** comprises or is comprised in the first data source **702**. In another embodiment, the third data source **710** comprises or is comprised in the second data source **706**.

The system determines a first set of computing prompts **712** associated with the second unstructured data. In some embodiments, the first set of computing prompts **712** associated with the second unstructured data may comprise at least one of: at least one question associated with a requirement of a system configuration, at least one question associated with a capability of the system configuration, at least one question associated with a setting of the system configuration, at least one question associated with a client associated with the system configuration, etc. The system receives second structured data, associated with a first computing format (e.g., JavaScript Object Notation (JSON) format), from a fourth data source **714**. According to one embodiment, the fourth data source **714** comprises or is comprised in the first data source **702**. In another embodiment, the fourth data source **714** comprises or is comprised in the second data source **706**. According to yet another

embodiment, the fourth data source **714** comprises or is comprised in the third data source **710**.

The system determines, based on the first computing format, a set of computing structures **716** associated with the second structured data. In some embodiments, the set of computing structures **716** comprises at least one example system configuration. The system transmits, at a first time, the first unstructured data **704** to an LLM **718**. In one embodiment, the LLM **718** comprises at least one of: GPT-4, LLAMA-3, BLOOM, PaLM, GPT-3.5, BERT, Gemini, LaMDA, Falcon, etc. In some embodiments the LLM **718** is hosted on a third-party server. In other embodiments, the LLM **718** is hosted on a local server. The system transmits, at a second time or at the first time, the first computing library **708** associated with the first structured data to the LLM **718**. According to some embodiments, the second time is prior to the first time. According to other embodiments, the second time is after the first time.

The system transmits, at a third time, at the second time, or at the first time, the first set of computing prompts **712** associated with the second unstructured data to the LLM **718**. In one embodiment, the third time is before the second time. In another embodiment, the third time is after the second time. The system transmits, at a fourth time, at the third time, at the second time, or at the first time, the set of computing structures **716** associated with the second structured data to the LLM **718**. In one embodiment, the fourth time is prior to the third time. In another embodiment, the fourth time is after the third time.

The system receives third structured data **720**, associated with a second computing format, from the LLM **718**. In one embodiment, the second computing format comprises JavaScript Object Notation (JSON) format. In one embodiment, LLM processing may comprise classification (e.g., the LLM classifies a system type, the LLM classifies a specific system type within the system type, etc.), and/or inference, and/or question answering, and/or traditional computer processing techniques, etc. In another embodiment, the LLM may choose certain computing prompts from the second set of computing prompts for a questionnaire based on the system type and/or the specific system type. In some cases, the first set of computing prompts **712** associated with the second unstructured data adds context to the documentation during the LLM processing. In some embodiments, the third structured data **720** comprises or is based on the first set of computing prompts **712** associated with the second unstructured data, a set of responses associated with the first set of computing prompts **712** associated with the second unstructured data, and a computing structure (i.e., a system configuration). Furthermore, the computing structure, according to some embodiments, is not comprised in the set of computing structures **716** associated with the second structured data.

In some cases, the computing structure comprises or is based on the first unstructured data **704**, the first computing library **708** associated with the first structured data, the first set of computing prompts **712** associated with the second unstructured data, and the set of computing structures **716** associated with the second structured data. In other cases, the computing structure comprises or is based on at least one of: the first unstructured data **704**, the first computing library **708** associated with the first structured data, the first set of computing prompts **712** associated with the second unstructured data, or the set of computing structures **716** associated with the second structured data. In some embodiments, the computing structure further comprises or is based on a second computing library. Furthermore, in some embodi-

ments, the second computing library is generated by the LLM **718**. In one embodiment, the computing structure comprises a system configuration in a configurable platform.

The system transmits the third structured data **720** to a first system (e.g., a computing system, a database, etc.) **722**. In one embodiment, the first system **722** may be one of the network systems **150-150n**. In some embodiments, the system may execute one or more of these instructions in a first stage and a second stage, such that fifth structured data associated with the second stage comprises or is based on fourth structured data associated with the first stage (i.e., firstly, the system configures a data dictionary, secondly, the system configures, based on the data dictionary, at least one configured form, and thirdly, the system configures, based on the data dictionary and the at least one configured form, at least one workflow).

Any of the embodiments, methods, systems, etc. described in this disclosure may be used in a third-party risk management system. The embodiments, methods, systems, etc. described in this disclosure may identify, assess, and minimize risk associated with third-parties (i.e., clients, vendors, service providers, suppliers, contractors, etc.). Furthermore, any files and/or documents referred to in this disclosure may comprise third-party risk management files and/or documents (i.e., audit documents, Service Organization Control (SOC) 2 reports, SOC 1 reports, 10Q reports, risk reports, etc.).

FIG. 9 shows an exemplary workflow for generating structured data from files comprising unstructured data, and transmitting recommendations, based on the structured data, to a system storing the files, using a large language model (LLM) in a system (e.g., application server **120** in FIG. 1) within a complex computing network such as the network of FIG. 1 according to some embodiments. The various blocks of FIG. 9 may be executed in a different order from that shown in FIG. 9. Some blocks may be optional. As shown in the figure, the system receives data, associated with a first computing format, from a first user (e.g., a computing system, a database, a user within the computing system, etc.) **902**. In some embodiments, the user comprises or is comprised in a first system. Furthermore, in one embodiment, the first system comprises or is comprised in one of the network systems **150-150n**. In one embodiment, the first computing format comprises JavaScript Object Notation (JSON) format. The system determines, based on the first computing format, a set of computing prompts **904** from the data. In some embodiments, the system (e.g., the application server **120**, an apparatus, etc.) may be located in one or more locations. In several embodiments, the set of computing prompts comprises a set of questions.

The system transmits a first computing prompt from the set of computing prompts **904** to a first LLM. In one embodiment, the first computing prompt from the set of computing prompts is a question (e.g., a questionnaire question, a survey question, etc.). In some embodiments, the first LLM comprises at least one of: GPT-4, LLAMA-3, BLOOM, PaLM, GPT-3.5, BERT, Gemini, LaMDA, and Falcon. According to one embodiment, the first LLM is hosted on a first third-party server. In another embodiment, the first LLM is hosted on the local server.

The system receives a first vector embedding for the first computing prompt from the set of computing prompts **904** from the first LLM. In some embodiments, the first vector embedding comprises or is based on a first semantic structure of at least some first content comprised in or associated with the first computing prompt from the set of computing prompts **904**. Furthermore, the first semantic structure of the

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at least some first content comprised in or associated with the first computing prompt from the set of computing prompts comprises a first conceptual meaning of the at least some first content comprised in or associated with the first computing prompt from the set of computing prompts 904. In some embodiments, the at least some first content comprised in or associated with the first computing prompt from the set of computing prompts 904 may comprise text, an image, a figure, a diagram, a graph, and/or a table.

The system transmits a second computing prompt, different from the first computing prompt, from the set of computing prompts 904 to the first LLM. In one embodiment, the second computing prompt, different from the first computing prompt, from the set of computing prompts is a question (e.g., a questionnaire question, a survey question, etc.). In some embodiments, the second computing prompt, different from the first computing prompt, from the set of computing prompts 904 is transmitted to the first LLM at the same time as the first computing prompt from the set of computing prompts is transmitted to the first LLM. In other embodiments, the second computing prompt, different from the first computing prompt, from the set of computing prompts 904 is transmitted to the first LLM before the first computing prompt from the set of computing prompts 904 is transmitted to the first LLM. In yet other embodiments, the second computing prompt, different from the first computing prompt, from the set of computing prompts 904 is transmitted to the first LLM after the first computing prompt from the set of computing prompts 904 is transmitted to the first LLM.

The system receives a second vector embedding for the second computing prompt, different from the first computing prompt, from the set of computing prompts 904 from the first LLM. In some embodiments, the second vector embedding comprises or is based on a second semantic structure of at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904. In one embodiment, the second semantic structure comprises or is comprised in the first semantic structure. Furthermore, the second semantic structure of the at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts comprises a second conceptual meaning of the at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904. In some embodiments, the at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904 may comprise text, an image, a figure, a diagram, a graph, and/or a table.

The system generates a first computing prompt group comprising the first computing prompt from the set of computing prompts 904 and the second computing prompt, different from the first computing prompt, from the set of computing prompts 904. In some embodiments, generating the first computing prompt group comprises clustering the first computing prompt from the set of computing prompts 904, with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904, based on a first similarity of the first vector embedding and the second vector embedding. According to some embodiments, the first similarity of the first vector embedding and the second vector embedding comprises or is based on a first semantic similarity of the first vector embedding

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and the second vector embedding. Furthermore, the first semantic similarity of the first vector embedding and the second vector embedding may be calculated based on one or more of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity. In one embodiment, the first computing prompt group comprises at least three computing prompts from the set of computing prompts 904.

The system first accesses the first database 906 at a first time. Based on first accessing the first database 906, the system determines first file data associated with at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904. In one embodiment, the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904, comprises first unstructured data. In some embodiments, the first unstructured data may comprise raw information or information without a predetermined structure or format.

In one embodiment, the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904 may at least partially match the first computing prompt from the set of computing prompts 904. In another embodiment, the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904 may at least partially associate with the first computing prompt from the set of computing prompts 904. According to one embodiment, determining the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904 may comprise or be based on a comparison between a first structure of language in the first computing prompt from the set of computing prompts and a second structure of language in the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904.

In some embodiments, the system receives filter data from a data source. Furthermore, the data source may be comprised in an external system. The external system may be one of the network systems 150-150n. In one embodiment, the filter data comprises document source specification data. According to some embodiments, the system may execute, based on the filter data, a filtering operation on the first database 908, thereby limiting entries in the first database 908. In some embodiments, the filter data may comprise at least one of: a nature of the file, a credibility of the file, a freshness of the file, a file quality indicator of the file, a name of the file, a third-party associated with the file, and a source of the file. In one embodiment, the filtering operation on the first database 908 may occur at the first time. In another embodiment, the filtering operation on the first database 908 may occur after the first time. In yet another embodiment, the filtering operation on the first database 908 may occur prior to the first time.

Furthermore, in one embodiment, the nature of the file comprises an indicator associated with a classification of the file. The classification of the file may be one of: audit from a reliable source, audit from an unreliable source, a policy or procedure document, and an unofficial document. In some embodiments, the indicator associated with the classification of the file is numerical. In another embodiment, the credibility of the file comprises an indicator associated with a source of the file. In some embodiments, the indicator associated with the source of the file is numerical. In yet other embodiments, the freshness of the file comprises an indicator associated with a creation time of the file. In some

embodiments, the file quality indicator of the file comprises or is based on the nature of the file, the credibility of the file, and the freshness of the file.

In some embodiments, the determining the first file data associated with the at least one first file that partially corresponds with the first computing prompt from the set of computing prompts 904, is based on a second similarity (e.g., semantic similarity) of the first vector embedding and at least one third vector embedding, wherein the at least third vector embedding is associated with or comprised in the first file data. Furthermore, the second similarity of the first vector embedding and the at least one third vector embedding may be calculated based on one or more of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity. In some embodiments, the at least one first file may comprise: an audit document, a SOC 2 report, a policy document, a 10K financial report, a technical description document, a SOC 1 report, a data security document, a corporate charter, an information technology procedure document, a financial report, a questionnaire, a 10Q report, a human resources document, or a screenshot of an internal system.

The system second accesses the first database 906 at a second time or the first time. In one embodiment, the second time is prior to the first time. According to another embodiment, the second time is after the first time. According to some embodiments, an entry in the first database 908 comprises a vector embedding and metadata associated with an indexed file. In other embodiments, an entry in the first database 908 comprises a vector embedding and metadata associated with a file comprising unstructured data. In yet other embodiments, an entry in the first database 908 comprises at least two vector embeddings and metadata associated with an indexed file. In still other embodiments, an entry in the first database 908 comprises at least two vector embeddings and metadata associated with a file comprising unstructured data. In one embodiment, the filtering operation on the first database 908 may occur at the second time. In another embodiment, the filtering operation on the first database 908 may occur after the second time. In yet another embodiment, the filtering operation on the first database 908 may occur prior to the second time.

Based on the second accessing the first database 906, the system determines second file data, associated with at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904. In one embodiment, the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904, comprises second unstructured data. In some embodiments, the second unstructured data may comprise raw information or information without a predetermined structure or format.

In one embodiment, the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904 may at least partially match the second computing prompt, different from the first computing prompt, from the set of computing prompts 904. In another embodiment, the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904 may at least partially associate with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904.

In some embodiments, determining the second file data, associated with the at least one second file that partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904, is based on a third similarity of the second vector embedding and at least one fourth vector embedding, wherein the at least one fourth vector embedding is associated with or comprised in the second file data. According to one embodiment, the third similarity of the second vector embedding and the at least one fourth vector embedding comprises or is based on a third semantic similarity of the second vector embedding and the at least one fourth vector embedding. Furthermore, the third semantic similarity of the first vector embedding and the at least one third vector embedding may be calculated based on one or more of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity. In some embodiments, the at least one second file that partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904 may comprise: an audit document, a SOC 2 report, a policy document, a 10K financial report, a technical description document, a SOC 1 report, a data security document, a corporate charter, an information technology procedure document, a financial report, a questionnaire, a 10Q report, a human resources document, a screenshot of an internal system, etc.

At a third time following the first time and the second time, the system transmits the first computing prompt group to a second LLM 910. In some embodiments, the second LLM 910 comprises at least one of: GPT-4, LLAMA-3, BLOOM, PaLM, GPT-3.5, BERT, Gemini, LaMDA, and Falcon. According to one embodiment, the second LLM 910 is hosted on a second third-party server. In some embodiments, the second third-party server comprises or is comprised in the first third-party server. In another embodiment, the second LLM 910 is hosted on the local server. In one embodiment, the second LLM 910 comprises or is comprised of the first LLM.

The system transmits, at the third time or a fourth time, the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904 to the second LLM 910. In one embodiment, the fourth time is before the third time, but still after the first time and the second time. In another embodiment, the fourth time is after the third time. At the third time, the fourth time, or a fifth time, the system transmits the second file data associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904 to the second LLM 910. In one embodiment, the fifth time is before the third time and the fourth time, but still after the first time and the second time. In another embodiment, the fifth time is after the third time, but before the fourth time. In yet another embodiment, the fifth time is after the third time and the fourth time. In still another embodiment, the fifth time is after the fourth time, but before the third time.

The system receives first structured data from the second LLM 910. In some embodiments, the first structured data comprises: the first computing prompt from the set of computing prompts 904, a first response associated with the first computing prompt from the set of computing prompts 904, a first citation associated with the first computing prompt from the set of computing prompts 904, a first file quality indicator associated with the first computing prompt from the set of computing prompts 904, the second com-

puting prompt, different from the first computing prompt, from the set of computing prompts 904, a second response associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904, a second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904, and a second file quality indicator associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904.

In some embodiments, the first file quality indicator comprises or is based on metadata associated with the first citation. In one embodiment, the metadata associated with the first citation comprises a first file quality score. In other embodiments, the second file quality indicator comprises or is based on metadata associated with the second citation. In another embodiment, the metadata associated with the second citation comprises a second file quality score. According to one embodiment, the first response associated with the first computing prompt from the set of computing prompts 904 may comprise an indication that the second LLM could not determine a response (e.g. a response of “I don’t know.”, a response of “Insufficient information to respond.”, a response in which no file from the first database is used in the response, etc.).

In other embodiments the first structured data may comprise: the set of computing prompts 904, a set of responses associated with the set of computing prompts 904, a set of citations associated with the set of computing prompts 904, and a set of file quality indicators associated with the set of computing prompts 904. According to one embodiment, the system may transmit the first structured data to the first user 902 and/or the first system. According to another embodiment, the system may transmit the set of computing prompts 904 and the set of responses associated with the set of computing prompts 904, to the first user 902 and/or the first system. In one embodiment, the first structured data is associated with a second computing format. Furthermore, the second computing format, in some embodiments, comprises JSON format.

According to some embodiments, the first citation may comprise: at least some first text from the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904, a file name corresponding with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904, and a page number corresponding with the at least some first text from the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904. In one embodiment, the first citation may further comprise metadata associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904.

In other embodiments, the first citation may comprise at least one of: at least some second text from the second file data associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904, a file name corresponding with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904, and a page number corresponding with the at least some second text from the second file data associated with the at least one second file that at least partially corresponds with

the second computing prompt, different from the first computing prompt, from the set of computing prompts 904. In one embodiment, the first citation may further comprise metadata associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904.

Furthermore, in one embodiment, the at least some first text from the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904 may comprise a first quote (e.g., indirect quote, direct quote, etc.) from the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904. The first quote may comprise one of: a first main idea of the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904, a first brief summary of the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904, or a first direct quote from the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts 904. In some embodiments, the term “text” anywhere in this disclosure may additionally or alternatively include other types of data, including any data described or not described in this disclosure. In some embodiments, the term “file” anywhere in this disclosure may additionally or alternatively be referred to as or include other types of documents, including any documents described or not described in this disclosure.

The system determines a first computing indicator 912 associated with the first computing prompt from the set of computing prompts 904. In some embodiments, determining the first computing indicator 912 is based on the first computing prompt from the set of computing prompts 904, and the first citation associated with the first computing prompt from the set of computing prompts 904. In one embodiment, the first computing indicator comprises a first confidence score.

In some embodiments, determining the first computing indicator 912 may be based on a fourth similarity (e.g., semantic similarity) of the first vector embedding and a fifth vector embedding, wherein the fifth vector embedding is associated with the first citation. In one embodiment, the fifth vector embedding comprises or is comprised in the at least one third vector embedding. In another embodiment, the fifth vector embedding comprises or is comprised in the at least one fourth vector embedding. Furthermore, the fourth similarity of the first vector embedding and the fifth vector embedding may be calculated based on one or more of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity.

The system determines a second computing indicator 912 associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904. In some embodiments, determining the second computing indicator 912 is based on the second computing prompt, different from first computing prompt, from the set of computing prompts 904, and the second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts 904. In one embodiment, the second computing indicator comprises a second confidence score.

In other embodiments, determining the second computing indicator 912 may be based on a fifth similarity (e.g., semantic similarity) of the second vector embedding and a

sixth vector embedding, wherein the sixth vector embedding is associated with the second citation. In one embodiment, the sixth vector embedding comprises or is comprised in the at least one third vector embedding. In another embodiment, the sixth vector embedding comprises or is comprised in the at least one fourth vector embedding. Furthermore, the fifth similarity of the second vector embedding and the sixth vector embedding may be calculated based on one or more of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity.

The system generates second structured data. In some embodiments, the second structured data comprises or is based on the first structured data, the first computing indicator and the second computing indicator. In other embodiments, the second structured data may comprise or be based on the first structured data and a set of computing indicators associated with the set of computing prompts **904**. According to some embodiments, the system may access a second database **914**. Furthermore, according to some embodiments, the system may transmit the second structured data to the second database **914**. In one embodiment, the second database may generate and/or transmit metrics to the first user **902** and/or the first system. In some cases, the metrics comprise a process trend report. Furthermore, in some embodiments, the process trend report may comprise at least one of: a trend in responses to computing prompts, a trend in file quality indicators of files used to respond to computing prompts, or a trend in computing indicators associated with the computing prompts. In other embodiments, the system may transmit at least some of the structured data to the first user **902** and/or the first system. In one embodiment, the second structured data is associated with a third computing format. Furthermore, the third computing format, in some embodiments, comprises JSON format.

The system generates, based on the second structured data, at least one recommendation **916**. In some embodiments, the at least one recommendation **916** is associated with the first database. In other embodiments, the at least one recommendation **916** is associated with at least one of the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts **904**, or the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts **904**. In yet another embodiment, the at least one recommendation **916** is associated with at least one third file associated with (e.g., stored in, managed by, etc.) the first database. According to some embodiments, the at least one recommendation may comprise suggestions and/or requests to improve at least one file quality indicator associated with the at least one third file associated with the first database (i.e., improve quality of files in the first database). In some embodiments, generating the at least one recommendation **916** may comprise using a third LLM.

The system transmits the at least one recommendation **916** to a first database system **918** and/or a second user associated with the first database **908**. In one embodiment, the first database system **918** comprises a second system managing the first database **908**. In some embodiments, the first database system **918** and/or the second user associated with the first database **908**, may, based on the at least one recommendation **916**, update or modify the first database **920**. For example, in one embodiment, updating or modifying the first database **920** may comprise or be

based on at least one request for an additional file to be inserted into the first database. Furthermore, in some embodiments, the additional file comprises or is associated with an improved file quality indicator (i.e., insert higher quality files into the first database). In other embodiments, the additional file comprises or is associated with data not comprised in the first database **908**. In yet another embodiment, the updating or modifying the first database **920** may comprise or be based on at least one request for an existing file in the first database **908** to be updated (e.g., audited) or replaced. In some cases, the updating or modifying the database **920** may be done automatically.

Exemplary Flowcharts

FIGS. **5A-5B** and **6A-6D** show exemplary flowcharts for methods, systems/apparatuses, and computer program products that implement generating an indexed computing file based on inserting vector embeddings and metadata, received from a large language model (LLM), into a configured vector database, and extracting structured data from a source comprising unstructured data, respectively, in an exemplary computing network such as the network of FIG. **1**.

FIGS. **8A-8B** and **10A-10E** show exemplary flowcharts for methods, systems/apparatuses, and computer program products that implement generating a computing structure based on both unstructured data and structured data using a large language model (LLM), and generating structured data from files comprising unstructured data, and transmitting recommendations, based on the structured data, to a system storing the files, using a large language model (LLM), respectively, in an exemplary computing network such as the network of FIG. **1**.

FIGS. **5A** and **5B** are block diagrams of a method for generating an indexed computing file in a complex computing network. The various blocks of FIGS. **5A** and **5B** may be executed in a different order from that shown in FIGS. **5A** and **5B**. Some blocks may be optional.

At block **502** of FIG. **5A**, the method comprises receiving, using one or more computing device processors, a file from a first file source, wherein the file comprises unstructured data. In some embodiments, the file may be an audit document, a Service Organization Control (SOC) 2 report, a policy document, a 10K financial report, a technical description document, a SOC 1 report, a data security document, a corporate charter, an information technology procedure document, a financial report, a questionnaire, a 10Q report, a human resources document, a screenshot of an internal system, etc. At block **504**, the method comprises extracting, using the one or more computing device processors, text from the file. At block **506**, the method comprises transmitting, using the one or more computing device processors, the text from the file to a large language model (LLM).

At block **508**, the method comprises receiving, using the one or more computing device processors, at a first time, metadata associated with the file from the LLM, wherein the metadata associated with the file comprises or is based on file quality data, wherein the file quality data comprises or is based on at least one of: a nature of the file, a credibility of the file, a freshness of the file, and a file quality indicator of the file. In some embodiments the metadata associated with the file further comprises a citation. The citation may further comprise at least some text from the file, a file name corresponding with the file, and a page number associated with the at least some text from the file. In some embodiments, the metadata associated with the file further comprises third-party source data.

In some embodiments, the file quality data comprising or being based on the at least one of: the nature of the file, the credibility of the file, the freshness of the file, and the file quality indicator of the file, further comprises or is based on at least two of: the nature of the file, the credibility of the file, the freshness of the file, and the file quality indicator of the file. In other embodiments, the file quality data comprising or being based on the at least one of: the nature of the file, the credibility of the file, the freshness of the file, and the file quality indicator of the file, further comprises or is based on at least three of: the nature of the file, the credibility of the file, the freshness of the file, and the file quality indicator of the file. In yet other embodiments, the file quality data comprising or being based on the at least one of: the nature of the file, the credibility of the file, the freshness of the file, and the file quality indicator of the file, further comprises or is based on: the nature of the file, the credibility of the file, the freshness of the file, and the file quality indicator of the file.

In some embodiments, the nature of the file comprises an indicator associated with a classification of the file. The classification of the file may be at least one of: audit from a reliable source, audit from an unreliable source, policy or procedure document, and unofficial document. In some embodiments, the indicator associated with the classification of the file is numerical. In other embodiments, the credibility of the file comprises an indicator associated with a source of the file. In some embodiments, the indicator associated with the source of the file is numerical. In yet other embodiments, the freshness of the file comprises an indicator associated with a creation time of the file. In some embodiments, the file quality indicator comprises or is based on the nature of the file, the credibility of the file, and the freshness of the file. In some embodiments, the file quality indicator of the file comprises or is based on the nature of the file, the credibility of the file, and the freshness of the file (e.g. an average of the nature of the file, the credibility of the file, and the freshness of the file, a median of the nature of the file, the credibility of the file, and the freshness of the file, etc.)

At block 510, the method comprises executing, using the one or more computing device processors, at a second time or at the first time, a chunking computing operation using the file, thereby resulting in a chunked file. At block 512, the method comprises transmitting, using the one or more computing device processors, text associated with the chunked file to the LLM. In some embodiments, the text associated with the chunked file comprises a word from the chunked file. In other embodiments, the text associated with the chunked file comprises a phrase from the chunked file. In yet other embodiments, the text associated with the chunked file comprises a sentence from the chunked file. In some embodiments, the text associated with the chunked file comprises a paragraph from the chunked file. In other embodiments, the text associated with the chunked file comprises the chunked file.

At block 514, the method comprises receiving, using the one or more computing device processors, at least one vector embedding for the text associated with the chunked file from the LLM, wherein the at least one vector embedding comprises or is based on a semantic structure of at least some of the text associated with the chunked file. In some embodiments, the semantic structure of the at least some of the text associated with the chunked file comprises or is based on a conceptual meaning of the at least some of the text associated with the chunked file.

Turning to FIG. 5B, at block 516, the method comprises configuring, using the one or more computing device pro-

cessors, a vector database to store vector embeddings and metadata, thereby resulting in a configured vector database. At block 518, the method comprises first inserting, using the one or more computing device processors, at a third time following the first time and the second time, the at least one vector embedding for the text associated with the chunked file, into the configured vector database. At block 520, the method comprises second inserting, using the one or more computing device processors, at the third time following the first time and the second time, the metadata associated with the file into the configured vector database. In some embodiments, the second inserting the metadata associated with the file in the configured vector database occurs at a fourth time. In some embodiments, the fourth time is before the third time. In other embodiments, the fourth time is after the third time.

At block 522, the method comprises generating, based on the first inserting the at least one vector embedding for the text associated with the chunked file into the configured vector database, and the second inserting the metadata associated with the file into the configured vector database, an indexed computing file.

FIGS. 6A-6D are block diagrams of a method for extracting structured data from relevant unstructured data in a complex computing network. The various blocks of FIGS. 6A-6D may be executed in a different order from that shown in FIGS. 6A-6D. Some blocks may be optional.

At block 602 of FIG. 6A, the method comprises receiving, using one or more computing device processors, data, associated with a first computing format, from a first data source. In some embodiments, the first computing format is JavaScript Object Notation (JSON). In some embodiments, the data comprises a questionnaire (e.g., a due diligence questionnaire, a risk questionnaire, a compliance questionnaire, a performance questionnaire, etc.). At block 604, the method comprises determining, using the one or more computing device processors, based on the first computing format, a set of computing prompts from the data. At block 606, the method comprises transmitting, using the one or more computing device processors, a first computing prompt from the set of computing prompts to a large language model (LLM). In some embodiments, the LLM comprises at least one of: GPT-4, LLAMA-3, BLOOM, PaLM, GPT-3.5, BERT, Gemini, LaMDA, and Falcon. In some embodiments, the transmitting the first computing prompt from the set of computing prompts to the LLM occurs at a first time.

At block 608, the method comprises receiving, using the one or more computing device processors, a first vector embedding for the first computing prompt from the set of computing prompts, from the LLM, wherein the first vector embedding comprises or is based on a first semantic structure of at least some first content comprised in or associated with the first computing prompt from the set of computing prompts. In some embodiments, the first semantic structure of the at least some first content comprises a conceptual meaning of the at least some first content.

At block 610, the method comprises transmitting, using the one or more computing device processors, a second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM. In some embodiments, the transmitting the second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM, occurs at a first time. In other embodiments, the transmitting the second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM, occurs at a second time. In some embodiments, the LLM

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may be a different LLM than the one used in the transmitting the first computing prompt from the set of computing prompts to the LLM. In other embodiments, the LLM may be the same LLM used in the transmitting the first computing prompt from the set of computing prompts to the LLM.

At block 612, the method comprises receiving, using the one or more computing device processors, a second vector embedding for the second computing prompt, different from the first computing prompt, from the set of computing prompts, from the LLM, wherein the second vector embedding comprises or is based on a second semantic structure or the first semantic structure of at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts.

Turning to FIG. 6B, at block 614, the method comprises generating, using the one or more computing device processors, a first computing prompt group comprising: the first computing prompt from the set of computing prompts, and the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the generating the first computing prompt group comprises clustering the first computing prompt from the set of computing prompts, with the second computing prompt, different from the first computing prompt, from the set of computing prompts, based on a similarity of the first vector embedding and the second vector embedding. In some embodiments, the similarity of the first vector embedding and the second vector embedding comprises a semantic similarity. In some embodiments, the semantic similarity comprises or is based on at least one of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity.

At block, 616, the method comprises first accessing, using the one or more computing device processors, a vector database. In some embodiments, an entry in the vector database comprises a vector embedding and metadata associated with an indexed computing file. In other embodiments, the entry in the vector database comprises a vector embedding and metadata associated with a file comprising unstructured data. In some embodiments, the vector database may be filtered based on filter data. In some embodiments, the filter data may be received from the first data source. In other embodiments, the filter data may be received from a second data source.

At block 618, the method comprises determining, using the one or more computing device processors, for the first computing prompt from the set of computing prompts, using the first vector embedding, based on the first accessing the vector database, at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, wherein the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, comprises first unstructured data, wherein the determining the at least one first file that partially corresponds with the first computing prompt from the set of computing prompts, is based on a similarity of the first vector embedding and at least one third vector embedding, wherein the at least one third vector embedding is associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts. In some embodiments, the similarity of the first vector embedding and the at least one third vector embedding comprises a semantic similarity. In some embodiments, the semantic similarity comprises or is computed using at least one of: cosine similarity, Euclidean

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distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity.

At block 620, the method comprises second accessing, using the one or more computing device processors, the vector database. In some embodiments, the second accessing the vector database may happen concurrently with the first accessing the vector database. In other embodiments, the second accessing the vector database may occur prior to the first accessing the vector database. In yet other embodiments, the second accessing the vector database may occur after the first accessing the vector database.

Turning to FIG. 6C, at block 622, the method comprises determining, using the one or more computing device processors, for the second computing prompt, different from the first computing prompt, from the set of computing prompts, using the second vector embedding, based on the second accessing the vector database, at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, comprises second unstructured data, wherein the determining the at least one second file that partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, is based on a similarity of the second vector embedding and at least one fourth vector embedding, wherein the at least one fourth vector embedding is associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts.

In some embodiments, the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, comprises at least one of the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts. In other embodiments, the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, comprises none of the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts. In some embodiments, the similarity of the second vector embedding and the at least one fourth vector embedding comprises a semantic similarity. In some embodiments, the semantic similarity comprises or is calculated using at least one of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity.

At block 624, the method comprises transmitting, using the one or more computing device processors, the first computing prompt group to the LLM. At block 626, the method comprises transmitting, using the one or more computing device processors, the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, to the LLM. At block 628, the method comprises transmitting, using the one or more computing device processors, the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM. In some embodiments, the transmitting the first computing group to the LLM may happen concurrently with the transmitting at least one first file that at least partially corresponds

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with the first computing prompt from the set of computing prompts, to the LLM, and/or the transmitting the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM.

Turning to FIG. 6D, at block 630, the method comprises receiving, using the one or more computing device processors, processed data from the LLM, wherein the processed data comprises or is based on the first computing prompt from the set of computing prompts, a first response associated with the first computing prompt from the set of computing prompts, a first citation associated with the first computing prompt from the set of computing prompts, a first file quality indicator associated with the first computing prompt from the set of computing prompts, the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second response associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, and a second file quality indicator associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts. In some embodiments, the processed data comprises the set of computing prompts, a set of responses associated with the set of computing prompts, a set of citations associated with the set of computing prompts, and a set of file quality indicators associated with the set of computing prompts.

At block 632, the method comprises determining, using the one or more computing device processors, a first computing indicator based on the first computing prompt from the set of computing prompts, and the first citation associated with the first computing prompt from the set of computing prompts. At block 634, the method comprises determining, using the one or more computing device processors, a second computing indicator based on the second computing prompt, different from the first computing prompt, from the set of computing prompts, and the second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts. In some embodiments, the determining the first computing indicator happens concurrently with the determining the second computing indicator. In other embodiments, the determining the first computing indicator occurs prior to the determining the second computing indicator. In yet other embodiments, the determining the first computing indicator happens following the determining the second computing indicator.

At block 636 the method comprises generating, using the one or more computing device processors, structured data comprising or based on the processed data, the first computing indicator, and the second computing indicator. At block 638, the method comprises transmitting, using the one or more computing device processors, the structured data to a first system.

FIGS. 8A and 8B are block diagrams of a method for generating a computing structure based on both unstructured data and structured data using a large language model (LLM) in a complex computing network. The various blocks of FIGS. 8A and 8B may be executed in a different order from that shown in FIGS. 8A and 8B. Some blocks may be optional.

At block 802 of FIG. 8A, the method comprises receiving, using one or more computing device processors, first unstructured data from a first data source. In some embodi-

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ments, the first unstructured data comprises raw information or information without a predetermined structure or format. In other embodiments, the first unstructured data comprises at least one of: text, an image, a figure, a table, audio, video, a graph, a diagram, etc. In yet other embodiments, the first unstructured data comprises at least one of: documentation of at least one system, documentation of at least one process, documentation of at least one application, documentation of at least one apparatus, documentation of at least one procedure, etc. At block 804 of FIG. 8A, the method comprises receiving, using the one or more computing device processors, first structured data from a second data source. In some embodiments, the second data source comprises or is comprised in the first data source.

At block 806 of FIG. 8A, the method comprises determining, using the one or more computing device processors, a first computing library associated with the first structured data. According to some embodiments, the first computing library associated with the first structured data may comprise at least one of: a second set of computing prompts, a set of attributes, a set of entities, a set of workflow task types, a set of configured objects, a set of resources, a set of functions, a set of scripts (e.g., code), etc. At block 808 of FIG. 8A, the method comprises receiving, using the one or more computing device processors, second unstructured data from a third data source. In some embodiments, the second unstructured data comprises raw information or information without a predetermined structure or format. In other embodiments, the second unstructured data comprises at least one of: text, images, figures, tables, audio, videos, graphs, diagrams, etc. In one embodiment, the third data source comprises or is comprised in the first data source. In another embodiment, the third data source comprises or is comprised in the second data source.

At block 810 of FIG. 8A, the method comprises determining, using the one or more computing device processors, a first set of computing prompts associated with the second unstructured data. In some embodiments, the first set of computing prompts associated with the second unstructured data may comprise at least one of: at least one requirement associated with a system configuration, at least one capability associated with the system configuration, at least one setting associated with the system configuration, at least one client associated with the system configuration, etc.

At block 812 of FIG. 8A, the method comprises receiving, using the one or more computing device processors, second structured data, associated with a first computing format, from a fourth data source. In one embodiment, the first computing format comprises JavaScript Object Notation (JSON) format. According to one embodiment, the fourth data source comprises or is comprised in the first data source. In another embodiment, the fourth data source comprises or is comprised in the second data source. According to yet another embodiment, the fourth data source comprises or is comprised in the third data source.

At block 814 of FIG. 8A, the method comprises determining, using the one or more computing device processors, based on the first computing format, a set of computing structures associated with the second structured data. In some embodiments, the set of computing structures comprises at least one example system configuration. At block 816 of FIG. 8A, the method comprises transmitting, using the one or more computing device processors, at a first time, the first unstructured data to an LLM. In one embodiment, the LLM comprises at least one of: GPT-4, LLAMA-3, BLOOM, PaLM, GPT-3.5, BERT, Gemini, LaMDA, Falcon,

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etc. In some embodiments the LLM is hosted on a third-party server. In other embodiments, the LLM is hosted on a local server.

Turning to FIG. 8B, at block **818**, the method comprises transmitting, using the one or more computing device processors, at a second time or the first time, the first computing library associated with the first structured data to the LLM. According to one embodiment, the second time is before the first time. According to another embodiment, the second time is after the first time. At block **820** of FIG. 8B, the method comprises transmitting, using the one or more computing device processors, at a third time, the second time, or the first time, the first set of computing prompts associated with the second unstructured data to the LLM. According to some embodiments, the third time is prior to the second time. According to other embodiments, the third time is after the second time. At block **822** of FIG. 8B, the method comprises transmitting, using the one or more computing device processors, at a fourth time, the third time, the second time, or the first time, the set of computing structures associated with the second structured data to the LLM. In one embodiment, the fourth time is before the third time. In another embodiment, the fourth time is after the third time.

At block **824** of FIG. 8B, the method comprises receiving, using the one or more computing device processors, third structured data, associated with a second computing format, from the LLM, wherein the third structured data comprises or is based on the first set of computing prompts associated with the second unstructured data, a set of responses associated with the first set of computing prompts associated with the second unstructured data, and a computing structure, wherein the computing structure is not comprised in the set of computing structures associated with the second structured data, and wherein the computing structure comprises or is based on the first unstructured data, the first computing library associated with the first structured data, the first set of computing prompts associated with the second unstructured data, and the set of computing structures associated with the second structured data.

In one embodiment, the second computing format comprises JavaScript Object Notation (JSON) format. In some embodiments, the computing structure comprises or is based on at least one of: the first unstructured data, the first computing library associated with the first structured data, and the set of computing structures associated with the second structured data. In one embodiment, the computing structure comprises a system configuration. According to some embodiments, the method further comprises initiating generating, using the one or more computing device processors, a second computing library using the LLM. Furthermore, in one embodiment, the computing structure comprises or is based on at least one of: the first unstructured data, the second computing library, and the set of computing structures associated with the second structured data.

At block **826** of FIG. 8B, the method comprises transmitting, using the one or more computing device processors, the third structured data to a first system. In one embodiment, the first system may comprise or be comprised in one of the network systems **150-150n**. In another embodiment, the first system may comprise or be comprised in a system associated with the endpoint device **125**. In some embodiments, the system may execute one or more of these blocks in a first stage and a second stage, such that fifth structured data associated with the second stage comprises or is based on fourth structured data associated with the first stage.

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FIGS. **10A-10E** are block diagrams of a method for generating structured data from files comprising unstructured data, and transmitting recommendations, based on the structured data, to a system storing the files, using a large language model (LLM). The various blocks of FIGS. **10A-10E** may be executed in a different order from that shown in FIGS. **10A-10E**. Some blocks may be optional.

At block **1002** of FIG. **10A**, the method comprises receiving, using one or more computing device processors, data, associated with a first computing format, from a first system. In some embodiments, the first computing format is JavaScript Object Notation (JSON). In some embodiments, the data comprises a questionnaire (e.g., a due diligence questionnaire, a risk questionnaire, a compliance questionnaire, a performance questionnaire, etc.). At block **1004**, the method comprises determining, using the one or more computing device processors, based on the first computing format, a set of computing prompts from the data. At block **1006**, the method comprises transmitting, using the one or more computing device processors, a first computing prompt from the set of computing prompts to an LLM. In some embodiments, the LLM comprises at least one of: GPT-4, LLAMA-3, BLOOM, PaLM, GPT-3.5, BERT, Gemini, LaMDA, and Falcon. In some embodiments, the transmitting the first computing prompt from the set of computing prompts to the LLM occurs at a first time.

At block **1008**, the method comprises receiving, using the one or more computing device processors, a first vector embedding for the first computing prompt from the set of computing prompts, from the LLM, wherein the first vector embedding comprises or is based on a first semantic structure of at least some first content comprised in or associated with the first computing prompt from the set of computing prompts. In some embodiments, the first semantic structure of the at least some first content comprises a conceptual meaning of the at least some first content.

At block **1010**, the method comprises transmitting, using the one or more computing device processors, a second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM. In some embodiments, the transmitting the second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM, occurs at the first time. In other embodiments, the transmitting the second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM, occurs at a second time. In some embodiments, the LLM may be a different LLM than the one used in the transmitting the first computing prompt from the set of computing prompts to the LLM. In other embodiments, the LLM may be the same LLM used in the transmitting the first computing prompt from the set of computing prompts to the LLM.

At block **1012**, the method comprises receiving, using the one or more computing device processors, a second vector embedding for the second computing prompt, different from the first computing prompt, from the set of computing prompts, from the LLM, wherein the second vector embedding comprises or is based on a second semantic structure or the first semantic structure of at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts.

Turning to FIG. **10B**, at block **1014**, the method comprises generating, using the one or more computing device processors, a first computing prompt group comprising: the first computing prompt from the set of computing prompts, and the second computing prompt, different from the first

computing prompt, from the set of computing prompts, wherein the generating the first computing prompt group comprises clustering the first computing prompt from the set of computing prompts, with the second computing prompt, different from the first computing prompt, from the set of computing prompts, based on a first similarity of the first vector embedding and the second vector embedding. In some embodiments, the first similarity of the first vector embedding and the second vector embedding comprises a first semantic similarity. In some embodiments, the first semantic similarity comprises or is based on at least one of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity.

At block **1016**, the method comprises first accessing, using the one or more computing device processors, a first database. In some embodiments, an entry in the first database comprises a vector embedding and metadata associated with an indexed computing file. In other embodiments, the entry in the first database comprises a vector embedding and metadata associated with a file comprising unstructured data. In some embodiments, the first database may be filtered based on filter data. In some embodiments, the filter data may be received from the first system. In other embodiments, the filter data may be received from a data source.

At block **1018**, the method comprises determining, using the one or more computing device processors, for the first computing prompt from the set of computing prompts, using the first vector embedding, based on the first accessing the first database, at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, wherein the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, comprises first unstructured data, wherein the determining the at least one first file that partially corresponds with the first computing prompt from the set of computing prompts, is based on a second similarity of the first vector embedding and at least one third vector embedding, wherein the at least one third vector embedding is associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts. In some embodiments, the second similarity of the first vector embedding and the at least one third vector embedding comprises a second semantic similarity. In some embodiments, the second semantic similarity comprises or is computed using at least one of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity.

At block **1020**, the method comprises second accessing, using the one or more computing device processors, the first database. In some embodiments, the second accessing the first database may happen concurrently with the first accessing the first database. In other embodiments, the second accessing the first database may occur prior to the first accessing the first database. In yet other embodiments, the second accessing the first database may occur after the first accessing the first database.

Turning to FIG. **10C**, at block **1022**, the method comprises determining, using the one or more computing device processors, for the second computing prompt, different from the first computing prompt, from the set of computing prompts, using the second vector embedding, based on the second accessing the first database, at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the at least one second

file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, comprises second unstructured data, wherein the determining the at least one second file that partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, is based on a third similarity of the second vector embedding and at least one fourth vector embedding, wherein the at least one fourth vector embedding is associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts.

In some embodiments, the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, comprises at least one of the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts. In other embodiments, the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, comprises none of the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts. In some embodiments, the third similarity of the second vector embedding and the at least one fourth vector embedding comprises a third semantic similarity. In some embodiments, the third semantic similarity comprises or is calculated using at least one of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity.

At block **1024**, the method comprises transmitting, using the one or more computing device processors, the first computing prompt group to the LLM. At block **1026**, the method comprises transmitting, using the one or more computing device processors, the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, to the LLM. At block **1028**, the method comprises transmitting, using the one or more computing device processors, the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM. In some embodiments, the transmitting the first computing group to the LLM may happen concurrently with the transmitting at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, to the LLM, and/or the transmitting the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM.

Turning to FIG. **10D**, at block **1030**, the method comprises receiving, using the one or more computing device processors, first structured data from the LLM, wherein the first structured data comprises or is based on the first computing prompt from the set of computing prompts, a first response associated with the first computing prompt from the set of computing prompts, a first citation associated with the first computing prompt from the set of computing prompts, a first file quality indicator associated with the first computing prompt from the set of computing prompts, the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second response associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second citation associated with the

second computing prompt, different from the first computing prompt, from the set of computing prompts, and a second file quality indicator associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts. In some embodiments, the first structured data comprises the set of computing prompts, a set of responses associated with the set of computing prompts, a set of citations associated with the set of computing prompts, and a set of file quality indicators associated with the set of computing prompts.

At block 1032, the method comprises determining, using the one or more computing device processors, a first computing indicator based on the first computing prompt from the set of computing prompts, and the first citation associated with the first computing prompt from the set of computing prompts. In one embodiment, the first computing indicator comprises a first confidence score associated with the first response associated with the first computing prompt from the set of computing prompts. At block 1034, the method comprises determining, using the one or more computing device processors, a second computing indicator based on the second computing prompt, different from the first computing prompt, from the set of computing prompts, and the second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts. In some embodiments, the determining the first computing indicator happens concurrently with the determining the second computing indicator. In other embodiments, the determining the first computing indicator occurs prior to the determining the second computing indicator. In yet other embodiments, the determining the first computing indicator happens following the determining the second computing indicator.

At block 1036 the method comprises generating, using the one or more computing device processors, second structured data comprising or based on the first structured data, the first computing indicator, and the second computing indicator. In some embodiments, the second structured data comprises or is based on the first structured data and a set of computing indicators associated with the set of responses associated with the set of computing prompts.

Turning to FIG. 10E, at block 1038, the method comprises generating, using the one or more computing device processors, based on the second structured data, at least one recommendation associated with the first database, the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, or the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts. In one embodiment, the at least one recommendation is associated with at least one third file associated with (e.g., stored in, managed by, etc.) the first database. According to some embodiments, the at least one recommendation may comprise suggestions to improve at least one file quality indicator associated with the at least one third file associated with the first database (i.e., improve quality of files in the first database). In some embodiments, generating the at least one recommendation may comprise using the LLM.

At block 1040, the method comprises transmitting, using the one or more computing device processors, the at least one recommendation to a second system, wherein the second system manages the first database. In some embodiments, the second system may, based on the at least one recommendation, update or modify the first database. For example, in one embodiment, updating or modifying the first

database may comprise or be based on a file quality assessment. In another embodiment, the updating or modifying the first database may comprise or be based on at least one request for at least one additional file to be inserted into the first database, wherein the at least one additional file comprises at least one improved file quality indicator (i.e., inserting files with improved quality that are relevant to the set of computing prompts (i.e., a questionnaire) into the first database).

Any of the embodiments, methods, systems, etc., described in this disclosure may be combined with any other embodiments, methods, systems, etc., thereby resulting in new embodiments.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to explain the principles of the disclosed subject matter and its practical applications, to thereby enable others skilled in the art to use the technology disclosed and various embodiments with various modifications as are suited to the particular use contemplated.

It is appreciated that the term optimize/optimal and its variants (e.g., efficient or optimally) may simply indicate improving, rather than the ultimate form of 'perfection' or the like.

Furthermore, the functions or operations described above can be implemented in digital electronic circuitry, in computer software, firmware or hardware. In particular, the disclosed techniques can be implemented using one or more computer program products. The computer program products, in some embodiments, comprises non-transitory computer-readable media comprising code configured to execute the disclosed approach, embodiments, methods, process flows, etc. Programmable processors and computers can be included in or packaged as mobile devices according to some embodiments. The processes and logic flows can be performed by one or more programmable processors and by one or more programmable logic circuitry. General and special purpose computing devices and storage devices can be interconnected through communication networks.

It will also be understood that, although the terms first, second, etc., may be used herein to describe various elements, these elements should not be limited by these terms. These terms are used to distinguish one element from another. For example, a first object or step could be termed a second object or step, and, similarly, a second object or step could be termed a first object or step, without departing from the scope of the disclosure. The first object or step, and the second object or step, are both objects or steps, respectively, but they are not to be considered the same object or step.

The terminology used in the description herein is for the purpose of describing particular embodiments and is not intended to be limiting. As used in the description of the disclosure and the appended claims, the singular forms "a," "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term "and/or" as used herein refers to and encompasses any possible combination of one or more of the associated listed items. It will be further understood that the terms "includes," "including," "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps,

operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

As used herein, the term “if” may be construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context.

Those with skill in the art will appreciate that while some terms in this disclosure may refer to absolutes, e.g., all source receiver traces, each of a plurality of objects, etc., the methods and techniques disclosed herein may also be performed on fewer than all of a given thing, e.g., performed on one or more components and/or performed on one or more source receiver traces. Accordingly, in instances in the disclosure where an absolute is used, the disclosure may also be interpreted to be referring to a subset.

What is claimed is:

1. A method for generating structured data from files comprising unstructured data, and transmitting recommendations, based on the structured data, to a system storing the files, using a large language model (LLM), the method comprising:

receiving, using one or more computing device processors, first data, associated with a first computing format, from a first system;

determining, using the one or more computing device processors, based on the first computing format, a set of computing prompts from the first data;

transmitting, using the one or more computing device processors, a first computing prompt from the set of computing prompts, to an LLM;

receiving, using the one or more computing device processors, a first vector embedding for the first computing prompt from the set of computing prompts, from the LLM, wherein the first vector embedding comprises or is based on a first semantic structure of at least some first content comprised in or associated with the first computing prompt from the set of computing prompts;

transmitting, using the one or more computing device processors, a second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM;

receiving, using the one or more computing device processors, a second vector embedding for the second computing prompt, different from the first computing prompt, from the set of computing prompts, from the LLM, wherein the second vector embedding comprises or is based on a second semantic structure or the first semantic structure of at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts;

generating, using the one or more computing device processors, a first computing prompt group comprising: the first computing prompt from the set of computing prompts, and the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the generating the first computing prompt group comprises clustering the first computing prompt from the set of computing prompts, with the second computing prompt, different from the first computing prompt, from the set of computing prompts, based on a first similarity of the first vector embedding and the second vector embedding;

first accessing, using the one or more computing device processors, a first database;

determining, using the one or more computing device processors, for the first computing prompt from the set of computing prompts, using the first vector embedding, based on the first accessing the first database, at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, wherein the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, comprises first unstructured data, wherein the determining the at least one first file that partially corresponds with the first computing prompt from the set of computing prompts, is based on a second similarity of the first vector embedding and at least one third vector embedding, wherein the at least one third vector embedding is associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts;

second accessing, using the one or more computing device processors, the first database;

determining, using the one or more computing device processors, for the second computing prompt, different from the first computing prompt, from the set of computing prompts, using the second vector embedding, based on the second accessing the first database, at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, comprises second unstructured data, wherein the determining the at least one second file that partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, is based on a third similarity of the second vector embedding and at least one fourth vector embedding, wherein the at least one fourth vector embedding is associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts;

transmitting, using the one or more computing device processors, the first computing prompt group to the LLM;

transmitting, using the one or more computing device processors, the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, to the LLM;

transmitting, using the one or more computing device processors, the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM;

receiving, using the one or more computing device processors, first structured data from the LLM, wherein the first structured data comprises or is based on the first computing prompt from the set of computing prompts, a first response associated with the first computing prompt from the set of computing prompts, a first citation associated with the first computing prompt from the set of computing prompts, a first file quality indicator associated with the first computing prompt from the set of computing prompts, the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second

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response associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, and a second file quality indicator associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts;

determining, using the one or more computing device processors, based on the first computing prompt from the set of computing prompts, and the first citation associated with the first computing prompt from the set of computing prompts, a first computing indicator;

determining, using the one or more computing device processors, based on the second computing prompt, different from the first computing prompt, from the set of computing prompts, and the second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second computing indicator;

generating, using the one or more computing device processors, second structured data comprising or based on the first structured data, the first computing indicator, and the second computing indicator;

generating, using the one or more computing device processors, based on the second structured data, at least one recommendation associated with at least one of: the first database, the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, or the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts; and

transmitting, using the one or more computing device processors, the at least one recommendation to a second system or a user associated with the second system, wherein the second system manages the first database.

2. The method of claim 1, further comprising transmitting, using the one or more computing device processors, the set of computing prompts and a set of responses associated with the set of computing prompts, to the first system.

3. The method of claim 1, further comprising:

accessing, using the one or more computing device processors, a second database; and

transmitting, using the one or more computing device processors, the second structured data to the second database.

4. The method of claim 1, wherein the first similarity of the first vector embedding and the second embedding, is based on a semantic similarity of the first vector embedding and the second vector embedding.

5. The method of claim 4, wherein the semantic similarity of the first vector embedding and the second vector embedding is calculated based on one or more of: cosine similarity, Euclidean distance, Manhattan distance, Minkowski distance, Chebychev distance, and dot product similarity.

6. The method of claim 1, wherein the second similarity of the first vector embedding and the at least one third vector embedding further comprises a semantic similarity of the first vector embedding and the at least one third vector embedding.

7. The method of claim 1, wherein an entry in the first database comprises a vector embedding associated with a file and metadata associated with the file.

8. The method of claim 1, the method further comprising updating or modifying, based on the at least one recommen-

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dation, the first database, wherein the updating or modifying the first database comprises or is based on at least one of: at least one first request for an additional file to be inserted into the first database, wherein the additional file comprises or is associated with an improved file quality indicator or the additional file comprises or is associated with second data not comprised in the first database, or at least one second request for an existing file in the first database to be updated.

9. A system for generating structured data from files comprising unstructured data, and transmitting recommendations, based on the structured data, to a second system storing the files, based on accessing a large language model (LLM), the system comprising:

one or more computing system processors; and

memory storing instructions that, when executed by the one or more computing system processors, cause the system to:

receive data, associated with a first computing format, from a first system;

determine, based on the first computing format, a set of computing prompts from the data;

transmit a first computing prompt from the set of computing prompts, to an LLM;

receive a first vector embedding for the first computing prompt from the set of computing prompts, from the LLM, wherein the first vector embedding comprises or is based on a first semantic structure of at least some first content comprised in or associated with the first computing prompt from the set of computing prompts;

transmit a second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM;

receive a second vector embedding for the second computing prompt, different from the first computing prompt, from the set of computing prompts, from the LLM, wherein the second vector embedding comprises or is based on a second semantic structure or the first semantic structure of at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts;

generate a first computing prompt group comprising: the first computing prompt from the set of computing prompts, and the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the generating the first computing prompt group comprises clustering the first computing prompt from the set of computing prompts, with the second computing prompt, different from the first computing prompt, from the set of computing prompts, based on a first similarity of the first vector embedding and the second vector embedding;

first access a first database;

determine, for the first computing prompt from the set of computing prompts, using the first vector embedding, based on the first accessing the first database, at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, wherein the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, comprises first unstructured data, wherein the determining the at least one first file that partially corresponds with the first computing prompt from the set of computing prompts, is based on a second simi-

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larity of the first vector embedding and at least one
 third vector embedding, wherein the at least one
 third vector embedding is associated with the at least
 one first file that at least partially corresponds with
 the first computing prompt from the set of computing
 prompts; 5
 second access the first database;
 determine, for the second computing prompt, different
 from the first computing prompt, from the set of
 computing prompts, using the second vector embed- 10
 ding, based on the second accessing the first data-
 base, at least one second file that at least partially
 corresponds with the second computing prompt,
 different from the first computing prompt, from the
 set of computing prompts, wherein the at least one 15
 second file that at least partially corresponds with the
 second computing prompt, different from the first
 computing prompt, from the set of computing
 prompts, comprises second unstructured data,
 wherein the determining the at least one second file 20
 that partially corresponds with the second computing
 prompt, different from the first computing prompt,
 from the set of computing prompts, is based on a
 third similarity of the second vector embedding and
 at least one fourth vector embedding, wherein the at 25
 least one fourth vector embedding is associated with
 the at least one second file that at least partially
 corresponds with the second computing prompt,
 different from the first computing prompt, from the
 set of computing prompts; 30
 transmit the first computing prompt group to the LLM;
 transmit the at least one first file that at least partially
 corresponds with the first computing prompt from
 the set of computing prompts, to the LLM;
 transmit the at least one second file that at least partially 35
 corresponds with the second computing prompt,
 different from the first computing prompt, from the
 set of computing prompts, to the LLM;
 receive first structured data from the LLM, wherein the
 first structured data comprises or is based on the first 40
 computing prompt from the set of computing
 prompts, a first response associated with the first
 computing prompt from the set of computing
 prompts, a first citation associated with the first
 computing prompt from the set of computing 45
 prompts, a first file quality indicator associated with
 the first computing prompt from the set of computing
 prompts, the second computing prompt, different
 from the first computing prompt, from the set of
 computing prompts, a second response associated 50
 with the second computing prompt, different from
 the first computing prompt, from the set of comput-
 ing prompts, a second citation associated with the
 second computing prompt, different from the first
 computing prompt, from the set of computing 55
 prompts, and a second file quality indicator associ-
 ated with the second computing prompt, different
 from the first computing prompt, from the set of
 computing prompts;
 determine, based on the first computing prompt from 60
 the set of computing prompts, and the first citation
 associated with the first computing prompt from the
 set of computing prompts, a first computing indica-
 tor;
 determine, based on the second computing prompt, 65
 different from the first computing prompt, from the
 set of computing prompts, and the second citation

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associated with the second computing prompt, dif-
 ferent from the first computing prompt, from the set
 of computing prompts, a second computing indica-
 tor;
 generate second structured data comprising or based on
 the first structured data, the first computing indicator,
 and the second computing indicator;
 generate, based on the second structured data, at least
 one recommendation associated with at least one of:
 the first database, the at least one first file that at least
 partially corresponds with the first computing
 prompt from the set of computing prompts, or the at
 least one second file that at least partially corre-
 sponds with the second computing prompt, different
 from the first computing prompt, from the set of
 computing prompts; and
 transmit the at least one recommendation to a second
 system or a user associated with the second system,
 wherein the second system manages the first data-
 base.
 10. The system of claim 9, wherein the first computing
 prompt group comprises at least three computing prompts
 from the set of computing prompts.
 11. The system of claim 9, wherein the first structured data
 comprises the set of computing prompts, a set of responses
 associated with the set of computing prompts, a set of
 citations associated with the set of computing prompts, and
 a set of file quality indicators associated with the set of
 computing prompts.
 12. The system of claim 9, wherein the LLM is hosted on
 a third-party server.
 13. The system of claim 9, wherein the LLM is hosted on
 a local server.
 14. The system of claim 9, wherein the system comprises
 or is comprised in one or more computing systems associ-
 ated with one or more locations.
 15. A method for generating structured data from files
 comprising unstructured data, and transmitting recommen-
 dations, based on the structured data, to a system storing the
 files, based on accessing a large language model (LLM), the
 method comprising:
 receiving, using one or more computing device proces-
 sors, data, associated with a first computing format,
 from a first system;
 determining, using the one or more computing device
 processors, based on the first computing format, a set of
 computing prompts from the data;
 transmitting, using the one or more computing device
 processors, a first computing prompt from the set of
 computing prompts, to an LLM;
 receiving, using the one or more computing device proces-
 sors, a first vector embedding for the first computing
 prompt from the set of computing prompts, from the
 LLM, wherein the first vector embedding comprises or
 is based on a first semantic structure of at least some
 first content comprised in or associated with the first
 computing prompt from the set of computing prompts;
 transmitting, using the one or more computing device
 processors, a second computing prompt, different from
 the first computing prompt, from the set of computing
 prompts, to the LLM;
 receiving, using the one or more computing device proces-
 sors, a second vector embedding for the second
 computing prompt, different from the first computing
 prompt, from the set of computing prompts, from the
 LLM, wherein the second vector embedding comprises
 or is based on a second semantic structure or the first

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semantic structure of at least some second content comprised in or associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts;

generating, using the one or more computing device processors, a first computing prompt group comprising: the first computing prompt from the set of computing prompts, and the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the generating the first computing prompt group comprises clustering the first computing prompt from the set of computing prompts, with the second computing prompt, different from the first computing prompt, from the set of computing prompts, based on a first similarity of the first vector embedding and the second vector embedding;

first accessing, using the one or more computing device processors, a first database;

determining, using the one or more computing device processors, for the first computing prompt from the set of computing prompts, using the first vector embedding, based on the first accessing the first database, first file data, associated with at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, wherein the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, comprises first unstructured data, wherein the determining the first file data, associated with the at least one first file that partially corresponds with the first computing prompt from the set of computing prompts, is based on a second similarity of the first vector embedding and at least one third vector embedding, wherein the at least one third vector embedding is associated with or comprised in the first file data;

second accessing, using the one or more computing device processors, the first database;

determining, using the one or more computing device processors, for the second computing prompt, different from the first computing prompt, from the set of computing prompts, using the second vector embedding, based on the second accessing the first database, second file data, associated with at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, wherein the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, comprises second unstructured data, wherein the determining the second file data, associated with the at least one second file that partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, is based on a third similarity of the second vector embedding and at least one fourth vector embedding, wherein the at least one fourth vector embedding is associated with or comprised in the second file data;

transmitting, using the one or more computing device processors, the first computing prompt group to the LLM;

transmitting, using the one or more computing device processors, the first file data, associated with the at least

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one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, to the LLM;

transmitting, using the one or more computing device processors, the second file data, associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, to the LLM;

receiving, using the one or more computing device processors, first structured data from the LLM, wherein the first structured data comprises the first computing prompt from the set of computing prompts, a first response associated with the first computing prompt from the set of computing prompts, a first citation associated with the first computing prompt from the set of computing prompts, a first file quality indicator associated with the first computing prompt from the set of computing prompts, the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second response associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, and a second file quality indicator associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts;

determining, using the one or more computing device processors, based on the first computing prompt from the set of computing prompts, and the first citation associated with the first computing prompt from the set of computing prompts, a first computing indicator;

determining, using the one or more computing device processors, based on the second computing prompt, different from the first computing prompt, from the set of computing prompts, and the second citation associated with the second computing prompt, different from the first computing prompt, from the set of computing prompts, a second computing indicator;

generating, using the one or more computing device processors, second structured data comprising or based on the first structured data, the first computing indicator and the second computing indicator;

accessing, using the one or more computing device processors, a second database;

transmitting, using the one or more computing device processors, the second structured data to the second database;

generating, using the one or more computing device processors, based on the second structured data, at least one first recommendation associated with the first database; and

transmitting, using the one or more computing device processors, the at least one first recommendation to a second system, wherein the second system manages the first database.

16. The method of claim 15, wherein the unstructured data comprises raw information or information without a predetermined structure or format.

17. The method of claim 15, wherein the first citation, associated with the first computing prompt from the set of computing prompts, comprises at least one of: at least some first text from the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, at

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least some second text from the second file data associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, a file name corresponding with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts or the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, and a page number corresponding with the at least some first text from the first file data associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, or the at least some second text from the second file data associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts.

18. The method of claim 15, wherein the first file quality indicator is based on metadata associated with the first citation.

19. The method of claim 15, wherein the determining, using the one or more computing device processors, based on the first computing prompt from the set of computing prompts, and the first citation, associated with the first computing prompt from the set of computing prompts, the first computing indicator, is based on a semantic similarity

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of the first vector embedding and a fifth vector embedding, wherein the fifth vector embedding is associated with the first citation.

20. The method of claim 19, wherein the fifth vector embedding comprises or is comprised in the at least one third vector embedding or the at least one fourth vector embedding.

21. The method of claim 15, wherein the one or more computing device processors are comprised in one or more computing systems, wherein the one or more computing systems are located in one or more locations.

22. The method of claim 15, wherein the at least one first recommendation associated with the first database comprises at least one of: at least one second recommendation associated with the at least one first file that at least partially corresponds with the first computing prompt from the set of computing prompts, at least one third recommendation associated with the at least one second file that at least partially corresponds with the second computing prompt, different from the first computing prompt, from the set of computing prompts, or at least one fourth recommendation associated with at least one third file that at least partially corresponds with a third computing prompt, different from the first computing prompt and the second computing prompt, from the set of computing prompts.

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